

Industry Specific Questions

Industry Specific Section Introduction

Before we begin the industry specific section, we wanted to give some words of advice.

It is rare to encounter industry related questions unless you're interviewing with a sector focused firm or for a specific group placement. Alternatively, you may encounter industry related questions if you bring up your personal interest in an industry on your own accord.

Each industry covered in this section contains questions related to:

- The recent trends and ongoing developments within the specific industry
- The unit economics of companies (revenue and cost drivers) within the specific vertical for financial modeling purposes

The ability to demonstrate an interest in an industry is one of the best ways to differentiate yourself.

Instead of expecting to be asked these questions word-for-word, identify a sub-industry or two that's of interest to you and strive to understand the underlying concepts, and then try to bring them up yourself when you're discussing your interest.

In addition, we have included questions on:

- Company-specific details on the industry's key players
- Historical events that helped shape the industry to how it operates today
- Technical terminology that you're not expected to know (especially if you're an undergraduate student); instead you'll learn this while on the job

If constrained on time, you should feel no obligation to review these extra questions. Instead, just ensure that you're aware of a few key industry trends.

Technology, Media & Telecommunications (TMT)

For our section on the TMT industry, we'll focus more on telecom and media as we have a separate section on the SaaS industry right afterward – albeit, there's a fair amount of overlap.

The two sections were separated as SaaS tends to involve metrics specific to early stage startups, whereas M&T interview questions are generally more about industry trends and the competitive landscape. Relative to SaaS companies, most telecom and media companies are valued in more traditional methods due to the industries being more mature and established.

SaaS companies, particularly those in the early-stages of growth, have their own set of metrics to evaluate their performance and use operational KPI based valuation multiples.

When it comes to following the technology news, sources such as [TechCrunch](#), [The Verge](#), and [Gizmodo](#) are great resources to stay up-to-date. But for more insightful commentary on the business side of technology and media, we suggest following Ben Thompson's blog, [Stratechery](#).

What are the most common types of business models in the telecom and media industry?

- **Traditional Equipment Sales/Wireless Services:** The first category comprises the traditional range of telecommunications equipment, networking products, and wireless services (e.g., Wi-Fi plans, cable television). As the oldest segment in TMT, recent revenue growth has been very low because this type of revenue is related to one-time equipment sales and then monthly plans, making revenue growth a function of geographic expansion (which has begun to stall). However, companies in this space tend to benefit from high operating leverage, barriers to entry, and lack of competition; hence, the anti-trust concerns. The incumbents are few and include companies such as AT&T, Verizon, and T-Mobile.
- **Subscription-Based Streaming:** The next category would include companies such as Netflix, Hulu, Spotify, in which business models are built around new customer acquisitions, minimizing churn, and increasing pricing by offering more value than their competitors. For companies in this segment, user count growth is the priority and many have no clear pathway towards becoming profitable without first achieving significant scale. Streaming has quickly become one of the most competitive spaces across all industries as traditional industries such as cable television and radio have been completely disrupted and new developments appear (e.g., cloud gaming, Esports)
- **Advertising:** This category would include companies such as Facebook, Twitter, and Google, which are companies that focus primarily on metrics such as MAU and DAU. Given their reliance on advertising, user engagement a key measure when assessing these companies' recent performance. While industry growth has been very strong for years now, particularly for the leading companies, the greatest hurdle looking ahead appears to be increased efforts by regulatory bodies to restrict user data collection and attempts to break-up these tech monopolies (citing anti-trust regulations).
- **Media Networks/Diversified Entertainment:** This category involves companies that create, acquire, and distribute programming content such as movies, films, theater events, and other media content. Through their broad reach, these companies can produce revenue from various segments such as advertising, affiliate fees, subscriptions, product sales, and licensing. Companies under this category could be best described as "media conglomerates" and examples include ViacomCBS, Disney, and NBCUniversal.

What are a few notable trends going on in the TMT industry right now?

- **Convergence of Business Models:** More companies are increasingly stepping outside into different segments either organically or through M&A. For instance, Google has countless subscription-based services, such as YouTube Music and Google Play Pass, as well as new ventures into communication services through Google Voice, home devices through Google Home, streaming through Chromecast, and fitness through its acquisition of Fitbit. Facebook has also been active by entering social commerce through Facebook Shops, music and video content creation through Creator Studio, and livestreaming through Facebook Pages and Facebook Watch Party. While this should come as no surprise for FAANG companies, this trend includes companies in low-growth segments as well, such as AT&T, which has continued its efforts to establish itself in streaming through AT&T TV, DIRECTV, U-Verse, and WatchTV. ViacomCBS is also planning to release its new streaming service Paramount+ in 2021.
- **Scale-Oriented M&A:** There have been many mergers of equals in recent years such as the merger between CBS/Viacom, Time Warner/AT&T, and T-Mobile/Sprint, which were all significant industry-changing mergers that raised regulatory concerns. Growth opportunities have run out in these areas, thus the pursuit of market leadership through consolidation of these low-growth industries has become more common due to the benefits from cost synergies, operating leverage, and diversification.
- **"Streaming Wars":** 2020 was a transformative year for the streaming video industry, as COVID-19 accelerated the adoption of video streaming with consumers streaming unprecedented amounts of video content through providers such as Netflix and Hulu. The competition has intensified as of late as more companies are increasingly moving into the streaming market, with some of the most notable entrants being Disney+, Apple TV+, HBO Max, Peacock, and Quibi just to name a few.
- **Hosted Online Events:** As lockdowns and social distancing measures prevented large social gatherings and events, many events have shifted to digital means. This has benefited media segments such as e-sports and livestreaming in particular. The number of major live musical or cinematic events hosted in real-time online has also increased with positive reception from consumers. The trend is expected to pick up and become more frequent in 2021 after a slow start following the initial outbreak in early 2020.
- **5G & Edge Computing:** 5G and edge computing are forecasted to be two of the highest growth segments, with adoption being led by large enterprises and increased B2B collaborations such as AT&T and Microsoft's multi-year alliance to work on the cloud, AI, and 5G. This trend is cited as being a central catalyst before 5G can reach mass adoption since 5G requires edge computing for its full capabilities to become more widely accessible given its reliance on the edge-computing infrastructure (likewise, edge computing requires more applications with 5G capabilities, particularly for mobile).
- **Consumer Usage of AI Assistants:** Sales in artificial intelligence (AI) assistants have increased amongst consumers and adoption is expected to increase, despite the security concerns. Many attribute this to work-from-home (WFM), which has led to consumers increasingly spending more on their homes (and thereby benefiting the "smart home" trend). This trend concurs with consumers' changed spending habits where more money is spent on their homes as seen in the record spending on gardening, home renovations, and various home-improvement projects.

Which valuation metrics are common to see for traditional telecom companies?

The category of traditional telecom companies would include mobile telecom service providers, convergent telecom service providers, cable operators, and data centers/infrastructure providers.

The traditional telecom industry is capital-intensive with high fixed costs, which are typically financed using debt as these companies operate in a highly concentrated, mature industry with minimal cyclicity. Because of the large amount of debt in their capital structure, telecom companies will incur substantial debt-related expenses. In addition, high amounts of depreciation will be recognized each year given their large fixed asset base. Another notable consideration is the tax incentives from the government provided for research & development (R&D), especially as the US attempts to build out its 5G infrastructure.

For the reasons stated above, accrual-based earnings are volatile year-over-year (YoY), making equity value based multiples like P/E less useful. Instead, the most commonly used valuation multiple is EV/EBITDA, but often EBITDA is on a run-rate basis or normalized over several years to account for inconsistencies.

EV/(EBITDA – Capex) is also common to see, as many consider it a better approximation of operating free cash flow given the capital intensive nature of the telecom industry and the need to adjust for capex. Telecom would be an example of an industry where trailing (LTM) and forward (NTM) multiples would be truly necessary to fully understand the valuation trends.

Unique to telecom companies, other multiples include EV/Data Centers, EV/Net PP&E, EV/Route Miles, EV/Fiber Miles, and EV/Access Lines in Service, EV/Broadband Subscribers, and EV/Broadcast Revenue.

As a side note, considering how diversified many of the traditional telecom companies have become, it may be necessary to perform a sum-of-the-parts valuation (SOTP).

What metrics would you use to measure user engagement?

In the context of media companies, user engagement is the level of involvement a customer has with a particular product, such as a website, application, or online platform. Higher user engagement rates imply users derive value from the product (leading to continued usage). This is of high importance for media companies because user engagement leads to customer retention and more recurring revenue.

User Engagement KPIs

- Daily Active Users (DAU) and Monthly Active Users (MAU)
- Active Subscriber Count
- Time Spent In-App Per Day or Week
- Pageviews/Website Hits
- Churn Rate (Retention %)
- Conversion Rate (Free → Paid Plan)

Which multiples are most commonly used to value modern media companies?

For high-growth media companies, operational KPI-based multiples are very common due to many of them being unprofitable or barely profitable. Traditional cash flow based metrics fail to capture the true value of many of these companies. By virtue of many of these companies having the objective of acquiring new customers, large losses will inevitably be incurred. Therefore, it would be unreasonable to assess a growth-oriented company based on profitability-based cash flow metrics.

EV/Revenue is often looked at, but it may not be the ideal multiple to look at if user growth is currently being prioritized over monetization. Instead, user count, growth in new customers, and the churn rate are the key value drivers used to assess performance. And multiples such as EV/MAU, EV/DAU, and EV/Monthly Subscriber Count may provide a better indication of the company's value.

For a company with a product meant for high frequency in usage, what is one way to assess user engagement?

A popular metric for measuring the level of user engagement ("stickiness") is the ratio between daily active users (DAU) and monthly active users (MAU). The DAU/MAU ratio, expressed as a percentage, is the proportion of monthly active users that engage with an application in a single day.

$$\text{DAU: MAU} = \frac{\text{Daily Active Users}}{\text{Monthly Active Users}} \times 100$$

For example, if the company's app has 500 DAU and 1,000 MAU, then the DAU/MAU ratio is 50%. This can be interpreted as the average user engaging with the app roughly 15 days in a 30 day month.

According to Sequoia, the standard DAU/MAU is between 10% and 20%, but certain apps such as WhatsApp can easily top 50%. Note, this metric would only be useful for products with daily use (e.g., social media, messaging platforms, mobile applications). It wouldn't be useful for products that don't require daily use. For example, this metric would be useful for companies such as Facebook, Twitter, and Snapchat, but not applicable for Airbnb, Uber, and Lyft.

What are economies of scale and could you give me an example?

Economies of scale occur when the per-unit costs of production decrease as output increases. These cost savings from greater scale will result in higher margins. The cost per unit decreases as more output units are produced because the costs are being spread over the increased number of goods. An example of a company benefiting from economies of scale would be Apple. Because Apple sells millions of iPhones each quarter, it can commit to component purchases at a massive scale with significant negotiating leverage that results in volume discounts (and a lower average cost per unit).

What effect does having high operating leverage have on the scalability of a business?

The operating leverage represents the proportion of a company's cost structure that consists of fixed costs, as opposed to variable costs. Thus, companies with a higher proportion of fixed costs in their costs structure have greater operating leverage.

- **High Operating Leverage:** If the company has high operating leverage, each additional dollar of revenue can be brought in at higher profits once the fixed operating costs are paid. Thus, each marginal unit is sold at a lesser cost, creating the potential for greater profitability since fixed costs such as rent and utilities remain the same regardless of output.
- **Low Operating Leverage:** If a company has high variable costs, each additional dollar of revenue may generate less profit as costs proportionally increase alongside increased revenue (i.e., the variable costs offset the additional revenue). If revenue were to increase, these costs would rise in tandem (or vice versa).

If a company has high operating leverage, earning outsized profits on each incremental sale is plausible.

Can you give me an example of a company benefiting from operating leverage?

An example would be a telecom business that has finished building out its network infrastructure. Initially, the business will incur substantial upfront capex to enable connectivity and network capabilities (e.g., equipment purchases, construction, security implementations).

But once the network has been built out and operations are running, each new customer acquisition comes at a low incremental cost, as the cost of adding one customer to an existing network is inexpensive. Most of the expenses incurred are mostly maintenance-related later on. The initial investment will eventually be earned back and what remains is a high margin business with recurring revenue.

When might having high operating leverage not benefit a company?

Companies with high operating leverage have the potential to earn more profits on each incremental sale as the business scales. The caveat being, if sales and customer volume turn out lower than anticipated, the company can end up in financial ruin, given the initial investment spent and fixed cost structure. This means companies with high operating leverage have more business risk (i.e., higher risk of insufficient profitability) because the break-even point has been moved higher. If sales were to decrease, a business with a variable cost structure could easily reduce costs, but a company with a fixed cost structure has very limited flexibility.

What are network effects and can you provide some examples?

The network effect is a phenomenon whereby the value of a product or service offering increases with each incremental user and increased adoption. Network effects compound once critical mass is attained, meaning past this inflection point, new customers' acquisition experiences a domino effect where less effort and monetary investments are required.

- Facebook is an excellent example of the network effect, as its advertising revenue took off once its user base grew and customer engagement increased. By virtue of being the largest, fastest growing social media platform, Facebook obtained a durable moat in the advertising market, which led to an influx of advertisers wanting to place ads on Facebook's platform (driving up ad revenue) and new opportunities for different products and services to be introduced.
- Another example would be Google's search engine, which provides more accurate results as more users use the platform. Here, the network effects led to more advertising revenue and a more effective product as more usage of the product by users increased the product's value for all users. This explains why Google has dominated the search engine market with a near ~87% market share.

What is pricing power and how would you measure it?

While there's no exact formulaic method to calculate a company's pricing power, a useful question to ask is: "If the company raised prices by ~25%, what would the impact on customer retention be?" It's a positive sign if customer retention remains relatively stable after raising prices.

If a company has pricing power, it can raise prices and not see a substantial customer churn increase. The amount of pricing power a company is determined by how essential a product is to the users, how unique the value it provides, and the availability of (or lack of) other alternatives in the market.

Pricing power is often the byproduct of a differentiated product, which is essential to the users and unique in the value provided.

What is the difference between a cyclical and secular trend?

- **Cyclical:** A cyclical trend will be less predictable in terms of timing than seasonality and can result in more severe implications on the financial performance of a company. Cyclical trends can be near impossible to time. An example would be the semiconductor industry, which is largely driven by GDP growth and spending trends by enterprises.
- **Secular:** A secular trend is a long-lasting, often permanent change in an industry because it's related to new technology developments or changing consumer behaviors. An example would be the rise of Netflix and how it benefited from the secular trend of over-the-top ("OTT") content. Most secular trends are here to stay and these shifts are often gradual with plenty of time for companies to make adjustments, but many still ignore ongoing trends until their products/services become obsolete (Blockbuster in this example)

What are some examples of barriers to entry that can help protect a company's profit margins?

- **Network Effects:** Network effects are the incremental benefits from more users joining a platform and once a platform has attained critical mass in users (i.e., reached the inflection point of product adoption), it becomes very difficult to take market share away.
- **Economies of Scale:** Improved cost structures from the increased scale can be a barrier to entry that deters competitors, as the existing incumbents have a clear advantage in profitability and thus have more cash flows to reinvest into the business. Since the unit costs of a product decline as the volume increases, new entrants would come in with a significant cost disadvantage right away.
- **Proprietary Technology:** By having a differentiated offering that nobody else has, competition would be non-existent (or minimal), especially if there are patents involved.
- **High Capex Requirements:** Capital requirements related to infrastructure, machinery, and R&D to get a business started can deter new entrants.
- **Switching Costs:** Unless the new entrant has a significantly better product/service than the current offerings, switching costs can serve as a barrier (i.e., the switching costs outweigh the benefits).
- **Regulatory Hurdles:** The government and regulatory requirements can serve as barriers to entry, especially in highly regulated areas such as healthcare. These requirements to receive approval to enter a market can deter new entrants but benefit existing incumbents.

Consistent profits that can be sustained over the long-term require differentiation in an industry with high barriers to entry.

What is 5G, and what benefits will it provide?

5G is the next-generation network that's much faster (at full capacity) than 4G that it has the promise of dramatically accelerating the development of smart cities, autonomous vehicles, drone technology, virtual reality, public transportations, and manufacturing, agriculture, health care, and many other industries.

The 5G communication system claims to provide 100 gigabits per second, which would be 100x faster than the speed of 4G communication. This coincides with the reduced latency and clearance of bandwidth issues associated with emerging new technologies (which 4G struggled to handle).

The nationwide deployment of 5G has come at a slower pace than advertised. Today, many users with 5G devices are actually on a slower form of 5G because many carriers are rolling out different tiers of 5G (i.e., technically no mobile devices are on "5G" as of the current date). While the major US carriers have been racing to launch 5G this year, successful implementation in some form or another has only been in select regions and has been inconsistent. For those in less developed cities, the step-up from LTE (e.g., the download speeds) has been marginal, and many users have been left disappointed by the barely noticeable change.

The shift towards 5G could take up to several years as this is the natural process of technology deployment, especially given its potential implications on society. However, 5G will be well worth the wait since it'll be the catalyst of what takes IoT to new levels and allows applications such as robotics, industrial automation, and autonomous vehicles to become a reality.

The term "latency" often comes up when discussing 5G. What does it mean?

To enable connectivity between devices, large amounts of data are being exchanged online. A delay in transmitting data is referred to as latency and can hinder the user experience. Thus, reduced network latency when transmitting data packages prevents irritation and frustration from users (i.e., comparable to being stuck in road traffic). Latency is directly related to a network's bandwidth, which is a network's maximum data transfer rate over a certain period. While the two are not interchangeable terms, they're closely interlinked.

Which companies do you expect to benefit the most from the increased adoption of 5G?

In the US, the companies that stand to benefit will be the leading wireless networks such as AT&T, T-Mobile, and Verizon, especially when you consider their collective market share in the wireless carrier space. Many semiconductor companies are also expected to benefit, such as Qualcomm, Broadcom, and Qorvo. Qualcomm recently announced a new portfolio of 5G infrastructure semiconductor platforms, Broadcom is a leading vendor of chips for 5G smartphones, and Qorvo is a supplier at the forefront of radio frequency (RF) solutions.

Many enterprises are expected to test private 5G deployments. What are the implications?

The increased deployment of private 5G networks by enterprises means more next-generation wireless 5G radio antennas and transmission resources will be dedicated to a specific enterprise (and become independent of cellular networks). In effect, private 5G networks will bring bandwidth to areas that carriers have neglected or locations with greater security requirements. In theory, these private networks should be more secure as the enterprise (the network operator) will set up and monitor its security policies instead of relying on a 3rd party, and it would enable data to be stored locally on the nearby premises of the facility.

"Edge AI" is a common buzzword when discussing artificial intelligence. What does it involve?

Edge AI is a combination of edge computing and AI. Using too many devices and traffic on a single network can often clog the network and slow down the speed. To ease these issues of latency and bandwidth that hamper performance, machine learning algorithms can be run on a local server (or device) with edge computing capacity as opposed to remote servers and data centers – this is known as “Edge AI.” To further explain, the algorithms are processed locally on a hardware device requiring no connection to a remote server. The chip uses data generated from within the device and then processes it to give immediate, real-time insights with no delay from network congestion. Edge AI's main advantage is that Edge AI-enabled devices can process data faster and be more responsive relative to centralized IoT devices, leading to better customer experiences.

What do over-the-top services refer to and can you name a few examples of “OTT” providers?

Over-the-top (“OTT”) is the delivery of film and television content through the Internet, bypassing traditional cable and satellite TV services. Examples of OTT providers include Netflix, Hulu, and Amazon Prime Video.

The trend of OTT is being driven by cord-cutters and trimmers, which refers to consumers increasingly cutting off their cable connections to switch to a lower cost subscription-based OTT provider. Simply put, digital transformation has re-defined the media industry and traditional business models are being replaced by consumer-oriented experiences. Many consumers nowadays don't see the benefit of having access to programs they don't watch and desire the customization that OTT enables them to have.

Tell me about the pricing models used in the OTT industry.

Broadly, there are three types of pricing models used by OTT providers:

1. **Subscription Video on Demand (“SVOD”):** The SVOD pricing model provides video-on-demand services that enable users to access an entire library of videos for a recurring fee under a subscription (most commonly a month). Once the user has paid the fee, they can watch unlimited content on any device using internet access. The most mainstream examples of SVOD include Netflix, Hulu, and HBO. At present day, SVOD is the most common type of VOD offered by media companies.
2. **Transactional Video on Demand (“TVOD”):** Under TVOD pricing, users purchase content on a pay-per-view basis. Rather than unlimited access under SVOD, the users are charged per video or video package rather than the entire catalog. Examples of TVOD would be Google Play, Amazon Prime Video, and iTunes.
3. **Ad-Based Video on Demand (“AVOD”):** AVOD refers to video content that's available for free to consumers. Here, the revenue brought in is through ad revenue rather than directly from the consumers. An example would be Hulu, which offers SVOD plans in addition to a free AVOD plan.

What is the “varied access” pricing model?

The varied access pricing model provides different content access levels to subscribers (i.e., different subscription plans based on their preferences).

For example, Netflix offers its users options to sign up for a single, one-screen plan or a family plan option with additional features such as higher resolution.

What are the two categories of programming we see in OTT?

1. **Original Content:** Offering original content can give an OTT service provider a significant competitive edge. Therefore, a company such as Netflix invests considerable amounts in content development (e.g., *Stranger Things*, *Bird Box*, *House of Cards*) to gain an edge over its competitors.
2. **Licensed Programming:** Licensed programming includes agreements to replay content previously aired elsewhere. The benefit of licensed programming is lower costs but at the expense of less control over the content (e.g., Disney pulled its content and Pixar content from Netflix’s catalog as it rolled out Disney+).

How would you forecast the subscription revenue for Netflix?

The revenue of an OTT provider such as Netflix would be based on its ability to minimize churn. Netflix has established itself as the market leader with the highest market share.

From now on, Netflix’s success (or failure) in being able to fend off new entrants such as Disney+ and Hulu will be very impactful on Netflix’s revenue forecast and anticipated churn.

1. **Renewal Rate:** The first metric to begin with would be the number of paid memberships at the end of the period. This is mostly a function of the renewal rate of Netflix’s existing customer base. Netflix doesn’t disclose its customer renewal rate or churn, but there are streaming services analytics reports to use as estimates. The key question is: “Can Netflix hold on to the new customers that subscribed during the stay-at-home period, as well as prevent its customers from leaving for its competitors?”
2. **New Subscribers:** Then, the number of paid net membership additions can be forecasted. For a mature market-leading company such as Netflix, this metric was relatively stable, but COVID and the emergence of new competitors have made this a challenging task. One consideration should be Netflix’s pricing increase in late 2020 at a time when the competition in the OTT space has increased substantially. But most consider the increase to be modest and the impact on churn will be negligible. Plus, its new plans for up-and-coming content will be important – this would have to be compared to the offerings of its competitors (Hulu and Disney+ in particular). For this reason, the churn forecasted should be a wide range as it depends on how customers react to the new content quality.
3. **Average Monthly Fees:** Next, the average monthly revenue per paying membership would be forecasted. This will be straight-forward and done based on historical averages and trends, as well as with the recent price increase in mind. But the pricing plans shouldn’t deviate much in the future, especially with the ongoing attempts to undercut Netflix’s pricing. Since the subscriber counts are in annual figures, you would need to annualize this fee amount.
4. **Total Subscription Revenue:** To wrap up this basic revenue build, you would take the projected subscriber count (net of customer churn) and multiply this figure by the annualized average membership fee to get Netflix’s revenue for the year, and then continue this for the rest of the forecast.

For pricing, why does Disney undercut its competitors?

Disney+ is viewed as the main competitor of Netflix for the foreseeable future, but interestingly, the service cost was priced at only \$6.99 per month, vs. Netflix's basic plan of \$8.99. However, Netflix's more popular plans are its standard and premium plans, which were increased to \$13.99 and \$17.99 per month in 2020.

There are a few reasons Disney has the flexibility to take this aggressive strategy:

- First, unlike Netflix, whose revenue generation and business model is entirely based on subscription sign-ups, Disney is essentially a diversified conglomerate with enough funds to take on this type of customer acquisition strategy. While Disney+ is likely to see an increase in price later on to become a more profitable division (in ~2024, according to the management), its current focus is to steal market share from Netflix. By strategically undercutting Netflix and providing its own original content pulled from Netflix, Disney has a significant advantage.
- Next, Disney+ is not a priority for Disney. The larger, overarching focus of Disney is its core businesses (e.g., Disney Theme Parks, Disney Studios, Disney Original Content). By controlling the distribution of its own content by going D2C, Disney can strengthen its relationships with its customers and create a network effect where each offering builds upon each other – for customers and families, in particular, this creates a fully integrated, cross-platform user experience not just in terms of online media but in-person visits. Rather than dispersing its content through various channels, Disney has now become a one-stop-shop for its content, especially given the lucrative opportunities in video streaming on the horizon.

Disney owns Disney+, Hulu, and ESPN+. Why does Disney not just bundle them in one offering?

Disney actually offers this bundled plan, but many users choose to opt for individual offerings. From an initial glance, it might appear odd that Disney has three different streaming services. But the controlling stake in Hulu brought immediate synergistic benefits to Disney. Right away, Disney+ became a part of the "streaming wars" conversation before their offering became more developed. Hulu was long considered the primary competitor of Netflix, and now it became Hulu and Disney+ in an instant.

For ESPN+, sports differ from streaming TV shows and movies because real-time, live events are the preferred consumption method by consumers (i.e., the traditional TV model). Thus, ESPN and Disney could develop an affiliate fee business model and offer a pay-per-view or pay-for-access type pricing for this arrangement. And given how sporting events have natural breaks in between, this creates an opportunity for commercials (and thus greater ad revenue).

Many investors initially dismissed Roku following their IPO. Why has the company far exceeded initial expectations to date?

Following their IPO, many investors had a negative outlook on Roku, given the ongoing shift away from hardware. Competitors such as Apple with its cult-following were expected to steal Roku's customers with ease. But to the surprise of many, Roku is one of the market leaders in 2020 for connected TV platforms and has outperformed expectations. Despite still using hardware (e.g., the stick users plugged into the TV), Roku made several strategic moves that enabled them to have a leading position.

Roku partnered with television manufacturers to make Roku OS come built-in the TV right out of the box. Despite having an arguably outdated design, its simple interface, ease-of-use, and affordability created a niche. Next, Roku established partnerships with Netflix, Hulu, and Disney+ to allow its users to stream any of the top media apps from their TV. This level of independence from not trying to get customers to use a certain app over another (i.e., from Amazon, Google, Apple) is a key reason it has become the choice of consumers – all Roku concerns itself with is providing an easy, affordable way for customers to stream content from their homes.

Why has Apple shifted its focus from hardware to software?

Computing hardware sales peaked in 2011 and have seen a contraction in growth every year since, and it has become increasingly challenging for Apple to provide an incentive for customers to upgrade to newer models. Since hardware products are periodic one-time purchases, rather than recurring revenue sources, Apple has shifted its focus to building out its software and related services. Software-based services have a larger total addressable market (TAM) and lead to higher customer retention rates (i.e., product “stickiness”), thus Apple has shifted its focus on strengthening its online service offerings and further integrating the Apple ecosystem (e.g., iOS App Store, iCloud, iTunes, Apple Music, Macs). Once a customer enters this ecosystem, it becomes more difficult to leave. Many of Apple’s products are intentionally not compatible with other competitors’ products (e.g., cannot download Apple apps on a Windows laptop; only the web-based versions are available).

What is enterprise collaboration software?

Enterprise collaboration software provides teams with tools to share information, coordinate activities, manage projects, and communicate. The collaboration market consists of many submarkets – for instance, employee communication applications (real-time chat, video-conferencing), file sharing and content collaboration, and project management.

The global workplace collaboration software market is expected to exceed ~\$18 billion within the next few years as the adoption of integrated collaboration and communication software increases. Amid the recent global coronavirus crisis, collaboration software applications have become an increasingly integral part of how businesses’ workflow is organized and carried out. As enterprises were forced to function remotely, executives deployed new collaboration tools company-wide to connect physically distant teams. Zoom (ZM) is considered a primary beneficiary of the coronavirus outbreak, as many employers and schools were left with no other option but to use remote video-conferencing software.

Tell me about a few of the ongoing trends in the enterprise collaboration software industry.

- **Platform Vendors vs. Specialist Players:** As the market matures, collaboration solutions that offer a suite of offerings under a unified platform will have a key competitive advantage.
- **Third-Party Application Integrations:** To compete with larger established vendors, specialized single-application companies must include more third-party application integration within the collaboration ecosystem to increase end-user satisfaction.
- **Upcoming Consolidation Phase:** The market remains highly fragmented with a low barrier to entry, making it ripe for M&A activity (e.g., productivity apps, note-taking apps, time management apps, employee tracking, and scheduling apps). Thus, more enterprise software providers will include social software and collaboration into their product mix.
- **Shift Towards Remote Work:** Before COVID, approximately ~70% of the global workforce worked remotely at least once a week. In 2020, remote work became completely normalized in our society, and many large corporations have stated they intend to continue working remotely for the foreseeable future.
- **Security → Customer Trust:** Privacy and security concerns will be a major topic of discussion in the future – earlier in 2020, Zoom came under scrutiny after major security flaws and platform deficiencies were uncovered. Thus, Zoom rolled out more encryption offerings and acquired Keybase to strengthen its security and regain users’ trust.

There are two distinct groups in enterprise collaboration software, platform vendors and specialist players. Could you tell me how they differ?

1. **Platform Vendors:** Platform vendors have an existing portfolio of software products, an established brand name, and a trusted customer base. For example, Microsoft is a trusted brand with established security infrastructures and a loyal customer following - therefore, their strategic focus is integrating Teams into Office 365 and onboarding existing customers.
2. **Specialist Players:** Specialist players are new entrant, high-growth companies with one service offering (e.g., Zoom). While platform vendors can afford to offer free option plans over the long-term, specialist players must urgently acquire new customers and monetize them. For specialist players, the acquisition of new customers is the focus - whereas platform vendors can leverage their existing user base that's already familiar with using their products and have gained their trust.

How were Microsoft Teams able to capture market share away from Zoom?

MS Teams took significant market share away from Zoom with relative ease because Microsoft had an existing customer base. Microsoft was not necessarily better at acquiring new customers than Zoom or more innovative; instead, it had existing long-term customer relationships, and it was a more convenient change from the perspective of consumers. Many of whom were already users of Office.

Would you consider B2B software to be non-cyclical?

Robert F. Smith of Vista Equity Partners once stated that "Software contracts are better than first-lien debt. You realize a company will not pay the interest payment on their first-lien until after they pay their software maintenance or subscription fee. We get paid our money first."

Today, many B2B software applications have become essential, embedded parts of companies - meaning, the removal of a software product from a business would leave its operations unable to continue functioning. Also, most software arrangements are structured as long-term contracts and include high switching costs. For these reasons, B2B software would be considered non-cyclical and resistant to downturns.

In the context of technology applications, what does product bundling mean and could you provide an example?

The concept of product bundling is taking complementary products that frequently get purchased together and selling them as a whole. There is typically a direct complementary relationship between the products such as Adobe's Creative Cloud offerings (Photoshop, Illustrator, and After Effects), whereas with others, there's a broad, indirect link. An example would be an Amazon Prime membership and Amazon Prime Video, in which a customer having both doesn't add more value to each other (other than the savings from the discount).

Name a weakness in Spotify's business model that explains its lack of profitability.

While Spotify's fixed costs (e.g., SG&A and R&D) are in line with its comparables, Spotify's problem is its marginal costs. These marginal costs are mostly related to the royalty fees it pays to record labels, songwriters, and publishers. In contrast to Netflix, Spotify has close to no original content, although they're negotiating new deals for lower rates and their mass adoption of subscribers provides an incentive for record labels to partner with Spotify to expand their reach. Spotify's value proposition to its customers completely depends on record labels - who ultimately control the rates charged.

To add insult to injury, competitors such as Apple offered free trials that lasted up to 3+ months. Similar to the Disney/Netflix relationship, Apple's revenue comes from diversified sources, unlike Spotify.

Despite its low margins, why do you suppose Spotify offers so many discount pricing models on its offerings?

Despite its often-cited struggles with profitability, Spotify offers a wide variety of discounted plans such as a very popular student plan and partnerships with Hulu, Showtime, Starbucks, and more. While this may decrease revenue in the short-term, this strategy is based on the focus on customer retention and the belief that this metric, for the time being, is more important than ARPU. These discounted plans may cut into their ability to make more revenue, but its impact on finding loyal customers has been positive. The churn rate for Spotify has been on the higher end, but over time this will decrease as the retained customers become a reliable base to sell other products and services (e.g., podcasts, audiobooks).

For Spotify to become more profitable, its primary competitive advantage is its customer count and network. Thus, this metric must be prioritized to provide an incentive for artists of all sorts to partner with their platform and not have all the leverage when negotiating contract terms (i.e., the artist would be missing out if their music or content is not on the Spotify platform).

What is demand-aggregation and could you name some companies capitalizing on this trend?

Demand-aggregation companies have business models built around establishing themselves as the middlemen between businesses and consumers. The two most well-known companies that enable this type of feature would be Groupon and Yelp. Groupon is known for providing discount coupons to consumers and connecting them with retailers/restaurants, whereas Yelp is known for its crowd-sourced reviews on businesses of all sizes. Another example would be Apple News+, which aggregates the news from various news outlets and blogs for the customer to create their personalized, curated news platform.

Tell me about the trend of livestreaming.

Consumers are increasingly following live video streaming apps to interact with friends, family, and other users. The popularity of live streaming first became mainstream in South Korea, but there has been a surge in popularity as many consumers enjoy watching content as it's happening in real-time and interacting with the content creator. The amount of livestreamed content and speed of adoption shows a clear desire for live content worldwide. Examples of companies capitalizing on this trend are Facebook Live, Periscope, and Twitch.

Where do the opportunities for monetization in the Esports industry lie?

There are four general monetization strategies for the Esports industry: merchandising, sponsorships, media rights, and gate revenues. But most industry revenue comes from advertising, which is a direct function of audience count and engagement hours, rather than just viewership. Therefore, advertisers see a large market opportunity in Esports, as these streams often have some of the highest levels of engagement, and there's a level of trust between an Esports athlete and his/her fans that's rare to find these days. Esports athletes (especially those that stream on video platforms) often stream for hours daily and interact with fans, thus creating a sense of community. However, Esports is often considered difficult to monetize as the Esports audience is heavily skewed towards the younger demographic, who have less discretionary spending power.

Why is the semiconductor industry known for being cyclical?

The semiconductor industry highly depends on the economic conditions (i.e., usually follows GDP growth rate), as IT spending fluctuates heavily based on how the economy is doing. Besides being tied to enterprises' spending trends, it depends on new device purchases such as smartphones and laptops by consumers, which are very cyclical and decrease substantially during downturns. Another aspect is how the industry's products have short life-cycles and can become obsolete quickly once better new developments occur, even from minor incremental improvements. Hence, the industry is known for inventory build-up and write-downs of inventory.

What are the five types of virtualization?

Virtualization refers to running multiple operating systems on a single system simultaneously, resulting in reduced IT expenses and increased efficiency for businesses.

1. **Desktop Virtualization:** Desktop virtualization is when a desktop operating system can run as a virtual machine on a physical server with other virtual desktops.
2. **Application Virtualization:** Application virtualization packages several applications into a single application, which is then separated from the operating system to run in a “sandbox.”
3. **Server Virtualization:** Server virtualization enables many virtual machines to run on one physical server, which leads to more efficient utilization of the physical server.
4. **Storage Virtualization:** Storage virtualization is physical storage being grouped to have a single storage device in a virtual format.
5. **Network Virtualization:** Network virtualization combines hardware appliances and software to enable management from a single external virtual network (i.e., aggregate various physical networks into one network).

Could you name an example of virtualized distribution?

An example of virtualized distribution would be cloud-based gaming. Often referred to as “gaming-as-a-service,” cloud gaming offers gaming through remote servers and streams them directly to a user’s device. Cloud gaming is a form of virtualized distribution instead of dependence on hardware (e.g., traditional disks). When it was initially released to the public and used by consumers, it was compared to Netflix due to its similarities in the consumer experience.

What impact did the success of Fortnite have on business models in the gaming industry?

The online video game Fortnite was unique in that it had a “free-to-play” business model, yet it generated well over \$2 billion in revenue across all platforms and became one of the most successful games that brought mainstream attention to Esports. Rather than selling its game for an upfront fee, Epic Games (the publisher of Fortnite) made the game free to download – betting on its ability to monetize content through offering optional in-game purchases of different costumes, skins, Battle Passes, and V-Bucks.

By offering the game for free, Fortnite amassed a cult-like following in a very short duration because of the minimal friction for a user to play with their peers. And this popularity and customer acquisitions led to opportunities for sponsorships, ad revenue, and live competitions. In effect, the success of Fortnite made video game incumbents realize the revenue potential besides the initial purchase (e.g., Esports, advertising, in-app purchases, priced unlocks).

What will the implications of Apple’s IDFA changes in iOS 14 have on advertising companies?

During its annual Worldwide Developers Conference (WWDC), Apple announced that following the release of iOS 14, developers and publishers will be required to receive permission from users before using the Identification for Advertisers (IDFA), which is Apple’s proprietary mobile ad ID used for ad tracking and measurement. Apple has effectively restricted one of the important elements of in-app mobile advertising. The result is that developers must now explicitly ask users to gather their data and track them across mobile apps and websites accessed on the iPhone/iPad. While originally the feature was slated for release in fall 2020, Apple delayed the privacy feature change to early 2021 to provide developers with sufficient time to adjust.

This privacy feature is one of the most aggressive policy changes a tech giant has made in recent years, and what Apple users should expect to see is a prompt requesting their IDFA – as one would expect, most users are expected to select the decline option. Advertising companies such as Facebook have publicly announced this change has the potential to impact its ad revenue negatively going forward.

What did Apple announce in 2020 that will negatively impact Intel?

In 2020, Apple's CEO Tim Cook officially announced Apple would part ways with Intel and confirmed the company would begin its transition towards using its own custom-built ARM processors in its laptops and desktops (i.e., to proprietary ARM-based processors developed in-house). Apple stated its "Apple Silicon" chips would allow for better, more powerful Macs, and its own chips would bring a new level of performance speed while being more energy-efficient. Apple claims the chip's advanced power management capabilities would enable maximized performance paired with industry-leading battery life. The first Mac with Apple Silicon will be released at the end of 2020, and Apple expects the split with Intel to take approximately two years.

Tell me about the controversial T-Mobile and Sprint merger that brought up concerns of a monopoly and how it ended up being approved.

The T-Mobile and Sprint merger was finally approved and completed in 2020, after the initial deal announcement in 2018 and delays related to regulatory scrutiny. The merger was controversial as it effectively brought down the number of major cell phone carriers in the US down from four to three. As part of the agreement for the acquisition to be completed, T-Mobile agreed not to raise prices for three years. Thus, T-Mobile and Sprint users cannot see any price increases until 2023.

One reason the Justice Department agreed to the T-Mobile and Spring merger was because of DISH Network's involvement. To secure the Justice Department's approval, the two carriers agreed to divest certain assets to DISH, most notably its prepaid subsidiaries (e.g., Boost Mobile), spectrum licenses, and many of Sprint's retail stores. DISH would become a viable 4th wireless carrier, allowing for healthy competition to be maintained.

How have the acquisitions of Time Warner and DirectTV by AT&T performed to date?

In the past five years, AT&T has spent billions to position itself well for the streaming wars. Its M&A spending spree began with its acquisition of DirecTV in 2015 in a \$49 billion deal, which appeared to be a well-planned, strategic acquisition when media companies were benefiting from a surge in online video consumption from consumers. Then the hugely controversial merger with Time Warner took place in 2018.

Since then, AT&T's streaming ambitions have appeared overzealous, and it currently owns many seemingly redundant streaming brands such as HBO Go, HBO Now, AT&T Now, AT&T TV, AT&T WatchTV, AT&T U-verse, and DirecTV. By the end of the fiscal year 2019, AT&T carried more than \$150 billion in debt on its balance sheet – which flowed down to its customers through increased pricing. In its FY 2019 annual report, AT&T not only lost traditional TV service customers but internet video platform users. Initially, AT&T had expected to benefit from the influx of new subscribers acquired from DirecTV and use this leverage in negotiations with programmers. Then, by merging with Time Warner and the acquisition of HBO, AT&T would have access to original programming and be the most well-rounded premium service offering in both quality and quantity.

While the original grand vision was to create a media conglomerate; instead, it saddled AT&T with massive debt, higher pricing plans than its competitors, and a lagging market position despite all their spending. The questionable M&A strategy of AT&T gave them a seat at the table for the streaming wars and made them a more modern, diversified company, but it came with such significant expenses that to this day – it's not yet clear whether those acquisitions were the right decisions (or if they were all necessary).

What is the FCC agency and what is its role as a regulatory agency?

The Federal Communications Commission ("FCC") regulates all interstate and international radio, television, wire, satellite, and cable communications in the US. The agency consists of a five-member panel appointed by the President of the US and confirmed by the Senate. The FCC aims to protect consumers from unfair increases in prices or be forced to accept lower quality services and prevent companies from becoming monopolies to where others cannot join the market from the barriers being too high.

What is net neutrality?

Net neutrality is the idea that internet service providers (ISPs) should treat all online data equally, without preference towards certain users or restricting access content or discrimination towards others – hence, the name “neutral.” If net neutrality were not to exist, ISPs could place restrictions on the kinds of access to content on the Internet (i.e., hinder downloads, uploads, application usage). Also, the ISPs could intentionally slow down or speed up the internet based on where the access is based (or the owner).

How did TikTok become such a popular app in an industry that's considered as being near a monopoly in terms of competition?

Owned by ByteDance, TikTok is a video-sharing social networking application that allows its users to make and share short ~15-second clips. At first glance, the platform doesn't come across as noteworthy, given how similar it sounds to the features of Snapchat, Instagram Stories, and Facebook Stories.

But the reason TikTok achieved such rapid adoption was because of its focus on user-generated content. More specifically, just about any person could go “viral” and garner millions of views on this app. Compare this to a platform such as YouTube, where the front page is filled with content created by well-known people (often sponsored partners). A similar application, Vine, was one of the first short-form video sharing applications to gain mainstream adoption and was acquired by Twitter. However, Vine was eventually shut down, which many attribute to its biased algorithm in which endorsed content was disproportionately promoted on the app (making the content repetitive as the same branded creators would be shown and often recycled in the feed). Given how TikTok's main appeal appears to be the aspect of users discovering new content by unknown people, it seems unlikely TikTok will make the same mistake as Vine, especially given how the company has stated its mission is to enable everyone to be a creator and share their creative expression through their videos.

While this question won't appear in an interview, it's useful to understand the niche of the leading technology companies within an industry. Likewise: How has Snapchat performed well to this date despite many believing Instagram Stories (practically a copy-cat of Snapchat's features) would kill the app's usage?

What major licensing deal did Spotify complete in 2020?

In May 2020, Spotify shocked the podcasting market after announcing that Joe Rogan, the host of the Joe Rogan Experience, had agreed to an exclusive \$100 million licensing deal. The Joe Rogan Experience is widely considered as being one of the most popular podcasts in the US, with each episode receiving millions of views.

Spotify CEO Daniel Ek has stated that he had realized that podcasts and audiobooks would be the future of Spotify, rather than music alone. These acquisitions and partnerships accelerate the path for Spotify to become the leading audio platform. The gradual change in strategy could be seen in the acquisition of podcast networks Gimlet Media, Anchor, and The Ringer, but the acquisition of JRE was by far the highest-profile strategic move by Spotify to establish itself in the podcast market.

Tell me about an ongoing trend in the news industry.

Nowadays, many writers, including reporters and journalists, are increasingly starting their independent online newsletters. These writers are creating mini-news platforms through an online email newsletter. With distrust of the media at an all-time high, a growing number of writers and readers have been gravitating towards newsletter platforms such as Substack. By doing so, creators have full autonomy over not only the pricing but their content. Also, rather than generating revenue through advertisements, these writers are paid through subscription plans that many of their readers pay each month. This trend of newsletter platforms is the decentralization of the news, which has increasingly become governed and censored by publishers.

Other examples include the continued digitalization of news to more mobile, news aggregation services (e.g., Reddit, Apple News), Twitter as a news platform, and daily/weekly curation newsletters.

Software as a Service (SaaS)

What are the advantages/disadvantages of being a vertical vs. horizontal software company?

Horizontal Software

- **Advantages:** Horizontal software companies provide complete, all-encompassing solutions for their customers, that can be used across a broad range of industries (e.g., Office 365, Salesforce CRM, QuickBooks). Simply put, these products each have larger TAMs and therefore have greater potential for total revenue. A company with a small fraction of a large market could generate more revenue than a market leader in a small market. By even by capturing a fraction of a large market, a company can make enough revenue to find its niche and refine its business strategy (even if it fails to become a market leader). Part of the reason these companies have more time to adjust their strategy is that the markets are large and take time to saturate. Thus, horizontal software companies can pivot and narrow down their target customer over time based on which end markets are most profitable.
- **Disadvantages:** The downside being SaaS tends to consist of "winner takes all" markets and only a few companies will end up dominating a market as they become the standard products used across most industries. As a result, competition in horizontal software markets tends to be more cut-throat, while involving larger, well-funded companies (often publicly-traded). As a result, sales & marketing spend is noticeably higher for horizontal software companies, given the extensive number of potential customers and the competitive nature of customer acquisitions (i.e., numbers game and a race to reach customers).

Vertical Software

- **Advantages:** Vertical software companies target specific niche segments. If a vertical software company comes in with a product that adds meaningful value, it can quickly establish itself as the industry leader. Once market leadership is established, the company can then create a tailored suite of solutions based on their understanding of their end market's specific challenges and needs. This further enables them to create a durable moat and even have the potential to change the industry as a whole. Hence, these companies experience lower churn (less threat from larger enterprises) and don't have to spend as much on sales & marketing.
- **Disadvantages:** The key drawback to vertical software is the amount of risk being undertaken as many of these markets are neglected for a reason (e.g., technical hurdles, specialization requirements, research & development costs). In the view of many, the potential revenue and market potential doesn't justify the expenditures. The development of a viable product has the potential to take longer and incur more expenses before reaching the market. So by entering this market, the company is making a high-risk bet that it can gain traction in this focused segment. Another drawback is, even if the company becomes a market leader, growth opportunities will eventually slow down and the company will be forced to pursue expansion opportunities in adjacent markets to expand its geographic reach and total addressable market (TAM), making the gap between S&M spending narrow at scale.

Vertical software companies develop their products to solve problems for a specific end market, which are typically neglected niche segments.

Tell me about the trend of verticalization in the SaaS industry.

In the early days of SaaS, vertical software companies received less interest and lower valuations from investors when compared to horizontal software companies. The implicit assumption being that their limited scalability would make multi-billion dollar exits unattainable.

Today, nearly every industry runs on software or is affected by it indirectly, which can be attributed to vertical solution providers ("VSPs"). By creating specialized products tailored to serve the needs of new or underserved markets, many of these VSPs have proven to redefine their target industries and obtain market leadership.

Since many of these segments served were slow to adopt technology and neglected by many horizontal software companies due to the level of customization necessary and technical requirements, there was an abundance of opportunities to bring value-add solutions specifically put together for their unique needs and challenges. The common pattern seen was one product being integrated into an underserved niche segment and then over time, companies would develop a platform of tools tailored for their customer base. Given their timing advantage and the lack of competition in the market, this enabled their developers to focus on product development and further improve upon their competitive advantage.

The number of vertical software providers has increased substantially, along with premiums in their valuations. Through offering specific, tailored solutions designed to address the unique demands and challenges faced by niche areas, VSPs had the advantage of increased value-add opportunities from customization, a loyal customer base with less churn, and more pricing power.

Which valuation multiples are used most commonly in SaaS?

For public SaaS companies, EV/Revenue remains the primary valuation metric. But as the industry matures, the market is rewarding companies with healthy EBITDA margins more, and EV/EBITDA has increased in usage for industries where the market leaders have become clear as seen by their market share.

When looking at unprofitable companies, EV/Revenue and EV/ARR are two of the most commonly used multiples, along with user-based multiples such as EV/DAU, EV/MAU, and EV/Monthly Subscription Users.

Given the prevalence of acqui-hiring in SaaS, two multiples looked at in those acquisition-type situations are EV/Total Funding Raised and EV/Total Employee Count.

What is product-market fit?

The concept of product-market fit (PMF) is one of the key factors that determine whether an early-stage startup start-up can succeed or not. From Marc Andreessen's blog, he described it as "Product/market fit means being in a good market with a product that can satisfy that market." So product-market fit is the validation of the product in the market, as confirmed from consistent consumption by existing users, organic new customer growth, and increased word-of-mouth promotion.

PMF is more of a qualitative trait as determining the degree to how a product satisfies a market's demand and the extent of how much a product resonates with a particular market is difficult to measure, let alone quantify. Often, PMF is described as one of those attributes that can be recognized from customer engagement and feedback, as well as exponential growth as the product begins to "sell itself" and marketing takes off on its own.

Read More → [Pmarchive: The Only Thing That Matters](#)

What does the LTV: CAC ratio represent and why is it an important metric to track?

The first and arguably most important metric for evaluating software businesses is the ratio of customer lifetime value ("LTV") over customer acquisition costs ("CAC"). The LTV is the total value that a single customer will generate for a business over its lifetime, whereas CAC is the total cost incurred to make the customer purchase the products and/or services being offered.

$$\text{LTV: CAC Ratio} = \frac{\text{Customer Lifetime Value}}{\text{Customer Acquisition Cost}}$$

Put together, the LTV: CAC Ratio ratio effectively shows what the lifetime value of a customer is to a company relative to the costs associated with the acquisition of that customer.

The question being answered here is: "Is the company deriving enough value from their customers to justify the costs spent to acquire them?"

The LTV: CAC ratio matters because how much a company spends to acquire a customer should be a function of how much value it receives in return.

How do you calculate the customer lifetime value (LTV) and customer acquisition cost (CAC)?

1. First, the customer lifetime would be calculated using by dividing 1 by the monthly customer churn rate. Customer lifetime is the implied duration a customer will remain with a company.

$$\text{Customer Lifetime} = \frac{1}{\text{Monthly Customer Churn Rate \%}}$$

2. We'll then calculate the customer lifetime value (LTV), but we must calculate the average revenue per account (ARPA) before we do. To reiterate, the periods used must be consistent, so the ARPA calculation will use MRR as the numerator, which is divided by the total number of accounts.

$$\text{Average Revenue Per Account (ARPA)} = \frac{\text{Monthly Recurring Revenue}}{\text{Total Number of Accounts}}$$

3. Next, the LTV will be calculated using the formula below since we have the ARPA, which will be multiplied by gross margin % and then divided by the MRR churn rate.

$$\text{Lifetime Value} = \frac{\text{ARPU} \times \text{Gross Margin \%}}{\text{MRR Churn Rate \%}}$$

4. Now that we have the lifetime value, we only need to calculate the CAC. CAC is the sum of all S&M expenses divided by the total number of new customers acquired.

$$\text{CAC} = \frac{\sum \text{Total Sales \& Marketing Expenses}}{\text{Total \# of New Customers Added}}$$

5. Lastly, we have calculated the two metrics needed to calculate the LTV: CAC ratio.

$$\text{LTV: CAC} = \frac{\text{LTV}}{\text{CAC}}$$

What is the ideal LTV/CAC ratio that software companies target?

- An LTV/CAC of >3.0x is considered sustainable and the typical target for most software companies seeking continual growth.
- An LTV/CAC of <1.0x is an unsustainable rate and implies the company is having difficulty monetizing its newly acquired customers.
- An LTV/CAC of >5.0x+ for an early-stage company means the management may need to spend more on new customer acquisitions.

Conceptually, what does customer lifetime value (LTV) represent?

Simply put, customer lifetime value (LTV) measures the total amount of revenue a company receives from a single customer on average. LTV is a measure of how valuable the average customer is to a company during the entire lifespan over their relationship, which is critical to track because it helps predict the future revenue that can be expected from each customer. This figure is then compared to the CAC to gauge its cost-efficiency.

How do you calculate customer churn?

Customer churn represents the rate that customers canceled their products/services agreement over a specific time horizon. The churn rate (also known as the attrition rate) measures the customers lost in a period as a percentage of the number of total customers at the beginning of the period.

$$\text{Churn Rate \%} = \frac{\text{Total \# of Churned Customers}}{\text{Total \# of Customers at Beginning of Period}}$$

How would you calculate revenue churn?

The two most common revenue churn rates calculated are:

1. **MRR Churn:** This represents the lost MRR from the churned customer as a percentage of MRR.

$$\text{MRR Churn \%} = \frac{\text{Churned MRR}}{\text{Churned MRR}_{\text{Previous Month}}}$$

2. **Net MRR Churn:** This represents the lost MRR but accounts for the MRR expansion from existing customers. If expansion MRR offsets (and outpaces) the lost MRR, the net MRR churn will turn negative.

$$\text{Net MRR Churn \%} = \frac{\text{Churned MRR} - \text{Expansion MRR}}{\text{MRR}_{\text{Previous Month}}}$$

What are bookings and are they recognized as revenue under GAAP?

Bookings represent the value of a contract a customer has contractually committed to spend, usually agreed to on an annual or multi-year basis. Although not recognized as revenue under GAAP yet, it's a key indicator of the company's prospects and is used for forecasting purposes as it indicates the direction the business is heading in terms of growth and momentum.

For software companies, why are bookings a better proxy than revenue to measure growth?

Bookings are a more useful proxy to measure a SaaS company's growth because it's a forward-looking indicator that doesn't understate the true value of customer contracts the way accrual-based revenue does. For this reason, bookings are more useful when evaluating a company's sales & marketing performance and one of the main metrics used to portray projected performance when raising venture funding.

Why would it be a mistake to use bookings and deferred revenue interchangeably?

Many mistakenly use the terms bookings and deferred revenue interchangeably. Bookings represent the contractual commitment of customers to use their products or services. The difference with deferred revenue is the customer has neither received nor paid for the products or services. For deferred revenue, the revenue is similarly unearned, but payment was received upfront.

In SaaS, what are billings and how does it differ from bookings?

Billings are the invoices due for payment from a customer. For example, if a company secured an annual contract of \$12,000 with billings agreed to be on a quarterly basis, the total billings for the first month would be \$3,000 while the remaining bookings would be \$9,000.

An early-stage startup has a very low churn rate. Why might this be misleading?

Many newer, growth-stage companies will have very low customer churn rates, but this can be misleading. The reason being an "early adopter" is the least likely customer type to churn, as most are passionate about testing out the latest technologies and innovations. Thus, the low churn rate would not be an accurate proxy for their churn rates later on because initial adopters are closer to "product testers" than real customers.

In the SaaS industry, what is a reasonable customer churn rate?

Within the software industry, a churn rate of ~5% annually is the norm. However, this rate must be compared to its closest competitors to understand the end-users and understand the expected churn.

Why is product revenue preferred over service-based revenue in the software industry?

When most of a company's revenue comes from product revenue rather than service revenue, it leads to higher gross margins and better scalability. In comparison, services-based revenue is less-recurring, has much lower margins, and is less scalable.

What are monthly recurring revenue and annual recurring revenue used to measure?

Monthly recurring revenue ("MRR") is a measure of the recurring, predictable revenue generated on a monthly basis by subscription SaaS companies. MRR should include monthly recurring revenue from subscription fees, upgrades, up-selling, and cross-selling, as well as revenue lost from downgrades, discounts, and churn. MRR is equal to the monthly recurring revenue at the end of each month and is computed by taking the MRR from the previous month and adding net new MRR.

In contrast, annual recurring revenue ("ARR") is just MRR annualized and calculated as $ARR = MRR \times 12$.

What does annual recurring revenue (ARR) mean in the SaaS context?

For the SaaS industry, annual recurring revenue (ARR) is a measure of a software business's recurring revenue components on an annualized basis. The proper calculation of ARR will exclude non-recurring, one-time fees such as professional service, consulting fees, installation, and set-up fees. ARR is a key measure that reflects the amount of total revenue that's recurring in nature.

What are the flaws of the annual recurring revenue (ARR) metrics?

ARR is used to estimate revenue for the upcoming year, based on the most recent MRR. Implicit in its assumptions is that there'll be no customer churn, upselling, or downgrades. And that the latest month is the best indicator of its future performance.

How do you calculate net new MRR for a given period?

Net new MRR is computed by taking the new MRR from new customers, adding new expansion MRR from existing customers, and deducting MRR churn from lost customers.

$$\text{Net New MRR} = \text{New MRR} + \text{New MRR Expansion} - \text{MRR Churn}$$

What are the different types of expansion MRR?

- **Upselling:** Convincing existing customers to spend more on an upgraded version of the products/services they're already using with additional features, add-ons, or capabilities.
- **Cross-Selling:** Offering customers complimentary products or services that enhance the product the customer is already using.
- **Add-Ons:** Providing opportunities to unlock more features or widgets currently not part of a customer's plan, so additional features must be unlocked.

Why is the net dollar retention an important metric to measure alongside the ARR?

ARR cannot be looked at alone, as a SaaS company can grow ARR 100%+ each year yet have poor net dollar retention (<75%). Net dollar retention is a core KPI to look at when assessing an early-stage SaaS company's health, as poor NDR will eventually catch up and slow down the ARR if the underlying issues are not fixed. Net dollar retention is expressed as a percentage of the revenue from customers retained compared to the beginning period after accounting for upsells, upgrades/downgrades, and customer churn.

$$NDR = \frac{(\text{Starting MRR} + \text{Expansion MRR} - \text{Downgrade MRR} - \text{Churn MRR})}{\text{Starting MRR}} \times 100$$

- **NDR > 100%:** Indication of an increase in recurring revenue from existing customers – the top-performing SaaS companies can exceed this figure by a large margin, but most will target ~100% or slightly above.
- **NDR < 100%:** Contraction in recurring revenue due to downgrades in user consumption and churn.

How are bookings accounted for when calculating MRR?

Bookings represent the value of a long-term contract that has been committed to by a customer. To account for bookings when calculating MRR, the amount must be amortized across the length of the contract (i.e. monthly).

What are some ways that software companies can grow their recurring revenue?

- **New Customers Acquisitions:** Increase sales & marketing spend, enter new channels/markets
- **Higher Average Selling Price (ASPs):** Requires pricing power over their customer base
- **More Cross-Selling:** Easier to sell complementary products/services to existing customers
- **Increase Upsell Rate:** Benefits from the additional revenue derived from existing customers

What is CMRR, and how does it compare to MRR?

Committed Monthly Recurring Revenue (CMRR) is calculated with MRR as the base metric, and then it accounts for new committed signed contracts, expansion opportunities, customer downgrades, and expected churn.

$$\text{CMRR} = \text{MRR} + \text{New Bookings} + \text{Upsell Bookings} - \text{Downgrade Bookings} - \text{Churn}$$

Unlike MRR, CMRR is a forward-looking metric with more discretion on what to include and the amount. CMRR is, therefore, more speculative as it tries to forecast changes in recurring revenue, but is arguably a better approximation of how a company might perform in the future.

In the SaaS industry, how would you measure sales efficiency?

The various sales efficiency metrics and "magic number" are all variations of comparing new recurring revenue generated during a given period due to sales & marketing spend. As a broad generalization, these sales efficiency metrics are answering: "For each dollar spent on sales and marketing (S&M), what was the amount of new revenue generated as a result?"

1. **Gross Sales Efficiency:** Simply divides Gross New ARR (doesn't account for churn) by the S&M spend.

$$\text{Gross Sales Efficiency} = \frac{\text{Gross New ARR}_{\text{Current Quarter}}}{\text{Total Sales \& Marketing Expense}_{\text{Prior Quarter}}}$$

2. **Net Sales Efficiency:** Net Sales Efficiency accounts for both new sales and lost customers

To calculate the net sales efficiency, the Net New ARR is first calculated. The Net New ARR calculation begins with the Net ARR from new customers, adds the Expansion ARR from existing customers, and then deducts the Churned ARR from lost customers.

$$\text{Net New ARR} = \text{Net ARR} + \text{Expansion ARR} - \text{Churned ARR}$$

Then, you'll divide the Net ARR as of the current quarter and divide by the S&M spend of the previous quarter.

$$\text{Net Sales Efficiency} = \frac{\text{Net ARR}_{\text{Current Quarter}}}{\text{Sales \& Marketing Spend}_{\text{Prior Quarter}}}$$

For SaaS companies, what is the "magic number" and why is it used?

The main weakness with the net sales efficiency metric was that not all public companies disclose the necessary figures required for the calculation. To bypass this, Scale Venture Partners developed its own "Magic Number" to enable meaningful comparison amongst companies. Net New ARR is replaced with the difference between the two most recent quarterly GAAP revenue figures and then annualizes it.

$$\text{Magic Number} = \frac{[(\text{GAAP Revenue}_{\text{Current Quarter}} - \text{GAAP Revenue}_{\text{Previous Quarter}}) \times 4]}{(\text{Sales \& Marketing}_{\text{Previous Quarter}})}$$

A Magic Number <0.75 is considered inefficient, between 0.75 to 1 is efficient, and >1.0 is very efficient.

Read More → [A Primer on SaaS Sales Efficiency](#)

What is the CAC Payback Period?

The CAC payback period measures how many months it takes for a company to recuperate its customer acquisition costs. The shorter the CAC payback period is, the sooner the company can break-even and become more profitable. In terms of the industry benchmark, the general aim is less than 12 months. The top, best-in-class SaaS companies average ~6-12 months, but most average ~18 months.

$$\text{CAC Payback} = \frac{\text{Customer Acquisition Cost}}{\text{Average MRR Per Customer} \times \text{Gross Margin \%}}$$

How do you calculate ARPA?

The average revenue per account (ARPA) is a measure of the revenue generated per active account, typically on a per month basis. This metric is often used interchangeably by the average revenue per user (ARPU). An ARPA that's growing is a positive sign that more value is being derived from each account as the business scales.

$$\text{Average Revenue Per Account (ARPA)} = \frac{\text{MRR}}{\text{Total Number of Accounts}}$$

Alternatively, ARR can be used instead of MRR to convert ARPA to a yearly metric.

What is the difference between total contract value (TCV) and annual contract value (ACV)?

- **Total Contract Value:** TCV is the total value of the contract and is independent of the time frame (i.e., the period can be a month, six months, twelve months, etc.).
- **Annual Contract Value:** ACV measures the contract's value over twelve months – and is thus the better metric to use when making comparisons across the industry since it's annualized.

How can a software company decrease its churn?

- **Focused Market Segmentation:** The first method is to segment the customer base and identify the lower churn customers to direct selling efforts. Through trial-and-error, the company should understand its market in more detail over time and adjust based on where demand appears to be the strongest (i.e., double down on the most profitable segments). Besides product validation, narrowing down the target customer market should be a key priority of any startup.
- **Improved Customer Interface:** The software company should continuously maintain contact with its existing customer base and provide value through additional insights and discounts to preserve and better their relationship. While interfacing with customers, feedback should always be collected, analyzed, and taken into consideration.
- **Upfront Payments:** The software company could make additional efforts to promote its annual payment plan (i.e., more discounts, benefits) to receive the full payment upfront. With upfront payments or any long-term contractual commitments, it's far less likely for a customer to churn if they're on an annual plan.
- **Variable Pricing Model:** Another option is to create a pricing model that introduces more payment options based on variables such as the number of seats, number of licenses, or amount of features. While this may lead to some downgrades from users, increased optionality helps reduce lost customers.

Negative customer feedback that's left unaddressed, will inevitably become churn for the company later.

If a SaaS company has net negative churn, what does this mean?

Net negative churn occurs when a SaaS company's expansion revenue from existing customers exceeds the revenue lost from existing customers, from cancellations and downgrades. A key distinction of this metric is revenue from new customers is not factored in. Therefore, net negative churn means recurring revenue from existing customers by itself can offset higher-than-normal churn. While customer attrition rates want to be minimized, being capable of weathering a sharp decline in new customer acquisitions confirms the value being provided to existing customers.

What is the importance of TAM in SaaS investing and how is it estimated?

To estimate the total achievable revenue and quantify the market size and opportunity, the total addressable market (TAM) is estimated. Market size is a critical factor for SaaS companies and an area where investors spend significant time evaluating.

The most insightful part of market sizing analysis is arguably segmenting the market, which involves identifying which areas are growing or contracting, customer categorization, and spending trends/behavioral patterns (i.e., cohort analysis).

In general, SaaS is a “winner-takes-all” market, which makes market share and churn critical metrics to track.

- **Bottom-Up Approach:** While there are various ways to size a market, the most common method is the bottom-up TAM approach, which involves identifying the target customer profile, the approximate number of potential customers that meet the profile criteria, the growth in the target customers, and the assumed penetration of the segment. At its core, bottom-up revenue forecasts are based on quantity sold multiplied by the product or service price (key data points collected from past selling efforts). Listed below are a few examples of the most basic bottom-up revenue forecast drivers:

B2C SaaS: Revenue = Total Number of Users × ARPU

B2B SaaS: Revenue = Total Number of Customers × ACV

Volume Based Build: Revenue = GMV or TPV × Take Rate %

- **Top-Down TAM Approach:** In contrast, a top-down approach uses broader assumptions and estimates the TAM based on a market share percentage-based assumption on the total market size. To support their market sizing assumption, industry research and reports would be used. Then, the top-down revenue forecast would involve the TAM being multiplied by the implied market share %. However, this is a “quick and dirty” method to forecast revenue, rather than being precise by any means.

What is the SaaS Quick Ratio?

Mamoon Hamid popularized his variant of the Quick Ratio, which compares MRR growth (from New MRR and Expansion MRR) to MRR loss (from churned MRR and contraction MRR) in the same timeframe.

The formula for calculating SaaS Quick Ratio is shown below:

$$\text{SaaS Quick Ratio} = \frac{(\text{New MRR} + \text{Expansion MRR})}{(\text{Churned MRR} + \text{Contraction MRR})}$$

For interpreting the output, a ratio less than 1 means MRR is being outpaced by the rate of churn, a ratio between 1 and 4 means MRR is above churn but still losing lots of growth potential, and then the optimal target Hamid mentions is 4.

There are four factors in the formula:

1. **New MRR:** MRR increase from new subscriptions following new customer acquisitions and bookings.
2. **Expansion MRR:** MRR increase from upselling and cross-selling opportunities.
3. **Churned MRR:** MRR lost from the churn, in which customers have canceled their subscription.
4. **Contraction MRR:** MRR lost from downgrades in subscription plans.

What is cohort analysis and its primary use case for SaaS companies?

Cohort analysis is a behavioral analytical tool used by companies to understand their customers, which helps them take actionable steps towards improving customer retention and customer lifetime value. A cohort is defined as a subset (or group) of customers who share a certain characteristic.

Some of the most common types of cohorts used are:

- **Time-Based Cohorts:** Customers are grouped based on the date of their purchase.
- **Segment-Based Cohorts:** Customers are broken down by purchase (e.g., a specific product, plan, tier).
- **Size-Based Cohorts:** Customers are split based upon size (e.g., small businesses, large enterprises).

Once the cohorts have been defined, it becomes easier to analyze user behavior by each customer type and spot trends in the compiled data. Based on the patterns identified, the company then tests the hypothesis to determine its validity and effectiveness. One of the key insights that can be derived from cohort analysis is understanding why certain customers churn, and then what measures can be taken to minimize churn.

In SaaS, what is the purpose of A/B testing?

Typically associated with the hypothesis testing stage of cohort analysis, A/B testing involves software being used to track user behavior and gather insights after implementing a change. In most cases, the experiment will compare two or more variants (i.e., product modifications) and compare the outcomes. This enables the company to change the user experience gradually and make the optimal decision following user testing.

What is the difference between gross margin and contribution margin?

- **Gross Margin:** Measure of how expensive it's to develop a product, consisting of both fixed and variable product direct costs.

$$\text{Gross Margin (\$)} = \text{Revenue} - \text{Cost of Goods Sold}$$

- **Contribution Margin:** Calculated on a per unit basis, contribution margin measures profit per unit without considering fixed costs. The contribution margin enables you to list out the different product lines of a business to identify which ones are the most profitable on a per-unit basis.

$$\text{Contribution Margin (\$)} = \text{Unit Revenue} - \text{Unit Variable Costs}$$

What is the "Rule of 40" in the SaaS industry?

In recent years, a popularized measure of growth is the "Rule of 40," which states that 40% of a company's growth rate added to their profit margin should exceed 40%. The rule suggests a company with low (or negative) profits can still be valued at a high multiple as long as its growth rate counterbalances the cash burn.

Most often, MRR or ARR is used for the growth rate, while the EBITDA margin in the same period is used for the profit margin. Opinions differ on what stage this rule becomes applicable and its usefulness as a metric, but it's a simple measure to make sure the trade-off between growth and spend is balanced (and prevent the "growth at all costs" mindset).

Financial Institutions Groups (FIG)

Walk me through a bank's income statement.

1. A bank's income statement starts with interest income, net of interest expense. This is called net interest income and represents the difference between how much a bank earns on loans and how much a bank has to pay out in interest to get customer deposits that finance those loans.
2. The next major line in a bank's income statement is an expense called "Provision for Credit Losses," representing an accrual for expected losses due to bad loans.
3. Net interest income minus the credit loss provision represents a bank's core operating profitability.
4. Below this line (often called "Net Interest Income After Provisions for Losses") are non-interest related income and expenses – Banks recognize income from other activities, such as fees, commissions, service charges, and trading gains. The most significant non-interest expense is salary and employee benefits. Other non-interest expenses include occupancy and equipment expense, technology expense, amortization, marketing, and insurance expense.
5. Lastly, there's the tax expense line item, and then you get to net income.

Walk me through a bank's balance sheet.

- **Assets:** The largest asset for a bank is the bank's loan portfolio. Loans represent the bulk of a bank's assets and are typically comprised of residential and commercial real estate, as well as other consumer and commercial loans. The next largest assets will be investment securities and cash.
- **Liabilities:** Deposits represent the bulk of a commercial bank's source of funds. Interest-bearing deposits contribute to a bank's interest expense, while non-interest-bearing deposits don't. The rest of a bank's liabilities are typically short- and long-term borrowings.
- **Equity:** Like any non-bank company, equity will consist of common stock and treasury stock, preferred stock, retained earnings, and other comprehensive income.

How are the financials of a bank different from a traditional company?

For a non-bank, traditional company, revenues, cost of goods sold, and SG&A typically represent the main elements of operating income, while non-operating items like interest expense and income, other gains and losses, and taxes are presented below operating income.

Separating revenues vs. non-operating items like interest income and expense is not feasible for a bank since its core revenue comes from generating interest income while its primary operating expense is interest expense. A bank's income statement is driven by the size and returns of its assets (loans) and the cost of capital of its liabilities (deposits) in a far more direct way than traditional companies.

What is the impact of an inverted yield curve on a bank's profits?

Banks lend long term (mortgages, car loans, business loans) and fund this by borrowing short term (like deposits and interbank loans). As a result, the larger the difference (spread) between the yields on long-term bonds and yields on short-term bonds, the more profitable a bank will be, and vice versa.

Thus, in the case of an inverted yield curve, bank profits would be expected to contract as long-term yields decline relative to short-term yields. Needless to say, different banks will have different levels of exposure to this duration risk based on their particular investment and loan portfolio mix. But as a general rule, bank profitability declines as the yield curve flattens or inverts.

How do you value a commercial bank?

There are significant differences between the valuation of a non-bank and a bank. The most common valuation models employed to value a bank include:

- Levered Discounted Cash Flow Analysis
- Dividend Discount Model (DDM)
- Residual Income Model (RI)
- Comps with Equity Value Multiples (e.g., P/B, P/E)

All these valuation approaches are levered approaches that value the equity directly. That's because disentangling operating value from nonoperating value for banks is impossible, seeing that the core operations are fundamentally tied to generating debt-related interest income, net of interest expense.

For this reason, multiples used for comps analysis must be equity multiples like P/B and P/E, as opposed to the EV/EBITDA, EV/Revenues, and EV/EBIT multiples (which you would use for non-banks).

Walk me through valuing a bank using a levered DCF.

You can't value a bank using a traditional unlevered DCF because it's impossible to disentangle operating cash flows from financing cash flows, a key requirement in the unlevered DCF.

However, you can value a bank using a levered DCF:

1. First, levered free cash flows are forecasted explicitly for 5-10 years.
2. Beyond the final explicit forecast period, the terminal value is calculated using either the growth in perpetuity approach or exit multiple method.
3. Then, both the explicit levered free cash flows and terminal value (TV) are discounted back to the present at the bank's cost of equity since this is a levered DCF.
4. Finally, the sum of the present value (PV) of levered CFs represents the bank's equity value.

Walk me through a bank valuation using the dividend discount model (DDM).

Because banks typically have large dividend payouts, a dividend discount model is a common valuation approach. At a high level, the DDM process usually involves arriving at an equity value by forecasting future dividends and discounting at the cost of equity. The 3-stage DDM is common for bank DDM models and breaks up the forecast period into three distinct stages:

1. **Development Stage (3-5 years):** Forecast dividends explicitly and discount to PV using the cost of equity as the discount rate.
2. **Maturity Stage: (3-5 years):** Rather than making explicit dividend forecasts, derive dividends based on the assumption that ROE and cost of equity converge. The logic here is that a mature company cannot sustain ROE significantly greater than its cost of equity forever.
3. **Terminal Stage:** The final stage represents the PV of all future dividends once the company has reached maturity, assuming a perpetual dividend growth rate or terminal price/book multiple.

Walk me through a bank valuation using the residual income model. Why is it arguably better than the DCF or DDM?

The residual income valuation approach works around a significant shortcoming of the DDM – namely, its sensitivity to terminal value. The residual income valuation calculates a company's equity value as the sum of two components: Book Value of Equity + Present Value of Residual Income.

The present value of the residual income is estimated by looking at the extra equity value above the company's book value. For example, if a bank with a 10% cost of equity and \$1 billion book value of equity is expected to

generate \$150 million in net income next year, the residual income for that year will be \$50 million (\$150 million – [\$1 billion x 10%]).

Conceptually, this represents excess returns above the cost of capital, and all excess returns are added to the book value of equity to arrive at the fair value of equity. By the terminal value period, the assumption is that excess returns go away, which leads to a terminal value of 0. This avoids the terminal value issue in the DDM and DCF approaches, which commingle value creation and value preservation.

What multiples are appropriate for valuing a bank?

- Price Per Share/Book Value Per Share (P/B)
- Price Per Share/Earnings Per Share (P/E)
- Price Per Share/Tangible Book Value Per Share (P/TBV)

Why is the unlevered DCF approach inappropriate for banks?

The unlevered DCF approach would not be feasible for banks because it attempts to calculate cash flows on an unlevered basis (i.e., before the effects of debt and leverage). That would not be a plausible option, as it's unreasonable because most of a bank's revenue and cost of goods sold comes in the form of interest income and interest expense, respectively. While it's useful to separate operating earnings from interest income and expense in a non-bank, it's a pointless exercise for banks.

How would you forecast the revenue of a boutique investment bank?

Since this is a boutique investment bank, its primary revenue sources will be from fees from mergers & acquisitions advisory and/or helping companies raise debt or equity financing.

One approach to forecast revenue would be to use revenue per employee as the main driver. Investment banking is a relationship-driven business, thus the amount of deal flow is a function of their senior bankers' ability to bring in and close deals. The other assumptions needed would be the average revenue per employee and growth in employee count. As a sanity check, the implied annual deal count can be backed out of using the forecasted revenue and an assumption for the average deal size and the average fee structure (i.e., % transaction fee). This could be done to ensure the deal count is reasonably in-line with the past deal flow. Other considerations would be the industry or product group focus, as these influence the potential deal count, growth rates, and cyclicity in revenue.

Understanding a bank's trajectory, how it ranks compared to its competitors, the area it specializes in, and the quality of its employees is required to create an accurate forecast. This is why when large investment banks acquire boutiques, the acquirer is paying for the senior bankers' client relationships, their industry reputation/track record, and expected deal flow to be generated by the new additions to the team. Hence, earn-outs are often structured to prevent senior partners from leaving and to align incentives.

How does a passively managed fund generate revenue?

The amount of revenue generated by an asset management firm is a function of its total AUM. The management fee, which is charged as a percentage of AUM, is intended to cover the fees to oversee the portfolio and cover all operational expenses. The percentage would increase the more active the asset manager is in the investment selection process and the amount of specialization required.

Often called the management expense ratio ("MER"), this metric represents the fee charged to manage the capital invested in a mutual fund. MER is the total of a fund's management fee, operating expenses, and taxes for a specific year, divided by the fund's total AUM. Thus, MER is expressed as a percentage of the average dollar amount of a fund investment.

How would you value an actively managed alternative investment management firm?

Alternative asset management firms such as hedge funds and private equity firms generate two types of cash flows: management fees and performance fees. From investing in riskier alternative asset classes, these types of funds can charge higher management fees traditionally around ~2% with an incentive fee in which ~20% of returns past a pre-specified threshold go directly to the GPs of the firm.

Each has different risk profiles and thus merit a sum of the parts approach.

1. **Management Fees:** The first type of cash flow, and the easier one to model, is related to management fees. Most charge management fees of 1-2% on total capital raised. The most common valuation approach is to apply a market multiple to the current or next year's forecast for management fees.
2. **Performance Fees:** The second cash flow comes from less predictable performance fees (often called "carried interest"), which tend to be around 15-20% of the returns. Since these fees are realized upon exiting investments, they tend to be lumpy and difficult to forecast. The typical approach here is to apply a multiple on these cash flows, but a much lower multiple than the multiple applied to management fees.

What types of multiples are used to value asset management firms?

The most common multiples used to value asset management firms are the standard P/E and EV/EBITDA multiples. EV/AUM is occasionally used as well, however, this metric would only be useful if peer group is closely related to one another.

For example, EV/AUM cannot be used to compare a passive asset manager and an actively managed equity mutual fund since the expected returns and fees as a percentage of AUM vary.

What is securities lending?

Securities lending is when a lender provides stock, bonds, derivatives, or other financial instruments to a borrower. And in return, the borrower will transfer collateral to the lender and pay a borrowing fee. The collateralized transaction is a temporary transfer, meaning the borrower is contractually obliged to return the securities if requested or on the agreed-upon date.

What are the three main types of securities lending models?

1. **Direct Lenders:** Some investors have the scale to build their securities lending operation – this would include large asset managers such as Vanguard and Blackrock.
2. **Custodian:** Bundled option of custody, fund administration, and securities lending – custodians manage large pools of assets for several investors (i.e., pool collateral investment).
3. **3rd Party Lending:** Either linked to a prime brokerage or specializes in securities lending, these separate custodians can operate a third option for assets not held in their custodian operation.

What are the two fee models in securities lending?

1. **Discretionary Lending:** This is the more traditional lending model, in which the lending agent (or principal) will negotiate each loan in the portfolio with the borrower. All lending is conducted on a "best-efforts" basis. Here, loans are negotiated based on fluctuating daily demand levels in the marketplace; therefore, the lender receives the market rate with a fee paid to the agent who managed the lending.
2. **Principal Exclusive:** Under this lending structure, the lender (or their representative) negotiates an exclusive arrangement with a principal counterparty. The borrower pays a flat fee for exclusive access to a portfolio (or portion). The steady lending fees are agreed upon at the expense of more upside potential. However, these financial arrangements are negotiated and could include profit-sharing past a threshold.

What is FinTech?

Financial technology ("FinTech") encompasses the modernized technology-enabled delivery of financial services. By leveraging technology, various types of services can be delivered faster and more efficiently than before. Everything from mobile banking on smartphones, investing in stocks from a mobile application, and being able to transfer money to a peer online are all part of FinTech.

What multiples would you use to value a FinTech company?

In FinTech, the most commonly used multiples appear to be EV/Revenue, EV/EBITDA, and P/E. For sectors with higher-growth, their LTM multiples will often not be meaningful and will thus be accompanied by one-year and two-year forward multiples. Besides the leading companies in the industry, EV/Revenue is used the most often as many FinTech companies are unprofitable or barely profitable.

Industry Specific Multiples

- **Digital Banking and Payments:** EV/Total Payment Volume (TPV), EV/Total # of Peer-to-Peer Transfers, EV/Total # of Premium Users
- **Transfer Services and Marketplaces:** EV/Gross Transaction Volume (GTV), EV/Total Origination Volume, EV/Average Revenue Per Transaction
- **Stock Brokerages:** EV/Daily Average Revenue Trades (DARTS), EV/Total # of Premium Subscriptions

How do most FinTech start-ups generate revenue?

The business model most commonly seen in FinTech start-ups is a brokerage-type structure, in which the amount a company makes is fee-based (i.e., a small percentage of each transaction). This categorization would include start-ups in verticals such as digital banking, payments, online money transfer services, stock brokerages, and lending marketplace. For example, a P2P online money transfer service platform may charge a 2-4% fee for international transfers of a particular size.

Other business models can include service fees such as Robo-advisors in which a flat subscription fee is paid each month or interest payments in the case of direct lenders that lend off their balance sheet.

Tell me about the trend of challenger banks.

Challenger banks are branchless, digital banks – over the past few years, these startups have acquired millions of banking customers, with many of them attaining unicorn status in terms of VC funding. The name "challenger" is due to competing against the traditional incumbent banks that many consumers criticize for slow processing speeds, hidden fees, and lack of transparency. The leading challenger banks include Chime, N26, Current, Revolut, and Varo.

From the consumer perspective, what is the benefit of a challenger bank over a traditional bank?

The primary appeal of challenger banks is their online presence and mobile app development that resonates with digitally native consumers. Their rapid adoption could also be due to the easy setup process of an account (can often take only a few minutes), transparent fee structure and no hidden fees, lower rates for premium services and features, and an overall better online experience. Many challenger banks such as Chime have an immediate crediting of paychecks, forgoing the typical two to three-day float period that traditional banks have.

Are challenger banks actually "banks" in the traditional sense?

Many challenger banks don't start with a banking license; instead, they rely on a partner bank to operate. Later, as more customers sign up for their banking services, the challenger bank will either continue to depend on their established partnerships with other banks or apply for their banking license.

What is the difference between a neo bank and a challenger bank?

The terms neo bank and challenger bank are often used interchangeably and categorized together. However, the key distinction is that neo banks are completely digital and have no physical presence (i.e., no brick-and-mortar) – instead, they serve consumers through their mobile apps and online web platforms. Neo banks don't possess a full banking license and instead rely on a partner bank to operate (e.g., N26 and Visa), but challenger banks have a full banking license to operate with a minor physical presence.

Given their current business model, do challenger banks have a pathway to becoming profitable someday?

Challenger banks bring attention to their transparent, reduced fees, and most offer free basic accounts. But over the past few years, challenger banks have increasingly rolled out more premium account options that offer their top customers premium services. For instance, N26 offers a basic account that offers a few free ATM withdrawals and foreign currency payments, and alternatively a “Black” option with travel, smartphone, and insurance services.

Similarly, the business model of Revolut offers a free basic account and two subscription-type premium plans – and the company claimed to break even in terms of profitability in early 2017. While many of these startups are most likely burning cash, these startups have a single-minded focus on growth, and this freemium business model (with the option for premium services) is the best strategy to continue acquiring more customers while simultaneously raising more funding from investors.

Why might consumer FinTech start-ups appear to be struggling to monetize their user base?

The majority of FinTech start-ups struggle to generate revenue because their focus is neither on revenue growth nor profitability, but user base growth. How these FinTech companies plan to monetize their users and become a sustainable business remains unknown. Many start-ups have begun to launch premium, pay-to-use services, but leading FinTech start-ups such as Monzo and Revolut have revealed deepening losses, employee layoffs/pay cuts, and complaints from customers about lower-quality service.

One explanation for their struggles is that the primary appeal of these FinTech apps for many consumers is that these applications offer greater control, visibility, and convenience – all while being free. For financial services applications such as N26, Revolut, M1, Brex, and Chime, the reason many of these users (heavily skewed towards the younger demographic) signed up was because of the free pricing, the sign-up process can take less than an hour, and the interface was simple and easy to use. To use challenger banks as an example, many expected to become profitable through the interchange income it receives when customers use their debit cards, but this type of business model has proven to be difficult. In addition, many of these customers don't actually use these neo/challenger banks as their primary banking service, but rather as a side application.

Tell me about the current state of the banking services start-up scene.

Consumer digital banking services, one of the most well-funded areas in FinTech, has quickly become overcrowded and concerns regarding how users will be monetized are now more apparent. The need to generate more revenue (and profits) has begun to show as increasingly more banking start-ups are restricting their free options, charging for usage past a certain level, and aggressively pushing for premium plans sign-ups. Certain challenger banks have even started charging overdraft fees, fees for replacement cards, and higher withdrawal fees, which contradicts the original consumer-friendly marketing pitch.

The reason being many traditional banks have adapted well to the changing environment. To make matters worse, Big Tech and large corporations have gradually been moving into the space as well. Google appears to have doubled-down on its FinTech ambition when it revealed Plex, a mobile-first savings and checking bank

account integrated into Google Pay. Similar to many challenger banks, Plex has no monthly fees, overdraft charges, or minimum balance requirements, and Google has partnered with 11 banks to launch this service.

2021 will likely be a "make or break" year for many FinTech start-ups and shake out many well-funded players as seen by the bankruptcy of Monedo. As user growth becomes more difficult and the pressure to produce more revenue and profitability increases, it's likely we'll see increased mergers between FinTech start-ups, partnerships to integrate various services, or exits to larger companies.

How have traditional financial institutions responded to the new trends of digital banking?

The rise of challenger and neo banks exposed the inherent issues within the traditional banking system, which initially appeared to be in the process of being disrupted. However, many traditional, legacy banks rolled-out improved digital applications that slowed down much of the momentum surrounding consumer banking apps. This was also aided by the interest rate cuts in early 2020, which made many consumers return to their original banking provider.

Given the outside threat from the new digital banking entrants and pandemic, the entire banking industry had to respond by improving upon their digital infrastructures and platforms. Under great pressure to adapt under a limited time horizon, and these banks did just that. Nearly all the large banks have made initiatives to build out their mobile interface better, improve their communication (e.g., immediate digital customer service, chatbots), and increase transparency.

COVID-19 and the sudden necessity to build upon their digital banking solutions, looks to have been beneficial catalysts for change in traditional banks.

Tell me about the controversy surrounding Robinhood and payment for order flow.

Robinhood was charged in December 2020 by the SEC for misleading statements that failed to disclose the payments received from high-frequency trading firms for routing customer orders to them. In short, Robinhood failed to fulfill its obligation to its customers to execute orders at the best, available terms. Previously, investors were under the belief that Robinhood's primary revenue sources were coming from their Gold service memberships, interest collected from customers' cash accounts, and by lending securities. Robinhood didn't disclose that most of their revenue actually came from offloading transactions to brokers and dealers (i.e., selling customer order flow to trading firms) and receiving compensation for doing so.

Simply put, market makers such as Citadel Securities can pay e-brokers like Robinhood for the right to execute customer trades, and then Robinhood is paid a fee for the shares that are routed. While frowned upon for the lack of transparency, payment for order flow is not illegal per se and nearly all brokerages do it. But the larger issue was how much this type of revenue contributed towards Robinhood's total revenue, in which more than half was from payment for order flow. If one of the leading FinTech unicorns on the verge of an IPO is struggling to generate revenue from their internal products, it brings into question whether other FinTech start-ups can.

Why did N26 exit the UK market in early 2020?

N26 announced it was shutting down its operations in the UK earlier in 2020 due to Brexit. Many Fintech startups in Europe take advantage of a process called "passporting," which lets them apply for a banking license to operate in a European Union member state and then expand to all other EU states.

But following Brexit, regulations surrounding passporting will change – which led to N26's decision to exit the UK banking market because it owns a European banking license through the central bank of Germany and thus could not operate in the UK with its European license.

What does open banking mean?

The ongoing trend of open banking is the leading factor in the innovation of the traditional banking system. Open banking refers to the allowance of 3rd party access to consumer banking and financial account data specific to the customer. Given how integration is one of the key themes in FinTech, you could easily understand why the approval of a customer to give data access to these 3rd party providers has opened up so many opportunities for startups to enhance the consumer banking experience.

What role do APIs play in open banking?

APIs serve as a key component and a necessity for Banking-as-a-Service (BaaS) to be a possibility. BaaS is defined as the end-to-end process that connects banking institutions with 3rd party startups. The connection is enabled directly through APIs as they help develop layers of offerings over the traditional base banking core product while safeguarding the valuable financial information of the customers.

Does open banking harm traditional financial institutions?

It would be the contrary. The trend of integrating 3rd party offerings with traditional banking systems has been in the works for a long time due to regulatory hurdles delaying the process. But now that open banking has taken off, the refusal of a traditional bank would cause their customers to move to other providers that are more modern. These 3rd party integrations are complementary to banking services and give them more opportunities for growth and additional revenue. The inability to have integrations available would cause a bank to fall behind its competitors – thus, there's an incentive to race to have as many integrations as possible.

What is the state of the payments industry landscape today, and how has it been changing?

The payment industry dynamics have been shifting as more businesses and consumers move from cash to digital payments. It seems inevitable that contact-less, digital payments will become the norm for online and in-person purchases (i.e., NFC mobile payments). In recent years, many tech companies have come into the payments industry such as Apple Pay, Samsung Pay, Amazon Pay, and Google Pay. Considering how security requirements are in higher demand such as biometric authentication, sensor technology, and smart chip technology, it appears the payments market will be one of the most competitive areas of FinTech in the years ahead led by existing industry leaders such as Square and PayPal and the new entrants mentioned above.

What impact do you predict COVID-19 to have on the usage of QR codes?

While QR codes have been around for decades, the trend of no-touch payments has been positively affected by COVID-19. The recent increase in usage by consumers has shown how reliable QR codes are for no-touch, digital payments. Previously, QR code payments had not yet taken off notably in the US, but COVID looks to be a tailwind for the industry and could lead to further widespread adoption.

Many merchants are now accepting QR code payments (often not accepting cash payments), and more mobile applications such as UberEATS have introduced “contactless ordering” and encourage the usage of QR code payments to minimize the risk for both the customers and delivery person.

What does peer-to-peer (P2P) lending involve?

Often referred to as “social lending,” P2P lending is alternative debt financing without an official financial institution serving as an intermediary. P2P lending enables consumers to get loans directly from one another, most often through mobile phones. In most cases, the lending is facilitated by a P2P platform, where an automated system will match lenders with borrowers. As most of these platforms have lower overhead costs and operations are run online, their costs will be significantly lower. However, the highest volume count of P2P transactions occurs via transfer apps (e.g., Venmo, Cash App, Apple Pay) that enable one to send money conveniently through mobile after connecting their bank account to the application.

Can you tell me about some of the lending models that have emerged over the past few years?

- **Peer to Peer (P2P) Lending:** P2P lending platforms match borrowers directly to investors, both parties are typically individuals or small businesses. The site determines the rates and lending terms of the specific transaction given the risk, which is obtained using big data or other technology programs. But regardless, these platforms aim to meet the needs of borrowers that require liquid funds with those willing to deploy their funds as a lender.
- **Point of Sale Transaction-Based Lending:** In POS lending, the consumer is offered customized financing options by the merchant when making a purchase either in-person or online (i.e., installment payments, rather than one-time purchase).
- **FinTech Partner Bank Lending:** FinTech data aggregator start-ups, marketplaces, credit reports/B2B financing, and other start-ups that focus on a particular niche such as SMBs or consumers can partner with a traditional bank for financing and banking products to be provided based on the transaction data collected on the risk profile of the borrower enhance credit analysis.
- **Supply-Chain Finance Solutions:** Specialized non-bank Fintech platforms with supply chain management platforms now offer working capital and supply chain solutions to underserved SMBs (e.g., dynamic discounting and reverse factoring solutions).
- **FinTech Enabled Marketplaces:** Online marketplace lending connects borrowers with investors willing to offer loans based on their application, which was processed internally to accurately assess the borrowers' creditworthiness with advanced underwriting using ML and other consumer data. Compared to P2P, marketplaces are larger-scale and involve institutional investors who are prepared to pay sizable fees for the convenience of the transaction along with the creditworthiness check (very popular in the real estate industry). The aggregator model focuses on providing the end consumer different lending options available in the market based on the exact need of the consumer (tenure, amount, prepayment mode, etc.).

What is point-of-sale lending?

Point-of-sale lending, also known as "buy now, pay later" (BNPL), enables consumers to make purchases based on scheduled installment plans at the time of purchase for a minimal flat fee. POS lending has already seen substantial adoption in the US, as seen by the increased payment optionality for making purchases today.

A newly developing trend is businesses are increasingly transitioning to cloud-based POS, which should lead to faster checkout times (an area of inefficiency for POS currently). By enabling regular installment payments, typically on a monthly basis, consumers benefit from greater flexibility in payment structures, which has resonated among millennials in particular. At present day, the leading POS lenders include Affirm (co-founded by Max Levchin), Klarna, and Afterpay.

Why have lending companies such as Square relied less on the FICO score in recent years when assessing a borrower?

One advantage that digital lenders such as Square have over traditional banks is the amount of consumer data owned. Many FinTech companies, including Square and Stripe, care less about the FICO score, the traditional way of assessing the credit-worthiness of a borrower. Instead, these technology-based lenders examine the payment history from their platforms using advanced algorithms to spot trends such as overall payment volume, percentage of recurring customers, and payment frequency to determine the borrower's credit health. The reliance on technology rather than lengthy applications, background checks, and eligibility requirement forms enables them to issue loans faster and become self-reliant in their lending decisions.

What are real-time payment (RTP) systems?

In 2017, The Clearing House (TCH) introduced a new payment structure in the US, marking a major shift in the existing payment system. The new system permitted real-time payment (RTP) clearing instead of the traditional Automated Clearing House (ACH) that took between one to two days. The widespread adoption of the RTP system, thanks to mobile usage and technological advancements, have contributed to an environment in which many consumers, merchants, and financial institutions are enabled to send and transfer payments with faster settlement period, setting a new payment standard worldwide.

What role does Stripe play in the payments industry?

Founded in 2010, Stripe has become one of the most influential FinTech companies in the US and globally. The San Francisco-based startup was formed to build the “economic infrastructure for the Internet,” and today, it serves millions of companies of all sizes, including startups, e-commerce stores, and Fortune 500 companies, including tech-giant Amazon. Using Stripe’s software and APIs, commercial customers can process online payments and transfers. In addition, Stripe’s platform handles the technical implementation, fraud prevention, and banking infrastructure required for secure payments.

What did Square accomplish in 2020 that was an important milestone for the FinTech industry?

Earlier in 2020, Square received approval from the Federal Deposit Insurance Corporation (FDIC) to operate as a bank through a conditional Industrial Loan Company (ILC) charter. Upon approval of its application, Square announced in a press release that it plans to open a bank in Utah called Square Financial Services, which is expected to launch later in 2021. Square is expected to offer small business loans to merchants using their technology to process the payments, as well as issue deposit products. For nearly three years, Square had been pursuing an industrial loan company charter, which was opposed by many banking lobbyists and regulators. The last time the FDIC approved a de novo ILC application was more than a decade ago.

What led to the downfall of Lending Club, a former FinTech unicorn?

Lending Club was founded on the idea of bringing individual borrowers and lenders together (i.e., P2P lending platform), and when it went public in 2014, it was valued at more than \$9 billion. Lending Club envisioned individuals could lend to one another, and they could be the facilitating platform. However, the Company experienced a loan scandal involving the CEO and realized that their algorithm to assess consumer credit risk was not working as intended. Since then, their valuation has dropped to below \$350 million.

Lending Club was once considered a leader in FinTech and pioneer in P2P lending, but it recently acquired Radius Bank and announced its P2P lending platform would shut down. Lending Club became a bank through this acquisition and shifted its focus to institutional investors, as it learned how difficult it could be to assess individual borrowers' credit risk. The belief that individuals would be more responsible for repaying their debt if the lender was another individual rather than an institutional bank turned out to be a false presumption.

What is a payday loan, and how does the process work?

A payday loan is a very controversial, short-term loan that will rarely last more than a couple of weeks. These loans are usually provided by lenders operating from storefronts, but the payment industry’s shift online has affected this small lending niche. Payday loans are meant for an emergency, short-term need, and the application can often be completed within a matter of minutes. Once a payday loan is approved, the funds will be deposited into the verified bank account. But the payday loan lender would require the borrower to write a post-dated check in payment of both the principal loan amount and the high amount of interest charged. The post-dated check ensures the lender will be paid back on the agreed-upon scheduled date. These borrowers will accept the post-dated check arrangement in a troubled situation, since their credit history was not rejected and/or even reviewed by the payday lenders.

What is InsurTech?

InsurTech refers to advancements in technology to make the traditional, outdated insurance business model more efficient. For example, chatbots and smartphone apps are increasingly being used to streamline the industry's back-end process. The insurance industry is known for its slow adoption and for being inefficient. Because of this trend, insurance companies no longer need to rely on hiring employees dedicated to customer service as in the past. Other use cases of InsurTech include risk modeling to predict future losses, demand modeling to estimate premiums, fraud detection through recognizing patterns of abnormal behavior, processing claims using automation, and using machine learning to speed up the underwriting process. The result is digital premium payments and digitizing claims processing (leading to higher margins).

A few companies at the forefront of the InsurTech industry include Oscar Health, Root Insurance, Bright Health, Clover Health, Lemonade, and PolicyGenius.

What is crowdfunding and what are some of its benefits?

Crowdfunding is a type of fundraising in which a relatively small amount of funds are raised from many individual investors from the public, typically through online platforms.

This financing method allows individual and institutional investors to invest in private, unlisted entities in exchange for ownership in the entity – which is often a new business venture of a smaller scale. Crowdfunding takes advantage of the reach and easy accessibility of the vast networks of potential investors through the internet and can be beneficial for society as it encourages entrepreneurship from the expansion of the pool of investors from whom funds could be raised.

What does card-not-present fraud mean?

Card-not-present fraud is a scam in which the customer doesn't physically hand the card to a merchant during a fraudulent transaction. These frauds occur during transactions conducted over the internet or the phone – and are becoming more common as scamming tactics become more sophisticated, and more payments are made electronically.

What are Robo-Advisors?

Robo-advisors provide automated portfolio management and balancing based on the individual consumer's money management goals. Betterment was one of the first Robo-advisors to become mainstream and remains among the leading Robo-advisors. As opposed to having a financial advisor manage their portfolio and asset allocation on their behalf, these users depend on online platforms based upon proprietary algorithms. The need for human interaction and supervision (and labor costs from the perspective of the firm/bank) is reduced. Hence, financial institutions are investing heavily in AI-powered Robo-advisors and customer service chatbots. While software-driven programs have caused frustration for many banking clients in the past, gradually, these bots are becoming "smarter" and more accepted as informants for help. From the customer's perspective, Robo-advisors come with fewer fees than financial advisors and have a low minimum required initial investment, enabling everyday investors to test out the latest technology at an inexpensive price.

An important distinction is that Robo-advisors automate passive indexing strategies (e.g., ETFs, mutual funds) and are intended for long-term investing, which comes with far less risk. Since Robo-advisors invest in ETFs and related investments, they represent low-cost, personalized portfolio management while automating the process of portfolio rebalancing in real-time and tax loss harvesting (selling at a loss to offset a capital gains tax, and then replacing with a similar asset to maintain the target portfolio risk/return allocation).

What does the “underbanked” refer to?

The “underbanked” refers to the people worldwide who don't have access to banking services or live in areas with no retail branches of banks or formal financial services nearby. The ability of FinTech to enable these people that lack the proper access to a bank account is considered one of the most important purposes of FinTech – specifically, to location-independent challenger banks.

What are cryptocurrencies?

A cryptocurrency refers to a digital store of value that uses cryptography for security under a decentralized system. This means that no central regulatory body can manage an individual's digital wallets. Instead, cryptocurrencies rely on blockchain ledger technology and smart contracts.

How would you value Bitcoin?

Like all currencies, Bitcoin has no intrinsic value and cannot be valued like a traditional cash-flow generating asset. Instead, the pricing of bitcoin depends on perceptions on the probability that it'll ultimately become a full-fledged currency and stable medium of exchange.

Warren Buffett said on national TV that “cryptocurrencies have no value and they don't produce anything.” Chamath Palihapitiya, of Social Capital, disagreed that Bitcoin has no value because its purpose is to serve as a hedge to the traditional existing financial structure and, for this reason, has value because it's not correlated to the rest of the market. Instead of viewing it as a cash-flow generating asset with intrinsic value, Chamath's counter-argument was that Bitcoin is a non-productive asset class worth what others in the market agree to pay for it, similar to how gold is priced. However, he believes Bitcoin has the potential to replace gold as the safe haven commodity of investors. A misconception is a belief that Bitcoin makes up a major percentage of an investor's overall portfolio and is expected to outperform market returns. But Chamath recommends all investors to have at least 1% of their assets in Bitcoin. While many early investors became wealthy from their Bitcoin holdings and its record surge in price made it go mainstream in 2017 during the “crypto-mania,” its actual purpose, in theory, is to serve as a hedge.

What is an ICO?

An ICO, which stands for an initial coin offering, is equivalent to an IPO for cryptocurrency projects. But the key difference is, an investor gets equity ownership in the company undergoing an IPO in return for their capital, whereas in an ICO, the investor receives digital tokens in exchange for their investment. The process is unregulated and prone to exit scams, which has led to regulatory scrutiny and a negative perception of the cryptocurrency industry by the general public.

What is the function of blockchain technology?

Blockchain technology is a secure, digital ledger that shows every single record of transactions. Each block contains data detailing each transaction, such as date, time, and dollar amount – and is theoretically designed to be impossible to alter. Each proceeding block is structured through links called a chain, and they're often used in cryptocurrencies – most notably, Bitcoin.

What does the trend toward “DeFi” signify?

Decentralized finance (DeFi) is an umbrella term for the shift towards financial applications by utilizing cryptocurrency and blockchain technology. The adoption of DeFi would suggest the disruption of traditional financial intermediaries and the removal of a centralized authority, which leads to more control over transactions by users since the middle-man is cut out. An example of DeFi would be lending platforms that rely on smart contracts rather than banks to oversee the lending.

What impact did Robinhood's zero-commission pricing have on the brokerage industry?

After Robinhood launched with zero-fee trading, major brokerages were forced to change their pricing structure to avoid losing users, and a wave of fee-eliminating announcements soon followed. Established players, Charles Schwab, TD Ameritrade, and E-Trade, were forced to adjust their business models by eliminating (or reducing) commissions on certain types of securities. The industry disruption by Robinhood is often credited for prompting the merger between Charles Schwab and TD Ameritrade.

What is the purpose of a FinTech sandbox?

A FinTech sandbox enables a company to test their newly developed technologies within an environment that mimics the actual real-life use case. However, there are safeguards in place to ensure there are no consequences to failure. The sandbox is essentially a testing ground to experiment and try out their new offering before becoming licensed and bringing the product to market when there are higher stakes and serious repercussions for failures (i.e., pilot testing).

What does Anti-Money Laundering (AML) refer to?

Anti-Money Laundering ("AML") is a set of regulations that prevent criminals from disguising illegally obtained funds as legitimate income. Over the past decade, financial institutions and governments have been pursuing ways to prevent money laundering, and AML laws require financial institutions to monitor their clients and report any suspicious activity related to money laundering.

What is RegTech?

RegTech is regarding the management of regulatory processes within the financial industry using software and related technology. The most common use cases of RegTech include regulatory monitoring, reporting, and compliance checks. The recent advancements in RegTech help financial services companies comply with regulations while enabling regulatory bodies to monitor and strictly enforce their rules.

What does "KYC" refer to in FinTech?

KYC is an abbreviation for "Know Your Customer." The term refers to the process in which a business will verify the identity of the client and is most commonly used by financial institutions (and often a legal requirement closely tied to AML purposes). The premise of KYC is to prevent identity fraud and to authenticate users via thorough identification and background checks to diligence the person.

Healthcare

When it comes to interviewing with healthcare investment banks, having a background in healthcare or studying biology, chemistry, or a health-related major is not necessary by any means. Healthcare investment banking groups hire undergrads with minimal medical knowledge all the time. If you can show a basic understanding of the healthcare business model and can discuss a few developing trends, that would be more than sufficient to convince an interviewer of your interest in healthcare.

Provide a broad overview of how the healthcare industry is segmented.

1. Life Sciences

- Biotech
- Large Pharma
- Specialty Pharma
- Generics

2. Healthcare Services

- **Other Services:** Contract Research Organizations, Pharmacy Benefit Managers (PBMs), Pharmacies, Distributors, HCIT
- **Payers:** Managed Care Organizations (Health Insurers), Centers for Medicare & Medicaid Services (CMS)
- **Providers:** Hospitals, Urgent Care, Inpatient Rehab Facilities, Ambulatory Surgery Centers, Long-Term Care Hospitals, Outpatient Rehab, Skilled Nursing Facilities, Home Health & Hospice

3. Med-Tech

- Medical Devices
- Medical Equipment

What are some ongoing industry trends in the healthcare sector?

1. Industry Consolidation via M&A and Alliances

- Payers acquiring or developing their own Pharmacy Benefit Managers (PBMs) – examples include the Cigna and Express Scripts merger and CVS's Caremark acquisition of Aetna
- Retail pharmacies aligning with distributors
- Pharmacies acquiring or creating alliances with payers

2. Integrated Delivery Networks (IDNs)

- Systems of healthcare providers and health plans within a geographic area – creating an alignment between the different sub-sectors
- IDNs be thought of as a shift towards an ecosystem where various healthcare services are provided by different providers but under a single brand

3. Value-Based Care vs. Fee-for-Service

- Recent concerns that care has been compensated based on quantity rather than quality
- The movement towards outcome-based payments for medical services is changing how healthcare providers are being reimbursed

4. Consumer-Centric Healthcare

- E.g., Urgent Care Centers, Increased Access to Personal Health Data, Telemedicine

What are the main differences between biotech and other types of pharma companies?

Broadly put, biotech companies use living organisms to create their drugs, whereas the other life sciences categories develop their drugs using chemical compounds. Biotech companies are typically very high growth and based around one product/therapeutic area. Other sub-sectors in the life sciences industry, such as large pharma, are far more diversified with more product lines across several therapeutic areas (e.g., Pfizer, Merck).

What do inpatient and outpatient refer to when discussing providers?

Inpatient and outpatient are two common ways to segment the services that healthcare providers offer.

1. **Inpatient Care:** Treatment where the patient stays overnight at a hospital or a comparable facility.
2. **Outpatient Care:** Refers to facilities where patients receive treatment but don't stay overnight.

Tell me the difference between acute and post-acute care.

- **Acute Care:** Acute care refers to receiving treatment at a hospital and is associated with shorter-term, often severe injuries and medical emergencies – e.g., urgent care, intensive care units (ICUs), and emergency room services.
- **Post-Acute Care:** Medical treatment received at non-hospital healthcare facilities is classified as post-acute care and can range from short-term rehabilitation to longer-term restorative care. These are all services beneficiaries receive after treatment at an acute care hospital – e.g., physical therapists, skilled nursing, rehabilitation facilities, and psychiatric care.

An illustrative sample care continuum would flow from the Primary Care Physician (PCP) → Hospital (Acute) → Rehab Facility, Skilled Nursing Facility (Post-Acute) → Physical Therapy (Post-Acute).

What are the two channels for pharmaceutical drug distribution?

1. Retail Pharmacy Channel

- In the retail channel, the product flows from the branded pharma or generics company to a distributor that's typically aligned with a particular retail pharmacy and then onto the retail pharmacy before finally reaching the customer.
- On the funds side, customers' co-pay just passes through the retail pharmacy onto the PBM. The amount the customer essentially pays and their premium goes from the payer to the PBM based on the negotiated drug prices for the drugs on the formulary (i.e., list of covered drugs).
- For funds flowing out of the PBM, the PBM will pay the retail pharmacies based on those negotiated prices, and the retail pharmacies will pay the pharma and generic companies based again on the negotiated prices. There is much negotiation involved throughout the chain and opportunities for PBMs or retail pharmacies to earn a spread by negotiating different prices with different parties.
- The result being industry consolidations with payers and PBMs merging and retail pharmacies and payers merging to reduce opportunities for "excess" profit from the spread and lead to better margins.

2. Mail Order Channel

- In the mail-order channel, the products go directly from the generic or branded pharma company to a typically PBM operated mail order facility and then onto the customer. The funds, the customer's co-pay, go directly to the PBM, and then the payers pay the PBM based on those negotiated drug prices.
- Then it's the PBMs who will pay the pharma and generic companies for the cost of that product based on the negotiated drug prices.

What is the biggest cost driver for pharma companies?

1. **Research & Development (R&D):** For pharma companies, clinical trials related to R&D are by far the highest cost. These trials typically take five to seven years going through the various stages of clinical trials (i.e., FDA approval for a product, regulatory hurdles, application acceptances).
2. **Production Costs:** R&D is followed by the costs of producing pharmaceutical products, which refer to active pharmaceutical ingredients (API) manufacturing.
3. **Sales & Marketing (S&M):** The third cost is sales & marketing, usually directed at physicians and patients – however, this trails the previous two by a large margin.

What roles do pharmacy benefit managers (PBMs) play in the industry?

Pharmacy benefit managers (PBMs) directly negotiate drug prices with retail pharmacies and influence the prices of the products – as their job essentially is to manage the pharmaceutical spend for insurers and to keep drug costs as low as possible. PBMs get paid on the spread between what payers pay them for drugs and what they payout to the retail pharmacies and pharma companies, which further incentivizes them to lower the drug prices. Once a doctor has prescribed a particular drug, the script is handed to a retail pharmacy, and the PBMs dictate which drug to dispense (i.e., a brand or generic). PBMs reduce costs by negotiating drug prices with drug manufacturers and retail pharmacies and driving generic utilization and mail order penetration.

Explain the purpose of rebates in the context of pharmaceutical pricing.

Rebates are essentially discounts offered to PBMs by pharma companies to incentivize them to include them in their list of covered drugs (called a formulary). Therefore, PBMs maintain on behalf of the payers a schedule of which drugs are covered for their customers and at what prices. Particularly when there's high competition for a drug product, a pharma company might give bigger rebates to the PBMs to ensure they remain on the list of the covered drugs – at the expense of reduced product revenue.

What type of services do contract research organizations (CROs) provide?

CROs operate clinical trials on behalf of pharma and biotech companies in an outsourced, contractor-type relationship. CROs are hired to manage and lead the medical company's clinical trials and to perform other specialized tasks to help bring a particular drug/medical device to the market and obtain regulatory approval quicker. Client organizations contract with CROs to use their expertise, which is required to carry out these high-stake trials safely and in a cost-efficient manner – since CROs are an alternative to having to hire permanent, dedicated staff members with the required skill set.

What is the difference between a branded and generic pharma company?

A generics company (e.g., Teva) develops and sells generic drugs, which are allowed for sale once the patents on the original branded drugs have expired. Given the commoditized nature of these no-brand name drugs, the generics business model and competition amongst one another is based on winning fixed volume contracts for mass production. Research & development costs are lower for generics companies since there are fewer clinical trials involved. However, sales & marketing costs targeted at retail pharmacies are higher since competition is based on securing those contracts with retail pharmacies. In contrast, branded pharma companies (e.g., Novartis) have patent exclusivity which enables them to compete on product innovation and reputation of quality, rather than pricing-based competition.

What are the two pathways for a generics drug being approved?

There are two pathways for approving a generics drug:

1. **Paragraph III:** First, there's the Paragraph III filing, often called the standard channel. This is when the generics companies wait out until a patent expires and then markets it at the same time as other generics.
2. **Paragraph IV:** The other channel is called a Paragraph IV filing – when a generics company challenges a patent before it expires. If this is approved by the FDA, the generic company can gain 180 days of exclusivity before other generics can come in. This is a pathway that most generics companies use if they think they have grounds to challenge a patent.

Let's say you're valuing a pharmaceutical company selling drugs approved for therapeutic usage. What would the basic drivers of revenue be?

For the typical pharmaceutical company, the most basic drivers of revenue growth would be the current number of patients using their drugs, the average price of the drug, and the average dosage of the medication recommended by the physician/medical professional.

What assumption would you look at to value a clinical-stage biotechnology company?

A biotech company typically consists of a portfolio of different experimental drug candidates. Because each experimental drug candidate is unique and specific, we must value them separately. We must determine the annual free cash flow of each experimental drug candidate, then use discounting principles to determine the net present value of each unique product. Biotech companies are very cash-dependent, so the company's current cash balance is also crucial when determining the full equity value of a biotech company.

Model Assumptions

1. **Number of Products in Pipeline/Development Stage:** In the first assumption, the number of products under development and the current development phase for each must be identified. Then, it must be determined whether the company has a discovery platform of pre-clinical models and research.
2. **Peak Opportunity:** Determining the potential peak revenue opportunity within a therapeutic area is often the most important assumption. Sales will grow modestly after the company has achieved peak revenue until patent expiry or loss of exclusivity. Loss of exclusivity means that a brand name drug will lose its exclusive rights to sell in a certain geography.
3. **Indication:** The indication refers to the use of that drug for treating a particular disease/condition. More specifically, an indication is a specific, identifiable condition that the FDA has cleared a therapeutic drug to treat and can be thought of as a subcategory to a broader therapeutic area.
4. **Estimated Time for Completion:** The estimated time for completion refers to how long it'll take for the biotechnology company to complete all necessary clinical trials.
5. **Launch Date:** Once you have determined the estimated time for completion, you can estimate a launch date for the drug. Once the product has officially launched, you can apply an uptake curve that grows each year until the estimated peak revenue is achieved.

What does the “probability of success” rate refer to in receiving FDA approval?

Depending on the targeted therapeutic area, certain drug candidates have a higher probability of success than others. But overall, research has shown that it's very difficult to achieve “approval” status from the FDA.

When modeling a revenue forecast build for clinical-stage biotechnology companies, the probability of success rate (called the “POS”) must be multiplied to each product line, excluding research & development. The risk-adjusted revenue would then be calculated as the unadjusted revenue multiplied by the cumulative POS rate.

Provide an overview of the path to becoming an approved therapy by the FDA.

FDA Approval Phases

- **Pre-Clinical Discovery:** Non-human testing to determine what a safe dosage for humans would be or if it even is safe to proceed with human testing (i.e., animal testing)
- **Phase 0:** An exploratory study usually involving fewer than 15 test subjects and dosage on the lower end due to safety concerns
- **Phase 1:** Human tests to determine if the treatment is safe while testing for the highest dose in humans without serious side effects (typically involves 20 to 80 test subjects)
- **Phase 2:** Continued human tests to determine if the treatment works and testing specific dosing to observe response and efficacy (can involve a few dozen to 300+ test subjects)
- **Phase 3:** Further human tests to determine if the treatment is better and testing the therapy against currently available therapies and treatments (ranges from several hundred to 3,000 test subjects)
- **New Drug Application (NDA):** Formal application submission asking for approval by the FDA – once an NDA is received, the FDA has ~60 days to decide whether to proceed with the process of review.
- **Phase 4:** Involves a thorough drug review by the FDA on all submitted data (e.g., studies conducted, labeling plan, and manufacturing facilities inspections) before making a formal decision on the application

How would you build a revenue model for a recently launched generic drug company?

Revenue Model Process

1. To begin, the two main revenue drivers would be recent product launches and the current pipeline.
2. For a generics drug, pricing and annual sales data on the branded drug before the patent expired will be looked to have a benchmark to reference. The branded annual sales can help estimate the total market size for the drug, and this proxy can project how the generic drug will perform (i.e., demand sizing).
3. In most cases, the revenue build will be a top-down model. The total market volume with the average quantity of capsules in each bottle and the estimated price for the bottle will be gathered. Then, the market growth will be projected based on historical growth rates, but this will usually be on the lower end, and most models will assume minimal annual growth YoY in line with population growth.
4. In the next step, the estimated generic market volume as a percentage of the total market will be calculated for each. Initially, the branded drug will retain some market share, but then gradually, this hold on the market will fade as PBMs drive the conversions to generics. Keep in mind, this is the capture rate by all generic companies, rather than just the company being valued. The assumption should be on the higher end (~85%) and then gradually increase each year once the patent has expired.
5. Next, the key variable, the competition, must be accounted for in the forecast. The question that needs to be answered is: "How many generics companies will there be in the market?" This will determine the estimated revenue for the target company and its specific market share. Thus, there should be a line item for the total number of generic companies that will compete with the company being valued. As more competitors enter each year, the revenue should decline from lower quantity and price declines. But once a certain number of competitors have entered the space, it no longer becomes profitable, and the number of new entrants will stagnate close to zero. A reasonable assumption is that when a new competitor enters the market, the price will drop by ~10%.
6. A simplifying assumption is that the market share will be evenly split between all competitors, but this will be driven by the contracts with the retail pharmacies. So generic companies with a history of being well-supplied and reliable can secure favorable contracts and grab more market share.
7. Lastly, now that we have the sale volume (% of total market share) of the company being valued and estimated drug price from sites such as GoodRx or the negotiated price with the pharmacies if publicly available, we can multiply the two to arrive at the revenue for the product.

On average, how does the pricing of generics compare to branded drugs?

Typically, the prices on generic drugs will be ~15-20% of the branded price.

How would projecting revenue from drugs in the pipeline differ from recently launched generics?

For pipeline products, it may be appropriate to apply a risk-weight to project revenue. The risk-weight will be based on the percentage likelihood that the product will make it through the development and receive FDA bioequivalence approval. This is intended to account for the risk the product may face manufacturing issues or run into quality control difficulties, especially if the particular drug is complex (e.g., extended-release, injected).

How would you build a revenue model for a hospital?

Model Assumptions

- 1. Number of Operating Hospitals & Freestanding Outpatient Surgery Centers:** The first key metrics would include the number of operating hospitals and the number of freestanding outpatient surgery centers. Then, a growth rate would need to be attached, which will depend on historical growth and M&A plans announced by management since nearly all growth in terms of location count is driven through acquisitions (rather than organic growth). The reimbursements for services are mostly out of a hospital's control, so growth through acquisition is the most reliable growth strategy.
- 2. Utilization Data:** Next, the utilization data of each hospital must be calculated. This includes the number of licensed beds (and the weighted average of licensed beds), average admissions (i.e., the inpatient volume), equivalent admissions, the average length of stay, and the average daily census.
- 3. Average Occupancy Rates:** Then, the average occupancy rate will be calculated using the weighted average number of beds and the average daily census.
- 4. Reimbursement Rates:** The reimbursement rates will differ by the payor, and thus the sources of revenue will often be separated by each type (e.g., Medicare, Managed Medicare, Medicaid, Managed Medicaid, Managed Care/Other Insurers).

Existing Hospitals Revenue Projections

- 1.** To start, calculate the average number of beds per hospital. Then, add a row for the implied outpatient volume, which is calculated as the equivalent admissions (a proxy for outpatient and inpatient volume) less the admissions (outpatient volume). Based on this metric, we can derive the average daily volume from admissions, beginning with the inpatient admissions, which will then be multiplied by the average length of the stay. This gets us to the inpatient volume adjusted for length of stay, and then we'll add the outpatient volume to arrive at the total annual volume.
- 2.** Since we are calculating the approximate average daily volume, we would then divide the total annual volume by 365 for the units to line up.
- 3.** Then, we'll project the number of hospitals owned for > 1 year and the beds per hospital from historical periods. That gets us to the total number of beds owned by existing hospitals, which will then be multiplied by the occupancy rate and then add a line for the annual improvement. Doing so gets us to the average daily census, which we'll then multiply by 365 to get to the total annual volume.
- 4.** Using the outpatient volume as a total % of the total annual volume, we can calculate the implied outpatient volume. We'll now calculate the inpatient volume adjusted for the length of stay by subtracting the outpatient volume from the total annual volume. To arrive at the inpatient admissions, we'll divide the inpatient volume by the average length of stay to get to the inpatient volume.
- 5.** To get to the equivalent admissions, we can add the inpatient volume to the outpatient volume. Now, we can apply the revenue per equivalent admission assumption metric to it. We'll assume a growth rate for this metric and then multiply the ending figure by the equivalent admissions to arrive at the total revenue generated by hospitals owned > 1 year.

Hospital Acquisitions

1. To forecast the revenue from acquired hospitals, the first assumption will be the new hospitals acquired multiplied by the beds per hospital to arrive at the total beds from acquired hospitals.
2. You'll then go through the same calculations as we did for the hospitals owned for over one year. In the first couple of years post-acquisition, there should be a lower occupancy rate to be conservative since these are new hospitals that have not built up brand recognition yet.
3. Again, the average daily census will be multiplied by 365 to arrive at the total annual volume.
4. We'll then make an assumption for the outpatient volume as a percentage of the total annual volume and calculate the outpatient volume. That allows us to calculate the inpatient volume adjusted for length of stay, which is the total annual volume less the outpatient volume. Then, inpatient admissions will be calculated by dividing the inpatient volume adjusted for length of stay by the average length of stay.
5. Ultimately, this allows us to calculate equivalent admissions, which, as a reminder, is the sum of the inpatient admissions and outpatient volume.
6. Lastly, the same metrics will be applied as the existing hospitals. The equivalent admissions will be multiplied by revenue per equivalent admission to calculate the revenue from new hospitals for the year. But since hospitals are not acquired precisely at the beginning of the year, we'll apply a mid-year adjustment by multiplying this figure by 0.5. Overall, the total revenue will be the sum of the revenue from the existing hospitals and acquired hospitals.

How do you calculate the equivalent admissions of a hospital?

The equivalent admission is a proxy for the combined outpatient and inpatient volume.

$$\text{Combined Outpatient \& Inpatient Volume} = \frac{(\text{Admissions} \times \text{Total Inpatient \& Outpatient Revenue})}{\text{Inpatient Revenue}}$$

What does the average occupancy rate represent, and how is it calculated?

The average occupancy rate is a key measurement of the profitability of a hospital that shows how well a hospital is utilizing its capacity. While the target rate will depend on the hospital and be specific to the location/services provided, an occupancy rate of ~75% is cited as the most profitable percentage whereas, for high-end hospitals in urban densely populated areas, a ~85% occupancy rate is considered "peak profitability."

$$\text{Average Occupancy Rate} = \frac{\text{Average Daily Census}}{\text{Weighted Average Number of Beds}}$$

What are the differences between Medicare vs. Managed Medicare and Medicaid vs. Managed Medicaid?

Medicare vs. Managed Medicare

- **Medicare:** Medicare refers to healthcare coverage managed directly by the federal government.
- **Managed Medicare:** Also known as Medicare Advantage, this plan is managed by a Managed Care Organization such as Humana or Aetna, which contracts with the federal government to offer healthcare coverage to Medicare-eligible individuals. Medicare Advantage plans have different cost and benefits structures and may include coverage for additional services not covered by Original Medicare, such as prescription drug coverage (known as Medicare Part D).

Medicaid vs. Managed Medicaid

- **Medicaid:** Often called Original Medicaid, this healthcare coverage is offered directly by the government. Medicaid is run at a state level, so the offerings vary by state.

- **Managed Medicaid:** Refers to plans provided by Managed Care Organizations (MCOs). The distinction is that many states have increasingly chosen to contract out their Medicaid program to MCOs because they have realized that the MCOs are more cost-effective and provide better services.

What are managed care organizations (MCOs)?

Managed care organizations (MCOs) are medical service providers who offer managed care health plans. MCOs refer to non-government payers in the US. MCOs make their money from the premiums and administrative fees from their customers (e.g., employers, individuals who purchase their health coverage plans).

What impact did Medicaid expansion have on hospital reimbursements?

The expansion of Medicaid eligibility to those with lower incomes was of the key focus points of the Affordable Care Act (ACA). Medicaid reimbursement rates have historically been the lowest of all the various payer types. So the Medicaid expansion that happened through the Affordable Care Act (ACA) had a positive impact on hospitals since more patients now had health insurance rather than being uninsured. The caveat being more oversight over hospital practices, outcome-based payments, and penalties for non-compliance.

How do you calculate the medical loss ratio (MLR) and why is it tracked?

The MLR ratio is used to measure the percentage of premiums an insurance company spends on claims and expenses that directly improve healthcare quality. Under the Affordable Care Act (ACA), this provision's intended purpose was to ensure a minimum percentage of the health insurance premiums were used to pay claims and encourage providing value to enrollees while limiting insurance companies' marketing and administrative expenses.

$$\text{Medical Loss Ratio (MLR)} = \frac{\text{Medical Costs}}{\text{Premium Revenue}}$$

For example, a medical loss ratio of 80% indicates that the insurer is using the remaining 20% of each premium dollar to pay overhead expenses.

What are some of the main health plans used today?

- Preferred Provider Organization (PPO)
- Health Maintenance Organization (HMO)
- Point-of-Service (POS)
- Indemnity Plans (Fee-for-Service)
- Health Savings Account (HSA)

What are preferred provider organizations (PPOs)?

Preferred provider organization (PPOs) refers to contractual arrangements in which healthcare professionals and facilities provide services to their subscribed clients at discounted rates. The PPO is structured as a subscription-based membership in which healthcare professionals have agreed with an insurer (or a 3rd party) to provide medical care services at reduced rates.

PPOs are by far the most common form of managed care in the US. Under PPOs, payers will steer patient volume towards certain providers in exchange for contracted reimbursement rates.

What are health maintenance organizations (HMOs)?

As a healthcare plan alternative to PPOs, an HMO is a network that provides health insurance coverage for a monthly or annual fee. The coverage is limited to the network of doctors and other healthcare providers that are under contract to the HMO, and participants are required to first receive medical treatment from a primary care physician (PCP).

HMO contracts have the benefit of lower premiums relative to traditional health insurance plans because the health providers have the advantage of having patients directed towards them; however, the downside is the restrictions related to coverage access.

What are point-of-service plans (POS)?

A point-of-service plan (POS) can be viewed as a hybrid managed healthcare care plan between HMO and PPO. Similar to an HMO, plan holders are designated an in-network physician as their primary care provider. However, the patient may go outside of the provider network for healthcare services like under a PPO plan. The benefits provided will depend on whether the policyholder uses in-network or out-of-network healthcare providers. Relative to HMO and PPO, POS represents a small fraction of the total health insurance market.

What do indemnity healthcare plans involve?

Often called “fee-for-service” plans, indemnity plans enable patients to direct their own healthcare decisions and enjoy greater flexibility. The benefits are greater flexibility in being able to visit any doctor or specialist at a hospital. The insurance company that provided the healthcare plan will then pay a set portion of the total incurred charges from the visit.

However, the patient must pay an annual deductible and out-of-pocket payment for services before submitting a reimbursement claim. Once the requirements are met, the insurance provider pays a percentage of the total costs based on the usual, customary and reasonable (UCR) charges of comparable providers.

What do usual, customary and reasonable (UCR) fees refer to?

The usual, customary and reasonable (UCR) fees are the fees paid out-of-pocket by insurance policyholders for healthcare services. The UCR is the amount paid for a medical service based on what nearby providers typically charge for the same (or similar) service and often varies depending on the geographic area in which the service was provided.

Could you explain what issue an integrated delivery network (IDN) is attempting to fix?

An integrated delivery network (IDN) is a network of healthcare facilities owned collectively under a single parent holding company. The term is used to define the connectivity of various healthcare specialists that provide a continuum of healthcare services (i.e., joint healthcare). The issues IDNs see in today's healthcare system is misaligned incentives, inefficiencies such as lack of coordination and poor communication leading to higher medical costs for patients, and fragmentation.

In the ideal healthcare system, there would be coordination and an alignment in economic incentives that increase the utilization of available resources, leading to better clinical outcomes and improving the delivery of healthcare services. IDNs promote the idea of organizations working collectively to combine their assets and specialties to deliver the most comprehensive, efficient healthcare services to the patients they serve.

The concept behind an integrated delivery network (IDN) is to provides a range of health care services across the continuum of care through vertical integration.

What does interoperability mean in healthcare?

In the healthcare setting, interoperability is the ability of different IT systems and software applications to communicate and exchange data. The implication being data on a particular patient may seamlessly be shared across different clinics, labs, hospitals, and pharmacies. This enables the more efficient delivery of healthcare services for individuals, enhances the overall patient experience, lowers medical costs, and reduces errors.

What is a physician-hospital organization (PHO)?

A physician-hospital organization (PHO) refers to ventures organized and owned under hospitals and physicians at medical practices that engage in paid contracting under managed care plan contracts and provide services in coordination with one another under a formal partnership. While integrations will vary by contract, there may be limitations in the cooperation between healthcare providers as the economic incentives are not completely aligned between hospitals and physicians, leading to disagreements.

What are physician organizations (POs)?

The physician organization (POs) model is the traditional system in which physical organizations and hospitals co-exist with no contractual integration. This type of model has faced recent pressure from patients (and regulatory bodies) and may not be sustainable in the long-term due to managed care and the trend of IDS. As more healthcare reform initiatives call for providers to integrate service delivery and align their economic incentives, a shift towards PHO models is expected.

What is a medical service organization (MSO)?

A medical service organization (MSO) provides management, administrative, and operational tasks on behalf of medical practices. While not licensed to practice medicine, MSOs encourage integration between hospitals and physicians by performing time-consuming responsibilities on their behalf. Some areas in which MSOs assist are financial/billing management, human resources (HR) management, recruiting, and regulatory compliance.

What is an independent practice association (IPA)?

An IPA, short for Independent Practice Association, is an organization of independent physicians licensed to practice. The purpose is to build a contracting relationship with HMOs and payers that provide discounts to patients under fee-for-service arrangements.

This type of integration is beneficial as it creates a funnel for patients to go from the IPA to the physician's office, which helps the physician generate more revenue while the IPA can process claims under the conditions outlined in the contract agreement.

What does a partially integrated medical group (PIMG) mean?

A partially integrated medical group (PIMG) is a step in the right direction towards IDS, but it's not a complete integration. In a PIMG, there's a medical group practice resulting from the mergers of multiple practices into a single legal entity. However, separate practices still retain independence in their operations.

While there's usually centralized governance, there are many non-integrated elements such as the total income allocation by each center and the retention of the individual identity of each facility.

What does a fully integrated medical group (FIMG) include?

A fully integrated medical group (FIMG) is one of the end goals of IDS. In a FIMG, the physical practice is listed as a single, legal entity and is managed under single, centralized governance similar to a PIMG. However, the practices under a FIMG operate under a common name, and there's complete integration in the organizational structure and management of workflow.

For example, the administrative operations, clinical systems, and income distribution are unified and completed under a single allocation system.

Explain the 3rd party business model in how providers are paid in the healthcare industry.

Hospitals and healthcare facilities get paid through a co-pay that the patient gives them at the time of care and through the fees from the payers. Once the medical care has been provided, a claim will be submitted to the payers describing what services were performed and what reimbursement is appropriate based on services

codes. Payers, to keep their costs down, will then send them to an internal payment integrity unit or an external 3rd party HCIT company to check for errors or services incorrectly billed. These adjusted claims are then sent back to the payer, and that's what the payers will choose to reimburse hospitals and other providers.

Payers reimburse providers based on contracted rates for the services rendered (e.g., with participation in a PPO, hospitals typically have a one to three-year contract stating reimbursement terms; for physicians, it's based on a set fee-schedule, often based on the Medicare rates).

Why is payment integrity such a point-of-conflict in the healthcare industry?

Payment integrity is ensuring claims were paid in the right amount. Payment integrity providers analyze disputes and adjust claims on behalf of managed care companies, Medicare, and others. Payment integrity companies earn a percentage of the errors they identify.

For example, if they believe a patient was incorrectly admitted as an inpatient when being treated as an outpatient would've been appropriate – then the payment integrity provider would file a formal dispute that the patient didn't need to stay overnight.

Today, most payment integrity programs include some level of automation. However, a significant amount of manual work is still required to verify each payment and identify potential misclassifications or fraud. The increased complexity in treatments and innovations in tests/services has made it even more difficult for payers to distinguish between what qualifies as medically necessary tests/services and those that don't meet the qualifications. The rise in the sheer volume of claims filed each year has made catching these mismatches and anomalies overwhelmingly challenging. Given the mounting amount of data, it would be very difficult to recognize patterns and fix them on time, but recent advancements in artificial intelligence (AI) have been promising as a potential solution for this inefficient payment integrity system.

What is telemedicine and how does the process work?

Telemedicine (or telehealth) refers to the treatment of patients provided remotely using modern HIPAA compliant video-conferencing tools. This type of remote treatment has gone mainstream in the past year as it enables patients to communicate with a healthcare professional virtually rather than physically visiting the medical office or hospital.

Through real-time videoconferencing, a patient can discuss their symptoms and medical issues with a qualified healthcare provider to receive a diagnosis and determine whether an actual visit is required. In effect, this reduces the healthcare bills of the patients as unnecessary visits are reduced, and it allows medical professionals to provide treatment conveniently.

During the COVID-19 pandemic, telemedicine was brought into the spotlight, with Teledoc being one of the main beneficiaries leading the movement. One reason telemedicine saw such staggering growth was because many non-COVID patients feared that going to a hospital or medical facility would make them catch the virus. Funding towards privately held telemedicine startups substantially increased in 2020 as virtual medical treatment became more normalized and accepted.

Telemedicine has shown to provide a multitude of benefits for patients such as improved clinical outcomes, increased patient engagement (from more frequency of communication), expanded patient access, reduced costs and improved efficiency (which leads to lower billings to patients), and enhanced care coordination as this integration can often have a ripple effect on other parts of healthcare operations.

What are some limitations of telemedicine?

Telemedicine applies only to non-urgent situations, not actual emergencies. Any serious medical condition that requires immediate hands-on treatment or x-rays to assess the severity of the situation cannot be treated over telemedicine. Instead, telemedicine is intended for non-emergency matters and follow-up consultations after

an in-person meeting. Telemedicine is also expected to free up more time for patients that require urgent treatment and give medical professions more time to rest, especially when the health of the medical providers themselves is a widespread concern due to insufficient sleep and constant stress.

What is utilization management in the healthcare context?

Utilization management in healthcare describes the periodic review of the hospital and healthcare facility utilization to manage costs better. In addition, the quality of care provided is assessed to understand which areas to adjust to provide better patient experiences while remaining financially efficient (i.e., understand demand curves better, recognize which areas to focus on, identify inefficiencies).

In the coming years, the trend of telemedicine and increased automated workflow is expected to have a very positive impact on the costs of hospital and related facilities.

What is the difference between onsite and near-site healthcare, and what benefits do they provide to employees and employers?

The health and well-being of employees have been a topic at the forefront of many regulatory conversations in recent days. Through onsite or near-site healthcare, employers can address increasingly rising employee healthcare costs and chronic conditions of employees, which both contribute towards time away from work for the employee and wasted non-productive time.

With an onsite clinic or a near-site clinic, employees can reduce the time spent away from work to receive preventive diagnoses or treatments. In addition, health benefit options can serve as an incentive for employees to join a certain company.

Other than the distance, the biggest difference between a near-site clinic and an onsite clinic is the required costs. Onsite clinics require significant investments to set-up, which is why it's usually only done by large corporations with considerable resources. Near-site clinics are managed by outside 3rd party organizations, which reduce the monetary investment substantially. Thus, smaller businesses opt for near-site clinics because they cannot afford to host an on-site clinic.

Contrary to common belief, these near-site clinics are arguably more convenient to employees since there are multiple location offerings. Since there are various locations to choose from, it's easier for employees to receive care regularly, whereas with onsite clinics, the employee would have to come near the workplace.

Talk to me about the rising trend of D2C startups in the pharmaceutical industry.

There has been a noticeable shift towards a healthcare system model where patients can conveniently go online, consult with a physician around a range of clinical and therapeutic topics, and receive a prescription for that specific condition. Well-capitalized startups within the digital health space include TruePill, PillPack (acquired by Amazon Pharmacy), Capsule, NowRx, Surescripts, and ZappRx (acquired by Allscripts).

The pharmacy retail market is undergoing significant disruption from startups like Amazon's PillPack, Capsule, NowRx, Hims & Hers, Ro, and Blink Health, which offer patients low-cost, fast-delivery prescription medication services. Thanks in part due to the de-stigmatization of certain healthcare areas (e.g., birth control, erectile dysfunction, hair loss, acne) and demand for increased privacy by consumers, the D2C model has taken off as one of the fastest-growing segments in healthcare.

Tell me about a healthcare startup you're closely following.

Founded in 2016, Truepill is a B2B telehealth startup that provides pharmacy API and fulfillment services of medications and is often referred to as the "AWS for pharmacies." Truepill works directly with D2C pharmacy brands (most notably Nurx, Hims, Lemonaid), digital health companies, and other healthcare organizations to enable them to improve upon and scale their platforms.

Truepill's team of licensed medical professionals can diagnose, provide consultations, and prescribe medications across various clinical areas via telemedicine and in-person clinics. In addition, Truepill operates pharmacies that can ship cash or insurance-billed medications directly to the doorsteps of the patient. Essentially, Truepill acts as the licensed pharmacist dispensing and shipping the medication prescribed through D2C startups. For D2C healthcare startups, Truepill handles all the logistics, shipping, and pharmacy fulfillment on their behalf, thus enabling them to benefit from Truepill's economies of scale and growing international footprint.

Truepill's API provides full transparency to its user base by providing data such as overall fulfillment volume, the number of orders going out on time, the delivery status of the order on a real-time basis, and reports on issues that caused delays.

Competition amongst D2C pharmacy brands has become crowded as more niche startups enter the market – however, Truepill provides non-partisan services (comparable to Stripe's role in FinTech) by functioning as a background infrastructure layer. Truepill's business model of being only B2B has previously never existed before, and co-founder Sid Viswanathan has stated that Truepill seeks to become “the pharmacy behind the scenes that's powering the entire ecosystem.” Thus, Truepill works alongside and negotiates agreements primarily with drug makers and PBMs, as opposed to competing with the industry intermediaries that sit between insurers and drug makers.

Telemedicine and on-demand virtual deliveries will have a significant role in the modern pharmacy, and Truepill looks to capitalize on this trend by leading the development of the digital pharmacy infrastructure.

From a financial perspective, how were hospitals and healthcare facilities affected by the COVID-19 pandemic?

Since the beginning of this year, healthcare workers have been at the forefront in the fight against the coronavirus outbreak, and the revenue and profitability in the healthcare industry have deteriorated due to the deferral of most surgical procedures for non-life-threatening conditions and the decrease in the number of accidents and non-coronavirus illnesses. Considering its high fixed cost nature, hospitals today are currently in an urgent, cost-cutting mode for expenses unrelated to COVID-19, as seen by employee layoffs and reductions in working hours for workers not relevant to COVID-19.

While the healthcare industry is notorious for being slow to adopt new technologies, a considerable portion of the forecasted total industry growth of IT spend is expected to stem from the vertical healthcare institutions such as hospitals and smaller-scale clinics that have been disproportionately affected throughout the coronavirus outbreak. The reason is that these medical facilities were using outdated technologies and have no choice but to adjust to the prevailing conditions. So while digital infrastructure, certain healthcare services (related to diagnosis, testing, and disease control), and telemedicine platforms experienced positive growth, many hospital holding companies and healthcare facilities have been seeing substantial margin contraction.

Consumer Products & Retail

Could you name some ongoing trends in the consumer goods industry?

- Ongoing Convergence between Traditional Brick-and-Mortar Stores and e-commerce
- Rapid Product Development & Consumer Testing
- Targeted Marketing using Data Analytics (e.g., personalized ads on social media)
- Direct-to-Consumer (DTC) Business Models
- Subscription-Based Payment Plans (i.e., the customer pays in fixed intervals)
- Increased Demand for Transparency on Labels (e.g., location, manufacturing method, health, and environmental sustainability)
- Same-Day/One-Day Click-and-Deliver and Click-and-Collect Feature in e-commerce
- Partnerships between DTC Brands with Brick-and-Mortar Stores

Which types of valuation multiples are often used for consumer and retail companies?

The valuation multiples used in consumer and retail companies are standard for the most part. Given the mature state of the industry, multiples such as P/E and EV/EBITDA are frequently used.

Specific to retail, one of the most popular multiples is EV/EBITDAR, with the "R" being rental and lease-related costs. This is done to better standardize the multiple since there are companies that rent their stores whereas others own them. Without the rent expense being added back, companies paying rent would be valued lower than those that own their stores.

A few industry specific valuation multiples based on operating metrics include EV/Total # of Stores, EV/Sales per Square Foot, EV/Sales Per Store, and EV/Same-Store Sales.

For e-commerce sellers or marketplace businesses, an important valuation multiple would be EV/Gross Merchandise Volume (GMV).

What unit economics drive revenue in the consumer goods industry?

For the consumer & retail sector, most companies that fall under this industry classification have revenue drivers, split broadly into volume and price.

- **Volume Drivers:** The volume drivers section could include the number of stores, the average number of daily customers visiting the store, the average number of products in each order, and the average number of customer transactions per day.
- **Price Drivers:** On the other side, the price per product sold will be a function of the product quality, brand reputation, and how it stands relative to its competitors. There are often significant constraints, and most retailers have very little pricing power (i.e., have to set near market rate).

Excluding premium retailers, most companies in the consumer sector will have a growth strategy oriented around achieving volume growth – this can be achieved by a better sales/marketing strategy, store count increase, and geographic expansion.

What is the most significant cost driver for consumer product companies?

Unique to the consumer products industry, a significant percentage of their total cost structure is spent on advertising and marketing campaigns. Advertising is a key channel to communicate with their customers, while marketing campaigns help end-users positively view brands.

A successful advertising and marketing strategy leads to an increase in sales and pricing power in this industry. For product sales, building an “emotional connection” with the end customers is necessary for brand loyalty (e.g., Patagonia's initiatives for environmental and social responsibility).

What are the two main retail channels to sell products to customers?

1. **Brick-and-Mortar Stores:** Brick-and-mortar means a company's store has a physical presence at a specific location, building, or retail outlet that customers can visit in-person.
2. **E-commerce:** e-commerce involves transactions via digital means (i.e., online businesses, mobile).

Nowadays, the two channels have converged, as seen by Walmart's online shopping platform's development to compete with Amazon and other e-commerce stores.

If you were a seller, would you prefer to sell your products on a marketplace or personal e-commerce store?

Both online marketplace platforms and e-commerce stores are related to digital commerce.

- **Marketplace:** In the simplest terms, a marketplace is a platform where various products come from multiple e-commerce sellers. Some of the most well-known examples of marketplaces would be Amazon, eBay, and AliExpress. The main benefit of making purchases in a marketplace is the platform's pre-existing reliability (i.e., intermediary and security infrastructure), website traffic, and buyers enjoy comparing similar product offerings side-by-side. There is less information asymmetry in what the price of comparable products is. The more diverse the marketplace is, the more convenient it's for customers to purchase items in different areas. For sellers, the effort to bring traffic and viewers to their products is minimal, which means fewer marketing expenditures are required. The downside of marketplaces is the commission fees on each transaction, which typically range between 5% and 15%.
- **Individual e-commerce Store:** For an individual e-commerce store, there are more responsibilities and work required to get the store running. To bring traffic to a new website requires spending on marketing and SEO, and even then, there's no guarantee the efforts will pay off. But while the process can be more time-consuming, the seller can customize the website to their liking and create a virtual shopping experience. In addition, no commission has to be paid to a 3rd party, but only a minor website hosting fee is paid. There are benefits to both, but certain brands with existing followers will avoid marketplaces so that their customers will have a differentiated online shopping experience in which the company's brand is the only focus. From the sellers' perspective, the saturation of brands in marketplaces, especially from lower-tier brands, makes the platform appear less "premium" to users and damages the online shopping experience (e.g., "sponsored" products shown next to a seller's products).

For a marketplace company, which KPIs would you pay the closest attention to?

- **Gross Merchandise Volume:** GMV is the total amount of sales made through all the transactions for merchandise on a platform over a specific time frame.
- **Average Order Value:** AOV is the average amount spent per transaction and is calculated by dividing total revenue by the total number of transactions.
- **Take Rate:** The take rate is the percentage of GMV collected by the marketplace (typically ranges between 10% to 30% for established marketplaces) and is how marketplaces produce revenue.

Explain how the "Amazon Effect" has impacted the entire retail industry.

The so-called "Amazon Effect" is the ongoing disruption of the retail market (and other online platforms/e-commerce stores), resulting from their sector dominance. Amazon single-handedly changed the competitive landscape of all retail and raised the bar for what customers expect during their online shopping experience.

- **Shipping Speed:** Amazon Prime's 1-2-day shipping raised the standards of the speed at which a product would have to be delivered to customers, which cuts into the margins of other online businesses.
- **Free Returns:** Online sellers have less leverage in return handling because of Amazon. In most cases, the seller will offer free returns to cover all shipping charges related to the return (along with free delivery).

Why have many e-commerce platforms opened brick-and-mortar locations?

While many people neglected the traditional brick-and-mortar business model, companies have started to re-think the amount of value provided by customized in-person shopping experiences – fueled by the trend of “click-and-collect.” Products sold online will have higher margins and be more convenient, but it’s often difficult to create an emotional connection with customers when sales are just online. Thus, many e-commerce platforms have opened brick-and-mortar locations recently.

Another reason is the convenience factor, as integrating an online sale with in-person pickups is the fastest method for product delivery. A recent example of this could be seen in the curbside pickup trends in 2020. Consumers want products fast and often prefer to go pick them up in-person immediately.

Lastly, the ability to return products purchased online at a nearby location can help a company save on shipping costs. Alternatively, many partnerships have developed between companies and even educational institutions. For example, Amazon returns can be accepted at a retail store such as Kohl’s and at universities with a department dedicated to the shipping and handling of product returns.

Many DTC companies have recently established partnerships with brick-and-mortar store chains and began opening up physical stores. What is the reasoning behind this trend?

Many DTC brands have realized how a traditional brick-and-mortar store could complement its products, and it’s expected to become a key part of their strategy increasingly. The multi-channel strategy to sell products on several platforms to maximize exposure to diverse consumer types. For example, Quip, the direct-to-consumer toothbrush brand, has partnered with Target to diversify its distribution channels and have a presence in brick-and-mortar stores.

Many customers prefer DTC, especially when there’s fast shipping. But certain customers prefer purchasing a product after seeing it in person, regardless of the option for free returns – commonly known as “buy online, pick up in-store” (BOPIS), which is not a new idea but a trend that appears to have returned. Besides partnerships with existing stores, many DTC brands such as Casper, Warby Parker, and Everlane opened physical locations over the past few years and have continued to grow their physical footprint.

What are some benefits that brick-and-mortar stores have over e-commerce?

- **Physical Shopping Experience:** Many customers enjoy the shopping experience, which helps develop an emotional connection to a particular brand if done properly. Many struggling retailers such as Macy’s have introduced strategies to offer a more “physical” customer experience through in-store VR, online integrations (e.g., mobile check-out), and nearby modern food vendors. Given Macy’s key demographic, a family-friendly shopping environment in their improved stores with better lighting/stylistic changes is expected to impact performance positively.
- **Fewer Product Returns:** Another reason is related to the percentage of product returns. Since customers see the product in-person, ~30% of products purchased online end up being returned, compared to ~9% for brick-and-mortar stores. And if a customer returns a product, it has become standard practice for shipping to be covered entirely by the seller.

Are multichannel and omnichannel marketing synonymous?

Both multichannel and omnichannel marketing strategies are mediums through which sellers communicate and pitch their products to prospective buyers. Examples of marketing channels would be TV advertisements, social media, billboards, and a brand’s website, whereas sales channels can be email campaigns, cold calling, live chats, and referral networks.

- **Multichannel Approach:** The multichannel approach is when a company uses multiple channels to spread their message about their brand and attempt to reach new customers. Each channel is separate, has its individual marketing message, and is not interlinked with the other channels.
- **Omnichannel Approach:** The omnichannel approach integrates different channels to create a more seamless online experience for customers (i.e., viewing the experience through the customer's eyes). The contextual engagement and marketing message will usually be the same (or similar) across all channels. However, omnichannel will have more of a focus on the customer experience.

Tell me about the significance of PayPal's acquisition of Honey.

Near the end of 2019, PayPal acquired Honey, a browser extension for consumers to find coupons and discounts while purchasing items online. Not only was this acquisition PayPal's largest acquisition to date, but it was the first of its kind in that a web browser extension could be acquired at such a high valuation. PayPal explained its rationale for the purchase by pointing to the possibilities of integrating Honey's browser extension and Smart Shopping Assistant mobile application with its PayPal/Venmo check-out services to create a more seamless, simplified shopping experience for consumers. For instance, consumers could make a single purchase from multiple different websites, all at the lowest prices possible.

Now, PayPal's Venmo a significant leg-up on its competition with Apple Pay and Square's Cash App, especially considering Honey's popularity with consumers. At the time of the acquisition announcement, Honey had more than 17M monthly active users. Considering Venmo's traction with consumers, acquiring Honey would further establish PayPal's footprint in the consumer market and provide a complementary portfolio of offerings to drive engagement. Hence, CEO Dan Schulman's statement that the acquisition of "Honey is amongst the most transformative acquisitions in PayPal's history."

Could you define what consumer-packaged-goods (CPG) and retail are?

- **Consumer Packaged Goods:** CPG refers to products that consumers frequently use and replenish routinely, typically in a short usage cycle (i.e., fast-moving consumer goods). A few common examples of CPG products would be toilet paper, paper towels, shampoo, conditioner, soda, and soap. As one would expect, the CPG sector is very competitive, as many companies compete for limited shelf space, and often products are highly commoditized. The success of a particular commoditized product depends on customer loyalty and familiarity with the brand. You'll often see "copycat brand" products (e.g., store brand soda placed next to Coca-Cola products at a slight discount).
- **Retail:** In contrast, retail focuses on the sale process of products to individual consumers. While CPG refers to a particular category of products, retail refers to selling the products. Often compiled each month and followed closely by investors as an indicator of the economy's health, retail sales data reports track consumer demand for durable and non-durable goods by calculating the purchases over a specified period.

What is social commerce, and how is it different from social media marketing?

The trend of social commerce refers to social networking websites such as Facebook, Instagram, Twitter, and Snapchat becoming sales channels for companies selling a product or service. A distinction from social media marketing is that a company doesn't use the social media platform only for promotional purposes. Instead of directing the customer to visit their company site to make the purchase, customers can make a product purchase within whichever social media site they were browsing. For example, Facebook recently rolled out Facebook Shops and Instagram Shopping, enabling businesses to set up an online store easily. There'll be an incentive for businesses to purchase ads on Facebook and Instagram's platform due to their user count, targeted ads (more consumer data), the prevalence of influencers, and modern features (e.g., live shopping, product launch videos, product tags). In addition, the shopping experience is seamless as Messenger, Instagram Direct, and WhatsApp allow sellers to answer questions and offer support through the chat.

How are chatbots being used in the retail industry?

Nowadays, most companies will have chatbots installed on their websites to assist customers. Chatbots are pre-programmed AI applications intended to simulate customer service interactions in lieu of the immediate need of a human representative.

The clear advantage here is that chatbots enable immediate responses to common questions and can help a company reduce costs over time once the program is developed. The caveat is that a human will usually be on standby, as the chatbot cannot answer all questions, and its capabilities will take time to develop and become less reliant on human aid. Many customers prefer speaking with an actual human due to the frustrating experiences that chatbots often provide, especially in the earlier days when they were first being rolled out.

What is the difference between a discretionary product and a non-discretionary product?

- **Non-Discretionary Products:** Often called consumer-staples, non-discretionary products are necessary for consumption every day by their users regardless of income levels or consumer type. Examples would include bottled water, soap, toothpaste, shampoo, and razors. Regardless of the economic conditions, consumers require these products, which makes them less cyclical. An example of a company that sells many non-discretionary products would be Unilever, which sells products related to nutrition, hygiene, personal care, and more.
- **Discretionary Products:** Discretionary products with higher price points are non-essential and are in greater demand when the economy is doing well. Examples would include purchases of new vehicles, TVs, and smartphones upgrades.
- **Distinction:** During recessionary periods, CPGs continue to sell relatively well, while most non-discretionary product spending will be put on hold by everyday consumers.

What are some common characteristics of CPGs?

- Low Average Selling Prices (ASPs)
- Essential, Necessary to Users
- Rapid Consumption and Disposal by Consumers
- Frequent Repurchases (Often Weekly/Monthly)
- Low Customization Required

What is the difference between CPG and durable goods?

- **CPG:** CPG products are used frequently and replenished quickly by consumers (i.e., short product replacement cycle).
- **Durable Goods:** Durable goods are purchased and replaced far less often than CPG and are usually significantly more expensive. Examples would include TVs, washing/drying machines, furniture, and refrigerators that provide benefits for long durations.
- **Distinction:** Besides the differences in product life cycle and pricing, a key difference is consumers' time when contemplating the purchase. For instance, a consumer can take weeks to decide which laptop to purchase, while CPG goods can often be decided within minutes.

What does it mean when a product has become commoditized?

A product has become commoditized when the product offerings have little differentiation from one another, and the competition is now done on pricing alone. The competitive dynamics of the industry become a “race to the bottom” to see who can sell at the lowest price.

Name a few KPIs that many retail chains report in their financial statements.

- **Sales Per Square Foot:** Using the sales per square foot metric, an investor could determine how well a company uses its floor space relative to its competitors. This KPI assesses the efficiency of a company at using the floor space in its stores.

$$\text{Sales Per Square Foot} = \frac{\text{Net Sales}}{\text{Sales Space by Sq. Foot}}$$

- **Sales Per Employee:** The sales per employee metric is used to determine whether a particular sales strategy is working and whether the right staffing decisions were made. Individual stores often track this to examine their employee list in more detail and see which are struggling to make sales.

$$\text{Sales Per Employee} = \frac{\text{Total Sales}}{\text{Total Employee Count}}$$

- **Conversion Rate:** The conversion rate is the proportion of customer store visits to the number of customers who made an actual purchase during their visit. The higher the conversion rate, the better – as it means that customers are not just walking in and leaving without making a purchase.

$$\text{Conversion Rate} = \frac{\text{Number of Transactions}}{\text{Total Number of Visitors}}$$

- **Same-Store Sales:** With the same-store sales metric, investors can compare individual stores' performance benchmarked against the same period in the previous year. By comparing an individual store over a specified time frame, a company can determine whether the store is performing above (or below) expectations relative to its comparable stores, and then try to figure out why that's the case.

$$\text{Same Store Sales} = \left[\left(\frac{\text{Current Period Sales}}{\text{Prior Period Sales}} \right) - 1 \right] \times 100$$

- **Average Transaction Value (ATV):** Average transaction value is a metric that shows the average dollar amount customers are spending on each order and can show a company's effectiveness in strategic product placement, promotions, discounts, and other sales strategies to increase the number of items purchased.

$$\text{Average Transaction Value} = \frac{\text{Total Revenue}}{\text{Total Number of Transactions}}$$

Tell me what the trend of voice commerce is and its use cases in retail.

An example of AI rapidly gaining traction, voice commerce uses voice recognition technology to find and order products online without dependence on hardware. For example, rather than having to search on your laptop/computer for the item, read articles and posts about which products are recommended, and then making the purchase – instead, a consumer can use voice commands to hear a list of products, specify their needs and listen to the differences, and then tell the voice-powered device (or assistant) to make the purchase.

What is the gig economy?

The gig economy is enabled by online marketplace platforms that facilitate short-term engagements of independent contractors to deliver services (“gigs”). These workers work flexible hours and based on their schedule, rather than being employed full time. Known as the “sharing economy,” this type of P2P facilitation of services is broad and encompasses freelance consulting services, ride-hailing drivers on Uber/Lyft, food delivery services through Grubhub/DoorDash, and temporary home rentals on Airbnb.

Tell me about an ongoing trend in e-commerce that you're following.

The pet foods and supplies industries were already experiencing high growth, given the increased humanization of pets worldwide by pet owners. Fueled by millennials, this trend has driven up demand for premium pet products and new product developments related to pet nutrition with greater transparency in the ingredients and overall quality. In the last few years, the pet products segment has been one of the fastest-growing consumer spending segments and has one of the highest order volume categories on online marketplace platforms.

Since COVID-19, demand for pet food delivery has been surging, as seen by the growth of pet e-commerce companies such as Chewy. Direct-to-consumer models in the pet products industry were already trending, but the pandemic accelerated this trend, which can likely be attributed to the increased amount of time pet owners spend at home with their pets. But other than consumers being forced to stay home, many shoppers realized how convenient it's to have subscription-based orders come in for their pets with minimal work on their end.

Could you give me an example of how augmented reality is changing the consumer experience in the retail industry?

Launched in 2010, Warby Parker has always been known as an innovative eyewear company from its founding days. Initially, Warby Parker disrupted the optical industry through its Home Try-On Program, where the company would ship sample frames to customers to try on at home. But now, it has transitioned to leveraging the camera capabilities of smartphones (the iPhone X and above, to be specific) to enable the Virtual Try-On feature where customers can see their appearance with a virtual frame through augmented reality. Based on an AI-generated map of the customer's face and the collected details on specific facial features, the app can recommend which types of frames would best suit the customer and make an automated list of recommendations for the customer to virtually "try on."

Warby Parker's virtual glasses show how augmented reality can be implemented and enhance the reality of online shopping, and many more startup products in different product categories with similar features have been developed, such as in furniture and clothing shipping. In addition, existing incumbents have rolled out and developed similar features for their mobile applications and websites.

While retail is known for being one of the most cyclical industries, can you name one resilient product segment?

One specific product segment in retail that's resilient to recessions is the cosmetics and beauty products segment. The phrase "insecurities sell" describes consumer spending patterns very accurately for the beauty products industry, as even higher prices don't appear to have much of an impact on spending volume.

This is because purchases based on "intangible benefits" such as beauty products, the price comes secondary. Intangible consumer benefits are more prevalent in items that matter more to customers (the benefits) – which is further enhanced by how the products are applied directly applied on the skin. Even during recessions, studies on past consumer spending behaviors have shown that beauty regimes are among the last to be cut. This explains why the cosmetics market has proven to be less discretionary than one may assume – as demand rises during economic expansions and remains steady during contractions. While everyday customers are often fickle and price-sensitive, that behavior doesn't show in this particular segment.

Why are luxury goods often not negatively affected during recessions?

While luxury goods are often considered to be significantly cyclical, the industry is actually not as cyclical as one might expect. The reason being the income level of customers that can afford luxury goods. Many of these high-income customers, even in a recession, don't change their spending habits. There might be less demand

from customers that can no longer afford the luxury goods, but those that can still afford these luxuries will continue to make purchases.

How does pricing power in retail differ from other industries such as manufacturing?

Pricing power arises from emotion-driven intangibles that are often unquantifiable with typical metrics, unlike what is necessary to gain pricing power for commercial and industrial products. Therefore, many companies emphasize nurturing customer trust in their branding and emotional appeal to their customers. For products in a technical industry such as manufacturing, the highest-quality product will come out on top over time. But in retail, a well-marketed low-quality product with more brand appeal can outsell a higher quality product.

What makes the beauty products industry not just resilient to recessions but recurring in terms of revenue?

For most consumer goods, there's a limit to how much can be purchased and used. However, skincare and cosmetics products' usage is essentially limitless as many of these products are non-reusable (e.g., face masks, skin lotion) while requiring constant replenishment. Hence, skincare products are among the most profitable categories of consumer goods with highly recurring purchases.

Besides cyclicity, why do many private equity investors stay away from retail?

Compared to price-based growth and upselling/cross-selling, volume-based growth is the least valuable since it entails continually increasing sales and adapting to the always-changing consumer demands. But from a different perspective, the ability to adapt, strong branding, and increased scale could be a sustainable competitive advantage given that the cost of producing each unit decreases and capex requirements decline, which have a net positive impact on gross margins. Regardless, most investments into retail are usually structured as VC/growth capital, not traditional LBOs.

Tell me about the trend of connected fitness.

Led by home-equipment cycling company, Peloton, the trend of connected gym experiences and virtual trainers has been positively affected by COVID-19. Rather than going to the gym to train with friends or a trainer, this type of equipment enables the ability to workout "on-the-go" enabled by new technologies and applications integrated into the equipment. An example of a recent M&A deal involving this trend would be Lululemon's acquisition of Mirror.

What is the trend of athleisure wear?

In recent years, the trend of "athleisure" has taken off, especially for the younger demographic. Athleisure is a hybrid between professional workplace wear and casual and exercise outfits. Increasingly, athleisure has become a larger portion of consumers' everyday wardrobe because it's more versatile and comfortable for users, as the need for formal wear is gradually decreasing (especially in Silicon Valley). One of the leading companies in this segment is the athletic apparel retailer Lululemon Athletica.

What does the phrase "fast fashion" mean?

"Fast fashion" is a strategy of placing clothing designs shown on the catwalk in retail stores as quickly as possible to meet consumers' demands. Trends change quickly based on changing consumer tastes and seasons, so modern retailers such as Zara recognized this and have led this movement of placing trendy items on their store racks as early as possible to take advantage of these short-term spikes in demand. The quantity will often be strategically limited, creating a rush of customers to purchase the items – thus, the rapid replenishment required of store stock and higher margins for retailers that have joined in on this strategy.

What is the secondhand market?

While not a recent development by any means, the secondhand market is being forecasted as one of the fastest-growing retail segments over the next few years, as the circular economy trend continues to expand across different industries. Helping this trend is the blurring of the line between new clothing purchases and secondhand purchases, in which the stigma of purchasing worn re-sold clothing is diminishing.

Two of the main beneficiaries of this trend have been platforms Poshmark and thredUP, which are social commerce marketplaces where everyday consumers can conveniently list their clothing and related products for sale, with the end function being similar to a closet clean-out.

Often retail-brand companies such as Supreme can sell clothing that will be resold for significantly more. Why does Supreme not just raise their prices?

Often, a product's resale value can be multiple times more expensive than the original price tag. The reason companies such as street-wear company Supreme (which has a business model based around the creation of scarcity and premium brand portrayal) don't raise their original prices is because increasing prices would decrease the number of customers that could afford the product.

Supreme's customer base primarily consists of the younger demographic. Therefore, having too high of a price point would make them "window shoppers" rather than customers lining up for days to purchase an item in one of their stores. For example, an upper-middle-class high school student could afford to purchase Supreme's product – and this ability to purchase the item (the only constraint being the supply) makes the product more marketable and has a positive impact on the brand image.

Therefore, absolute, upper-tier premium brands are relatively unknown to most consumers. But Supreme, which is considered a premium/luxury brand by most customers, is well-recognized by consumers of all income levels. This is the marketing strategy and niche Supreme has carved out, which eventually led to a 50% equity sale to the Carlyle Group at a \$1 billion valuation.

What is scrambled merchandising?

Scrambled merchandising is a product assortment strategy used by retailers to serve as many customers as possible by offering a diverse product line. In effect, the retail store can portray itself as a "one-stop" shop. An example of scrambled merchandising would be a pharmacy retail store (e.g., CVS) having a section dedicated to selling electronics such as headphones and speakers.

In the retail context, what does "channel surfing" mean?

Channel surfing in retail, also known as "showrooming, is when customers try on products and get product information in-person at stores, but rather than making the purchase there, the customer will order the product online for a lower price and take advantage of online-only discounts, or pay less in taxes."

What type of differentiated business model did Dollar Shave Club have?

Dollar Shave Club (DSC) was one of the first-movers in the trend of subscription-based B2C business models. In the example of DSC, its value proposition almost immediately took away Gillette's leading market share as customers enjoyed the inexpensive, convenient subscription plans, as well as the brand transparency that it offered. However, Gillette soon launched its subscription-based kits, and new competitors emerged, such as Harry's in men's grooming.

Tell me about the trend of retailers using DOOH for their advertising.

Digital out-of-home (DOOH) is an advertising medium where their content is displayed on outdoor electronic screens and displays on buildings, highways, subways, and bus stops. More retailers are allocating their advertising spend towards DOOH due to the higher ad engagement and conversions based on customer survey

data. Similar to traditional out-of-home (OOH) advertising, the two are based on the same idea of placing advertisements in highly populated, urban places for people such as those commuting to work could see. But the key difference is the utilization of data to reach the right viewers strategically. DOOH can be viewed as data-driven OOH. Many of the trends that fall under AdTech, such as geo-fencing and personalization, are used in DOOH. For example, imagine a retailer such as Nordstrom having an advertisement on a billboard that automatically adjusts the clothing being promoted based on the current weather condition.

What is a private label brand?

A private label brand is when a third party company manufactures a product but then sold under a more established brand. These private-label brands can be seen in various industries but are most prevalent in food, cosmetics, and consumer merchandise. For example, Target has its “Open Story” brand launched in 2020 to sell premium luggage. Like most private brand labels, this line of luggage is sold at a discount and often mimic a market leader. Today, Target has over 45+ private label brands to choose from, which has boosted its revenue growth and margins (hence, its strong performance from 2019 to 2020).

Tell me about some trends related to the food industry.

- **Food Delivery/Takeout Platforms:** These platforms allow for food to be ordered from restaurants and food shops using a mobile application, which can then be delivered by a gig worker or be picked up by the customers (e.g., Uber Eats, Grubhub, DoorDash, Postmates).
- **Meal Kit Services:** Similarly, meal kit service companies deliver pre-prepared food; however, the difference is the customers cook the food at their home at their convenience. These companies' business model is usually based on a subscription plan with enough meals to last several weeks (e.g., HelloFresh, Blue Apron Home Chef).
- **Plant-Based Substitutes:** Consumers are looking for plant-based alternatives to meat and dairy for the supposed health benefits and environmental sustainability (e.g., Beyond Meat Impossible Foods, Amy's Kitchen).
- **Lifestyle-Based Menu Options:** Many companies are becoming more aware of the popular diet trends being followed by consumers and attempt to offer menu items to cater to their specific needs (e.g., Paleo and Keto Bowls from Chipotle).
- **Greater Transparency/Sustainability:** More consumers are demanding products to be sold with transparent labels, sustainable packaging, and ethical production.

Tell me about some trends in the beverage industry.

- **Cannabidiol (CBD Oil) Infused Beverages:** CBD Oil was the clear winner in 2019 for its supposed health benefits for anxiety, pain-alleviation, and insomnia. Products ranging from CBD infused drops and drinks were flying off shelves while drug/food regulatory bodies were scrambling to classify and regulate CBD.
- **Health-Based Hybrid Beverages:** More health-conscious consumers opt for beverages that support their lifestyle decisions and allow them to derive multiple benefits from one product. An example would be coffee that comes pre-mixed with protein powder (e.g., Kitu Super Coffee, REBBL).
- **Hard Seltzer:** Hard seltzer is a seltzer (carbonated water) spiked with alcohol. The sudden burst in popularity is due to the variety of flavors while being low calorie, which appealed to the younger demographic (e.g., White Claw, Vizzy).

Financial Restructuring

Restructuring is unique, requiring an understanding of credit, valuation, and bankruptcy law. On the investment banking side, you'll encounter these questions at restructuring advisory firms or M&A boutiques with a significant restructuring practice. You'll also see these types of questions in corporate banking, private equity, and hedge funds that invest in distressed debt.

If you know that restructuring is where you want to be, in addition to reviewing the questions below, invest the time and watch our [free online course on the basics of restructuring](#).

What are the most common reasons for a company to become distressed?

- **Liquidity Problems:** The catalyst for most restructuring is from the debtor defaulting on its debt obligations, meaning that an interest payment or principal repayment was missed. The reason for this diminished liquidity is most often due to an unexpected deterioration in financial performance.
- **Accelerated Payments:** Certain high yield or straight debt instruments come with accelerated payment provisions, designed to compensate the investor for the entire principal, applicable premium, and interest prior to first call/maturity. The enforceability of so-called "make-whole" provisions has been a controversial topic, but the terms were specified in the indenture and meant to protect the lenders' yield.
- **Failed LBO:** Another reason for bankruptcies is failed leveraged buyouts (LBOs), which are associated with a company being over-levered and financial underperformance, leading to the inability to support the debt load. A recent example would be Toys "R" Us when it filed for Chapter 11 in 2017.
- **Missed Suppliers and Vendors Payment:** A non-debt related example would be the inability to pay suppliers or vendors. Suppliers and vendors are typically more accommodating to help keep the company afloat, albeit they can discontinue working with the debtor if they feel it's a lost cause.
- **Covenant Violation (and Forced Call):** A breached covenant could also force a borrower to declare bankruptcy, as the creditors have an agreement stating that failure to comply with the debt covenants would trigger a forced liquidation (e.g., cannot receive a downgrade from a credit rating agency).
- **Industry Disruption:** Structural shifts in an industry can lead to traditional products/services becoming obsolete (e.g., the OTT trend led by Netflix causing Blockbuster's downfall and bankruptcy).
- **Black Swan Event:** Unforeseeable events with significant macroeconomic impacts that could trigger mass bankruptcies. Examples include the subprime mortgage crisis and most recently, the COVID-19 pandemic that led to the bankruptcies of companies such as J. Crew, Brooks Brothers, and Ruby Tuesday.

What are a few typical signs that a debtor is planning to file for bankruptcy soon?

A debtor that's aware it'll eventually have to file soon will attempt to time the filing date to conserve cash and give themselves as much flexibility as possible. Oftentimes, the filing will be done right before a large payment has come due. Some common indications that a debtor intends to file soon are the revolving credit line will have been drawn to maximum capacity, debt covenants have already been breached (or will be soon), and payments to suppliers or vendors have been delayed.

Other than declaring bankruptcy, what alternatives does a company have to stay afloat?

- **Straight Refinancing:** For the distressed company to replace the old debt issuance with a new issue would be an ideal scenario, but unrealistic as it'll be difficult for a lender to refinance the debt. Another consideration is, the additional debt may help them in the short-term, but then once again, they haven't solved the problem. Instead, the company has just delayed it (or maybe even worsened it) unless they have a turnaround plan bound to work.
- **Raise Additional Equity:** Another option would be to raise capital through share issuances, but very few equity investors would purchase the stock of a company nearing bankruptcy. In the case the company declares bankruptcy, equity holders would receive nothing until debt holders are paid in full (e.g., a recent example was when car-rental company, Hertz, attempted to raise new equity after filing for bankruptcy, which was approved before being suspended shortly after; and now de-listed from the exchange).
- **Out of Court Restructuring:** The most practical option is to re-negotiate the debt terms with existing creditors and work something out out-of-court. The new arrangement could involve temporarily delayed payment dates, extending the maturity of debt (and reducing the due coupon amounts), or switching to PIK interest. Sometimes, the creditors may agree to a "bondholder haircut" in which the debt principal and interest are reduced so that the company can continue operating and avoid filing for bankruptcy.

As a company nears becoming distressed, how might its accounts receivable, deferred revenue, and accounts payable balances change?

- **Deferred Revenue:** For companies with business models based around long-term contracts or subscriptions, deferred revenue tends to increase as the debtor tries to get some much-needed liquidity in return for delivery of a good or service on a later date (often with steep discounts).
- **Accounts Payable:** Days payable outstanding (DPO) will often extend significantly due to the company being incapable of paying the suppliers, even if the company wanted to make the payments.
- **Accounts Receivable:** The inability to collect A/R might be one of the reasons for the lack of liquidity. Regardless, since A/R is a part of the borrowing base, a company nearing distress may provide more services on credit to increase its borrowing base. Customers that are aware of the situation may take advantage of this by intentionally delaying payments in case the company goes bankrupt.

What does operational restructuring involve?

Once becoming distressed, many companies see no chance in an achievable turnaround and are simply liquidated. But others often have a short period to get in front of this problem and try to solve the underlying issues. During operational restructuring, the company will try to improve its operations through cost-cutting, improving working capital management, and headcount reduction. One of the main parts of operational restructuring is asset restructuring, which is the sale of unproductive, non-core assets (or divisions) to re-focus on the core business. If it's an option, the company's management team may retain restructuring advisors to help them navigate through these operational issues or hire 3rd party consultants.

What is the difference between a voluntary and involuntary petition?

In most cases, bankruptcy begins when the debtor files a petition themselves (called a "voluntary petition"). Being the filer provides certain advantages, such as selecting the jurisdiction. Thus, the debtor almost always ends up filing preemptively. Although a rare occurrence, bankruptcies could be initiated if creditors file a "non-voluntary petition." This is when the creditors themselves force the distressed company into Chapter 11 bankruptcy, often done due to a technical default (e.g., broken covenants, violation of terms).

Contrast an out of court restructuring with an in-court restructuring.

Out of Court Restructuring

- An out of court restructuring is the process in which a company negotiates with creditors (both lenders and trade claims) without a court overseeing the process. In an out of court restructuring, all creditors must agree to any modifications to contracts and agreements, and nothing can be imposed on creditors.
- An out of court restructuring's primary benefit is that it can be done more quickly and is less costly than a formal bankruptcy. The process can also remain private, thus mitigating the disruption to the operations of the business related to the bankruptcy's negative publicity. For example, one risk of declaring formal bankruptcy is that vendors will require immediate cash on delivery and stop accepting delayed payments, creating even more stress on the distressed company.
- However, because out of court restructurings require all creditors to agree to changes, it's difficult to get consensus the more parties involved (i.e., leading to holdout problems). In addition, access to debtor in possession (DIP) financing is unavailable, as it's usually difficult to convince new lenders to provide fresh capital without the lender protections that only bankruptcy can provide.

In-Court Restructuring

- Through an in-court process (bankruptcy), a judge oversees the process and is the only way changes to (and outright rejections) contracts like leases, pensions, and debt agreements can be imposed on creditors.
- In-court restructuring addresses many of the issues associated with out of court restructuring by providing protection from creditors who are owed payments ("automatic stay"), addressing the holdout problem by lumping various creditors into various classes and paying all members of the same classes their pro rata share of the recovery, and empowering a judge to reject or modify specific unfavorable contracts
- The distressed company is given access to debtor-in-possession (DIP) financing. DIP lenders are protected via priority (sometimes "super-priority") status.

Out of Court Restructuring	In-Court Restructuring
If successful, an out-of-court restructuring far less expensive than in-court bankruptcy.	Legal fees can drain cash from the business at a critical time when liquidity is a major concern.
Historically, out of court restructurings have been much quicker than in-court bankruptcy processes.	The process can take years, but "pre-packs" have accelerated the timeline as the restructuring plan is agreed upon by all parties before a formal declaration.
The process can remain private and create less disruption to the operations of the business.	All creditor claims, lists of company assets, liabilities, and recoveries are publicly disclosed.
Creditors can continue collection efforts, eviction proceedings, etc.	The company gets an "automatic stay," temporarily protecting the debtor from creditors.
Suffers from the holdout problem, as reaching a consensus with many parties is difficult.	Recoveries agreed to as part of the plan applies to the entire class.
No access to financing as lenders don't want to lend to distressed businesses without the protection afforded to them in a bankruptcy.	Debtor-in-possession (DIP) financing becomes available, which is only an option for companies that filed for bankruptcy protection under Chapter 11.
Leases and contracts cannot be nullified (i.e., reject or reassign leases and contracts).	Non-financial liabilities such as pensions, unfavorable leases, and contracts can be rejected.

What is the purpose of the automatic stay provision?

The automatic stay provision in Chapter 11 temporarily prohibits creditors (e.g., lenders, collection agencies, government entities) from attempting to collect owed claims through legal action. This injunction halts debt collectors' ability to affect the company that has declared bankruptcy negatively. Importantly, the automatic stay provision is designed to enable the company to continue to operate.

The timing of the petition is important as it creates a distinct line between "pre-petition" vs. "post-petition," which prevents creditors from collecting pre-petition claims without approval of the court beforehand.

Liabilities incurred pre-petition are subject to compromise, whereas post-petition liabilities receive admin status priority treatment over pre-petition liabilities.

Which claim holders get priority in a bankruptcy?

1. Creditors with secured claims (i.e., claims secured by liens on a company's assets) have priority in a bankruptcy and usually get the highest recoveries. However, if new debtor-in-possession (DIP) financing has been approved by the Court and put in place, these claims will have a "super-priority" status, even higher than regular secured claims.
2. Below secured claims are administrative claims. These are usually legal, advisory, and other professional fees accruing as part of the bankruptcy process. For the bankruptcy process to function, these claims must be given a higher priority over other unsecured claims.
3. Below secured claims are the unsecured claims. Unsecured claims include all the bondholders and trade claims. As a group, they're called the general unsecured claims ("GUCs"). GUCs are typically the trade creditors and unsecured lenders.
4. There are some unsecured claims to whom the court grants priority over the GUCs. These claims may include accrued employee salaries and taxes owed.
5. Any value remaining after all these claims are satisfied belongs to equity holders. But often there won't be enough assets to make all other claims whole, so existing equity usually gets wiped out in a bankruptcy.

To recap, the waterfall structure is: "Super Priority" Claims → Secured Prepetition → Administrative Claims → Priority Unsecured → General Unsecured Claims (GUCs) → Preferred & Common Equity

Following a bankruptcy filing, will the management team be replaced or remain on board?

In most cases, the same management team will stay in place post-petition. The exception would be if there was proven fraud or any type of misconduct that warrants replacement. Otherwise, the existing management should manage day-to-day operations to turn things around as the management team is the one that likely made the poor decisions that led to this current situation. Thus, they understand where they went wrong and why better than anyone else. But to provide further support (or as an "insurance policy"), often an outside consultant that specializes in turnaround restructuring will be brought on.

In Chapter 11, under which circumstances might a Court appoint a trustee to replace the existing CEO?

In some cases, a court-appointed trustee will displace the management team that placed the company in its current state of financial distress. While this is a less than ideal situation, as more time is required to become familiarized with the company, the Chapter 11 trustee will become in charge of day-to-day operations if deemed appropriate. This appointment of a trustee would be made under two justifications: 1) the management team participated in fraudulent behavior, dishonesty to stakeholders, or criminal misconduct, or 2) the Court has determined it's in the "best interests" of the company and all stakeholders for the current management to be replaced. This is a less objective decision that would usually be based on signs of gross mismanagement (beyond the typical level) and mistrust from all stakeholders.

Can a decline in a company's share price and valuation cause it to become distressed?

By itself, a decline in a company's share price and valuation cannot lead to the company becoming distressed. But the decline reflects the market's anticipation of the company becoming distressed, which may be justified. If the company can continue making its debt-related payments and not breach any covenants, it should continue to operate. A side impact of this negative market perception can be suppliers, vendors, and customers' reluctance to continue working with the company, and the increased difficulty of raising additional capital.

From an advisory standpoint, what are the two sides to a restructuring deal?

A restructuring (Rx) banker can advise the creditor side or the debtor in a restructuring deal.

- **Creditor Side:** The creditor side refers to the loan or bondholders. Rx creditor mandates often involve working alongside many holders of debts within a certain class that have banded together. The more parties involved on the creditor side, the longer coming to a formal agreement will take. On this side, the Rx banker advises creditors/lenders, which often includes hedge funds or private equity firms.
- **Debtor Side:** The debtor is the distressed company. Rx debtor mandates involve dealing with solely one party: the company at risk of bankruptcy unless changes aren't made soon. From this advisory role, the Rx banker gains a deep understanding of the business and works on the "turnaround."

Broadly, what are the two main types of bankruptcies?

There are two main types of bankruptcies: Chapter 7 and Chapter 11.

1. **Chapter 7:** A Chapter 7 bankruptcy refers to the pure liquidation of a distressed company in which all assets are liquidated and then disbursed to stakeholders based upon the absolute priority rule. Following the waterfall schedule, those with higher claims on company assets are made whole before any proceeds can flow down to the claim holders lower in the capital structure.
2. **Chapter 11:** During a Chapter 11 bankruptcy, a company's reorganization is overseen by the court. This process involves putting together a Plan of Reorganization (POR) to identify the impaired classes. Only those who have been classified as impaired have a say in the matter and vote on the terms. Once the POR is agreed upon, the troubled company will then re-emerge under a new capital structure.

What is a pre-pack in a bankruptcy?

A pre-packaged bankruptcy is a type of restructuring procedure where a plan is agreed upon amongst all the relevant parties before a company declares its insolvency. When Chapter 11 is filed, it's accompanied by a disclosure statement, the proposed POR, and an agreement to support the legally binding plan.

The approval will be based on meeting the sufficient number of applicable votes (~85-90% participation rate required). The process has become more common in recent years and can sometimes take as little as 45 days.

Contrast the pre-pack vs. the traditional Chapter 11.

In Chapter 11 bankruptcy, the company can either have a pre-pack or a traditional Chapter 11.

- **Pre-Pack:** A pre-pack involves a pre-planned POR that has been agreed upon by the relevant, impaired creditors before its official filing. The key stakeholders have already been working on the processes together and agreed upon a POR before entering bankruptcy – thus, as you would expect, this streamlines the process considerably and enables the company to emerge quickly from Chapter 11 in a few months.
- **Traditional Chapter 11:** In the traditional Chapter 11, the process starts from scratch and can take years to complete. There can be contention between creditors, as the process can be hectic, and not each creditor is on the same page (often called a "free-fall"). The caveat to the often inefficient, long-lasting traditional Chapter 11 is the pre-pack.

What information is contained in a plan of reorganization (POR)?

Once a debtor files for bankruptcy, the priority is an emergence from bankruptcy via an agreed-upon plan of reorganization (POR) between the company and all relevant claim holders. A POR is a document detailing what will happen to the debtor's assets, liabilities, and equity upon exit from bankruptcy. Most notably, the expected recoveries and the form of consideration for the claims are all identified in the plan of reorganization.

What are the four tests representing the minimum standards of fairness for a POR to be approved?

1. **Best Interests of Creditor Test:** Recoveries under the POR must be proven to be greater than under Chapter 7 liquidation.
2. **Good Faith Test:** POR is required to have been prepared and proposed in "good faith."
3. **Feasibility Test:** Ensures the POR will not result in further restructuring or subsequent liquidation.
4. **Consent or Cram-Down:** Even if an impaired class votes to reject the plan, the proponent may ask the court to adopt the plan despite the objection.

What type of creditor might prefer a Chapter 7 liquidation over a Chapter 11?

A secured creditor that's reasonably assured of recovering 100% under both Chapter 7 and Chapter 11 might prefer a Chapter 7 due to the process being faster. Although recoveries tend to be higher under Chapter 11, this has little relevance to the creditor since 100% recovery is expected from either option.

What role does the US Trustee have in Chapter 11 bankruptcies?

In Chapter 11 bankruptcies, the US trustee will be appointed to monitor and provide oversight on how the process is progressing. The primary responsibility of the trustee is to ensure the debtor in possession is in compliance with operating and reporting rules, as well as to supervise that all reports and documentation requests are filed on time. Overall, the US trustee protects the integrity of the bankruptcy system by overseeing the administration of the bankruptcy case and address any concerns regarding fairness from either the debtor or creditors' side.

How does valuation play a role in bankruptcies?

Valuation often plays a contentious role in bankruptcies because it's subjective but carries very real implications on the recoveries by the claim holders.

Example Bankruptcy Scenario

- Imagine a bankruptcy with \$300 million in face value secured claims, \$200 million in unsecured claims, and the POR submitted to the court by the company proposes to have the \$300 million in debt of the secured creditors reinstated. In contrast, all unsecured claims get their claims converted to equity.
- If the court assumes a value for the company's assets to be \$400 million (based on a combination of a DCF and a comps analysis), there are \$500 million in total claims against a business with a value of only \$400 million. Here, the unsecured claims get 100% of the equity, and current equity owners will get wiped out.
- That's because the equity value of the restructured company is \$100 million, and since it's less than the \$200 million unsecured claim, it must all be given to the holders of the unsecured claims (under the absolute priority rule, unsecured claims need to be made whole before equity gets anything).
- But if the court instead assumed a value of \$700 million for the company, the equity value of the restructured company is now assumed to be \$400 million (\$700 million - \$300 million). The \$200 million unsecured claims are considered to get full recovery when given only 50% of the equity (\$200 million/\$400 million), leaving the remaining 50% of equity to trickle down to the old equity.
- This example shows just how important valuation can be in deciding who gets what in a bankruptcy. Inevitably it leads to arguments about valuations, and the bankruptcy framework encourages negotiation and horse-trading between the various classes to come to an agreement.

What is the relationship between a liquidation analysis and the "best interests" test?

One of the main reasons a liquidation analysis is completed is to compare it to the recoveries received in a proposed plan of reorganization (POR). In other words, for the POR to pass the "best interests" test, it must provide greater recovery to creditors under its proposed Chapter 11 turnaround solution than a Chapter 7 liquidation. So the liquidation analysis essentially sets the floor for the recovery amount of the creditors (i.e., if the recoveries to the creditors are above this floor, the POR is in their best interests).

During Chapter 11 restructuring, what does the cash flow test require?

For a company to emerge from Chapter 11 restructuring, the cash flow test must be passed (i.e., the "feasibility test"). The cash flow test represents the debtor's projected future financial solvency under the proposed POR. In the forecast, it must be demonstrated the new capital structure post-emergence will be sustainable. Even if the plan benefits all stakeholders, the Court may reject the plan if the company will eventually have to be liquidated or require further restructuring in the foreseeable future.

The test will consist of a projection model of a company's cash flows, a debt schedule under the new capital structure and debt terms, and the ending balance of cash for each period. To receive approval, this test must show the company will remain solvent not only in the base case but even if it underperforms.

What is the absolute priority rule, and why is it important?

The absolute priority rule ("APR") is a legal principle in bankruptcy that states that nobody in a lower-priority class gets repaid until the higher priority class gets paid in full. Classes are ranked based on priority and receive recoveries in accordance with the APR, such that creditors in a claims class shall receive full recovery before any value trickles down to a lower claims class.

But in reality, the recoveries by each class deviates from the APR because unsecured creditors or equity get some recovery despite higher priority creditors not receiving 100% recovery. This occurs because claims by the lower classes can be made to hold up the process and litigation can be threatened, which further complicates the process. To prevent the potential hassle, lower classes are sometimes given partial recovery.

In bankruptcies, which class usually holds the largest claim?

Typically, the class with the largest claim will be the general unsecured claims (GUCs), which include pre-petition trade creditors (not given priority claims), unsecured lenders, and all lenders the company owes money to on an unsecured basis (e.g., accounts payable due to suppliers). This class will often include riskier investors of unsecured bonds and unsecured notes.

What does the priority unsecured class involve?

Priority unsecured receives priority over general unsecured claims. These include accrued employee salaries – these get some preference in the bankruptcy process even though they're unsecured. Other groups receive certain priorities in the bankruptcy code. For example, the government might give itself priority status if the company owes them taxes.

How would you value a distressed company?

Companies under distress are often valued both as a going concern and as a liquidated business.

- **Going Concern Approach:** The going concern valuation uses the DCF approach, usually with a higher-than-normal cost of capital assumption to reflect the added risks of investing in a company under distress. A comps analysis may also be used, but since non-distressed comparable companies are used, the multiples derived from the analysis will probably be too high.
- **Liquidation Analysis:** Companies under distress can also be valued using liquidation analysis, which looks at the value of the business as a collection of assets and estimates a liquidation value.

What is the intent of liquidation analysis?

Liquidation analysis values a company based on the assumption it'll go out of business soon, and its assets will be sold. The liquidation value will include only its tangible, physical assets such as its real estate, PP&E, and inventory, while its intangible assets are excluded. Because of the "fire-sale" nature of liquidations, the assets being sold are usually sold at a discount to their fair market value because of the necessity to collect cash within a short time period to return to lenders (and investors).

When using the traditional valuation methodologies, why might valuing a distressed and non-distressed business differ?

The key distinction is regarding the going concern assumption, as much of the total valuation will come mostly from earlier years in the forecast than the terminal value.

To calculate the terminal value (TV), the forecasted perpetuity growth assumption will be very low or may even be a negative rate. A negative perpetual growth rate implies the company will eventually progress towards liquidation each year until dissolving. If using the exit multiple approach to calculate TV, the exit multiple assumption would be on the very bottom-tier of the comparable universe.

While the standard valuation methodologies will be used, there'll be a heavy bias towards being conservative in the financial projections to account for the distressed company's current risks. For example, the discount rates used in restructuring scenarios are higher (15% to 25%+), and relative valuation will use the lower range of the multiples derived from the peer group to reflect the higher risk.

What is distressed debt?

Distressed debt refers to bonds that are trading significantly below face value (below 80 cents on the dollar is a rough benchmark) or very high spreads over LIBOR in the case of loans ($L + 1000$ is a rough benchmark). The causes of distress are usually due to an impending or a perceived need for financial restructuring, often due to operational challenges and high leverage. Often, distressed debt investors will buy the debt from the original debt holders (institutional investors, traditional banks).

Would you rather own bank debt or bonds in a bankruptcy?

That depends. Assuming the bank debt is secured, and bonds are not, the secured bank debt will get higher priority when determining recoveries. Secured creditors often get full recoveries, while unsecured creditors get only partial recovery. If neither claim is secured, in theory, they'll both be treated equally (*pari passu*).

However, there are a lot of other factors that could alter the relative recoveries.

For example, if the bonds are subordinated to other debt through an inter-creditor agreement, the bonds in question would have lower recovery.

In addition, the maturity date of the debt can influence recoveries – for example, if a sizeable chunk of bank debt principal is due prior to the bankruptcy petition date, bank debt recoveries might be higher should the company decide to make the payment.

Lastly, there's another dimension to question from the perspective of a potential investor in the bank debt vs. the bonds, which is the price. For example, let's say secured bank debt is trading at 95, while unsecured bonds are trading at 50.

If I believe the bonds will get 70 cents on the dollar as an investor, I would prefer to buy the bonds because I view them as underpriced.

How does a distressed debt investor make money?

While there are various ways that distressed debt investors make money, at a high level, distressed debt investors generate returns by investing in bonds, loans, or even trade claims that are trading below their likely recoveries. Distressed debt investors will often attempt to accumulate enough of these securities to gain influence in the negotiation with the company and affect the restructuring in a way that increases the chance of recovery in the securities they hold.

For example, a bond trading at 50 cents on the dollar implicitly reflects a market expectation of a 50% recovery.

A distressed debt investor might analyze the company's capital structure and discover that:

- A credible argument can be made to the court that a seemingly more senior claim is not, in fact, senior due to where it sits in the corporate structure or questions related to its lien
- The recovery to the bondholders would be less risky and come more swiftly if the company liquidated (Chapter 7) instead of attempting to restructure (Chapter 11) by taking on more obligations via a DIP facility and continuing to pay exorbitant bankruptcy fees
- The value of some assets has been undervalued

Here, the distressed debt investor might decide to accumulate the distressed debt at 50 and get a seat at the table to influence the judge and other creditors towards their view.

What is a fulcrum security in a restructuring?

The fulcrum security is known as the most senior security that will most likely convert into equity ownership post-restructuring. In the event of a reorganization, the fulcrum security represents the last piece of the capital structure that received partial recovery (i.e., less than par value).

Said another way, the fulcrum security's placement is the point in the capital structure in which the "value breaks." While this may be an oversimplification, the secured, senior class of creditors above the fulcrum security were in high likelihood paid in full (or compensated sufficiently), whereas all claims junior to the fulcrum security theoretically should receive nothing.

The fulcrum security is the most senior security that will not receive a full recovery; thus, it's the tier of claims most likely to convert into equity.

Since the fulcrum security is the part of the capital structure most likely to be converted into equity, the holder is in a position with the most leverage to lead the company's restructuring plan. For this reason, distressed investors focus on identifying the fulcrum security not just because it has the highest probability of converting into or receiving equity in the newly emerged company but as a strategy to have control over the new entity.

What does the phrase "debt to control" mean?

"Debt to control" is when an investor seeks to gain control over a company through bankruptcy or reorganization. Under this strategy, the investor will purchase the distressed company's senior debt and bonds at steeply discounted prices. Thereby, the investor will become a major shareholder if the company emerges from bankruptcy and debt obligations will be exchanged for newly issued equity as part of the agreement. Compared to the strategy of attempting to identify the fulcrum security and having leverage from being the priority shareholder post-emergence, this strategy is considered to be higher risk since influence over the company is obtained through becoming a major creditor of a company and then obtaining a controlling stake in the new equity. Since more capital must be invested at a higher risk, the firms that specialize in this type of investing strategy tend to be of larger scale and institutional firms that invest across the entire capital structure and different asset classes (e.g., Apollo Global), as opposed to distressed hedge funds.

What are “super-priority” claims?

“Super priority” refers to secured DIP claims that have priority over other administrative claims and usually pre-petition secured claims. This is capital that was obtained once the distressed-situation of the company was recognized. Thus, it has repayment priority over even senior debt lenders because its approval was contingent on being beneficial to all the stakeholders, as determined by the Court.

What does priming mean in restructuring financing?

Priming refers to a situation in which a lender high in the capital structure (e.g., secured loan) is superseded by another lender. Thus, priming is when a new tranche is added to the existing capital structure – but the key difference is, this tranche is higher than the others, including the senior secured lenders. In a restructuring, DIP financing gets a superior claim to all others out of Chapter 11.

What is DIP financing and why is it considered as being safe for the lender?

Debtor-in-possession financing (“DIP financing”) is a special type of financing only available to companies under distress that have already filed a bankruptcy petition. DIP financing is placed at the very top of the capital structure. Legally, this form of financing has super-priority administrative over all other existing stakeholders. In most cases, DIP financing comes from those who had previously provided loans to the company in the past. DIP financing is considered very safe for lenders due to being secured, having collateral backing, and coming with very restrictive covenants.

What conditions must be met for DIP financing to be approved?

The Court will only approve DIP financing if there's adequate collateral to protect pre-petition lenders. DIP financing that primes existing secured creditors must be deemed to make pre-petition secured creditors better off than in a liquidation. Once it has been proven not to impair the secured holders and is determined to bring enough positive benefits to the company, the DIP financing loan will help the distressed company with its business operations and reorganization.

Could you explain the purpose of the TWCF model in restructuring?

The thirteen-week cash flow model (“TWCF”) is a required submission in a bankruptcy process to track the liquidity of a company vulnerable to bankruptcy. The TWCF is used in turnaround situations once a company enters or is on the verge of becoming financially distressed.

The model provides visibility into the company's short-term liquidity to assess its options for short-term cash management. While quarterly and annual models will be created for the troubled company, the priority is to ensure the company can stay afloat in the short-term – hence, the TWCF's importance. Besides the transparency to creditors and its usage as a tool by management to make critical decisions, the TWCF is used to determine the borrowing needs during the reorganization period. If the ending cash balance falls to a level too low, a drawdown of the revolver or DIP financing will be necessary.

How does the TWCF model differ from typical financial models?

- **Cash-Based Accounting:** TWCF is a cash-basis analysis, meaning it uses cash flow accounting instead of accrual-based accounting. This direct method for cash flow accounting is more insightful for the restructuring advisors, as the real cash balance matters and represents a more accurate portrayal of the company's liquidity. The company's cash inflows/(outflows) are projected carefully in a weekly and monthly cash flow budget to provide the highest transparency level for management and creditors.
- **Weekly Rolling Forecast:** Notably, TWCF models are based on rolling forecasts and updated continuously by Rx bankers weekly, enabling granular comparisons between the projected numbers vs. the actual

numbers. The variance can also be assessed to better understand the company's financial health and the borrowing needs (DIP) during the reorganization period.

Walk me through the basic structure of a TWCF model.

1. **Operating Cash Inflows:** The operating cash receipts will first be counted, representing the company's cash collections from customers. Then, operating cash disbursements such as payroll, insurance, and inventory purchases will be deducted to arrive at operating cash inflows.
2. **Net Cash Inflows/(Outflows):** Next, non-operating disbursements such as interest payments, principal amortization, and professional fees are subtracted to calculate the period's net cash inflows/(outflows).
3. **Ending Cash Balance:** In the final step, the ending cash balance will be calculated by subtracting the net cash inflows/(outflows) calculated in the previous step from the beginning cash balance.

What is an "amend and extend" agreement?

An "amend and extend" is when the debtor has come to an agreement with the lending parties for the debt to remain in place, but with the maturity extended to a later date. In return for the extension, the cost to the borrower is usually an increased interest rate, an amendment fee, a different amortization schedule, or other newly negotiated debt terms that favor the lender.

How do you determine the value of a secured claim?

Collateralized claims are secured to the extent the pledged collateral covers the claim. If the value of the collateral is less than the claim, this shortfall is treated as unsecured. Therefore, the amount of value a secured claim has is limited to the minimum between the secured claim and the creditor's collateral value. To the extent a secured creditor has collateral worth less than its claim, this deficiency would be treated as general unsecured claims (GUCs).

The claim of a secured creditor was bifurcated into two claims. Could you explain when this might take place?

If the collateral value is greater than the claim, then the claim is fully secured. However, the secured creditor's claim being bifurcated means there was inadequate collateral. The secured claim's value will be up until the collateral value, while the other claim is unsecured and treated just like any of the other unsecured claims. Despite being secured by the debtor's collateral, secured pre-petition claims only have priority over admin and other unsecured claims to the extent of the collateral value. Therefore, if the collateral is less than the claim, the under-collateralized portion will be treated as an unsecured deficiency claim.

How does the fiduciary duty of management change during a bankruptcy?

During a bankruptcy, a shift in the management's fiduciary responsibility occurs once a company appears insolvent (or close to being so). Specifically, management's fiduciary duties must change from equity holders to creditors when the company enters the "zone of insolvency."

What does a negative pledge provision restrict a borrower from doing?

In a negative pledge provision, the borrower agrees to not pledge certain specified assets as collateral. From the perspective of the lender, prohibiting the borrower from granting a lien to an additional lender protects them from the risk of the available collateral being reduced and prevents a new creditor with higher priority. In effect, negative pledges make it more difficult for the borrower to raise debt.

What is the purpose of adequate protection?

Adequate protection limits what a firm can do with the collateral pledged and is put in place in an agreement for secured creditors to be assured their interest in the collateral assets are protected (and not sold off or damaged). The lack of adequate protection can be grounds for relief from the automatic stay, which means

creditors have the right to seize the debtor's assets or property based on the contractual terms. Thus, as long as secured creditors are adequately protected, the debtor may use the cash collateral (as well as any other assets with liens on them). The key benefit is that these adequate protection liens are granted “super-priority” status – above post-petition admin claims.

What is the main difference between liquidation done under Chapter 7 and 11?

Liquidation can occur in both Chapter 7 and 11. The key distinction is that under Chapter 7, the management team will be removed, and instead, a liquidating trustee will be appointed to oversee the bankruptcy process.

What is a plan support agreement?

Plan support agreements obligate creditors that sign it to vote for a plan consistent with the terms outlined in the agreement. These agreements reduce the overall costs and duration of bankruptcy proceedings while creating less disruption to the business and its employees/customers.

What is the difference between consent and a waiver?

Both refer to creditors giving their consent to the company breaching covenants. However, a waiver is temporary (e.g., temporary breach due to an unforeseeable event), whereas consent is permanent.

What are debentures?

Debentures (or unsecured bonds) are unsecured obligations with covenants that are much less restrictive than those of bank loans. If the troubled company undergoes bankruptcy, there are no assets the holder of the debentures can claim and are paid only once secured bondholders are compensated, assuming these lenders, which are low in the capital structure, even receive any form of compensation.

What does a sale under Section 363 of the US Bankruptcy Code involve?

A Section 363 Sale is when a distressed company sells its assets and provides a potential opportunity for buyers (often called “stalking horse bidders”) to purchase the assets at a bargain. A key feature of a 363 Sale is the stalking horse offer bid is subject to higher bid offers within a specified period and would require the Bankruptcy Court's approval beforehand. Perhaps the most significant risk to bypass in this process is an improperly conducted Section 363 Sale that fails to receive approval from the court.

Would a Section 363 Sale hurt creditors because of the loss of collateral?

The contrary would be true as the sale cannot be completed without the creditors' approval, the key stakeholders in the bankruptcy process whom the sale process should benefit above all else. Thus, a completed sale would imply that no objections were presented to the Court.

For example, if the debtor is attempting to sell an asset at a discount far below the fair value, the Court must listen to the creditors' objections before granting approval/rejection – but again, the creditors' interests carry more weight than the debtor.

In addition, 363 sales enable secured creditors to place credit bids, which can help cancel a portion of the debt amount owed by the debtor. If eligible, the secured creditor can opt to bid on the debt owed to them on which the asset is collateral (i.e., acquire the asset solely using credits and with no cash outflow).

What is credit bidding?

Credit bidding is a feature of acquisition in Chapter 11 that allows secured lenders to use their claim as acquisition currency without actually having to put up new cash (only secured claims can credit bid). Notably, credit bidding enables investors, who are often financial sponsors, to acquire secured debt from existing creditors at a discount and credit bid the full claim to purchase the underlying collateral.

What is the purpose of bondholder committees?

During an out-of-court bankruptcy, the various lending parties will often band together in a bondholder committee to negotiate with the borrowing company with more leverage.

These committees work alongside one another to increase the probability that the borrower will agree to their negotiation terms in the out-of-court restructuring scenario. If needed, the committee may hire a 3rd party Rx advisory shop to help them conduct their analysis and solidify their pitch.

Why might certain distressed funds be hesitant to join a bondholder committee?

The downside to joining a bondholder committee is the disclosure requirements – this obligation to disclose security positions is typically why certain distressed funds refuse to join a committee. Once the fund becomes a part of the committee, they must sign an agreement to disclose when they buy or sell securities.

Why does the holdout problem occur in out-of-court restructurings?

A company cannot unilaterally provide individual creditors with partial recoveries out-of-court – creditors must agree. This agreement becomes increasingly challenging to get a consensus when there are more parties involved. In these cases, especially when there are many bondholders, bondholders may calculate that if enough other creditors participate, they'll be better off not participating and insist on a full recovery. In contrast, for in-court restructuring, the recoveries agreed upon applies to the entire class.

What are administrative claims?

Otherwise known as post-petition unsecured claims, administrative claims represent the necessary costs and expenses of preserving business operations' continuance. Examples of administrative claims would include employee wages/salaries, payments for goods and services received post-petition, and professional fees (taxes, legal fees, turnaround consulting services).

Name some specific add-backs to EBITDA in the context of restructuring.

- Professional Fees (Restructuring Advisory, Accounting, Legal Fees, Turnaround Consulting)
- Employee Severance Costs
- Legal Settlement Expenses
- Inventory or PP&E Write-Downs
- Goodwill Impairments
- Uncollectible Accounts Receivable ("Bad AR")
- Gains/(Losses) on Sales of Assets and Facility Closures

What is a critical vendor motion?

A critical vendor motion is when vendors deemed "critical" receive their pre-petition claims to ensure their ongoing affiliation with the debtor. The borrower cannot operate in any capacity without this vendor. Therefore, the Court has approved the filed motion that allows the company to pay this vendor.

Courts will sometimes approve a company's request to make payments to certain critical vendors during the bankruptcy before adopting the reorganization plan – reasoning that not paying will degrade the company's value in a way that hurts the higher priority claims even further.

What is a fraudulent conveyance?

Fraudulent conveyance is when the debtor, when the company was considered to be already "insolvent," made a cash, property, or other asset transfer intending to defraud its creditors. The lienholder that claims a fraudulent transfer must prove the company was insolvent when making the sale and that the sale was made to delay or avoid its due obligation to its creditors. If successful, the lienholder can claw back some proceeds.

Give me an example of a spring forward provision.

An example of a spring forward provision would be a term loan with a longer maturity than a senior note lower in the capital structure. The term loan lending agreement could include a provision allowing the payment date to “spring forward” if a specific triggering event is met. This contingency could be based on the senior note being refinanced by a pre-specified date, the revolver being drawn too frequently or past a stipulated percentage, a credit downgrade from a rating agency, or if an upcoming debt swap offer was not accepted. The cause could also have been related to the traditional financial covenants, such as leverage ratio surpassed a certain level. In most cases, the spring forward provision will pull the maturity of the debt ~90 days earlier than when the other corresponding debt would come due.

What are voidable preferences?

Voidable preferences occur when a monetary (or asset) transfer is processed less than ~90 days before the official bankruptcy filing. This often refers to disputes where senior creditors have accused the debtor of an unwarranted payment right before declaring bankruptcy to a creditor lower in the order of payment. If the creditors making the accusation successfully prove the payment violated the waterfall schedule, the payment would thus be voided and be legally required to be returned to the rightful creditors.

What does a cramdown refer to in a restructuring scenario?

In the so-called cramdown provision, a plan of reorganization (POR) is confirmed and imposed on an impaired class that had voted to reject the plan. Despite the objections, the Court has the authority to approve a proposed plan regardless if one or more classes of creditors assert that their claims would be impaired further if the plan is confirmed.

What is an equitable subordination?

Although a rare occurrence, secured creditors should be aware of the very unfavorable scenario that the courts could unilaterally equalize all claims under certain circumstances (called “equitable subordination”). The type of situation that would invoke such an uncommon decision and is typically associated with the misconduct of secured creditors with proof of wrongdoing (i.e., “acting in bad faith”). For example, an insider could be acting on the secured creditor’s behalf to influence the debt to the detriment of other creditors.

How is a capitalization table used for restructuring purposes?

For restructuring, a capitalization table (“cap table”) will be put together and continuously updated by the Rx banker to assess the debtor’s liquidity. All the claims on the company will be listed, but with a focus on the debt securities that the company has issued. Next to each debt tranche, the maturity dates, coupon rate/pricing, credit ratings, and market price (if publicly traded) will be listed, along with the key credit metrics calculated, such as the leverage and interest coverage ratios. By assessing the maturity dates of all the looming debt maturities and the dates of the interest payments coming up, the Rx bankers can gauge the urgency of the situation regarding the company’s liquidity and need for DIP financing.

Do equity shareholders receive any form of compensation during Chapter 11s?

Theoretically, both preferred and common equity holders will get wiped out and receive nothing during Chapter 11s. This is intuitive since equity is at the bottom of the capital structure; however, equity holders are occasionally given a “tip” (or called “tipping equity”) where they’re given 1-2% of the newly reorganized company’s equity or a nominal payment.

Real Estate (REIT)

REITs are a unique type of investment. If you're interviewing at a REIT or in the Real Estate group at a Bank, make sure you read [this guide](#) before proceeding so you have a general understanding of what REITs are all about.

Do REITs pay taxes?

REITs are “pass-throughs,” meaning REIT profits are not taxed on the corporate level and instead get passed through to the shareholder who gets taxed at the individual level on the REITs' dividends.

However, one requirement for qualifying for this tax-advantaged REIT status is that at least 90% of the REIT's profits must be distributed as dividends, so a REIT can't hoard profits like a C Corp and not issue any dividends.

Also, a REIT's dividends are “non-qualified,” meaning they're taxed at the ordinary income (higher) tax rate, not the capital gains (lower) tax rate. In contrast, C-corporations have a corporate level tax, followed by another tax on dividends, but C-Corp dividends are qualified, meaning they're taxed at a lower capital gains tax.

REITs don't pay corporate-level taxes, but REITs pay property taxes on the properties they own. These taxes are included as part of direct real estate expenses on a REIT's income statement.

Walk me through the income statement of a REIT.

Primary Line Items

- On the income statement, rental income is the primary source of revenue. In addition, there may be ancillary income, such as management fees, which are included in revenue.
- Direct real estate expenses include utilities, marketing/advertising expenses, payroll, maintenance, and property taxes.
- Revenues – Direct Expenses = REITs Net Operating Income, which represents a REIT's property-level profitability.
- Next, we subtract corporate overhead, SG&A expenses, and depreciation expenses to get to GAAP operating profit. Under US GAAP, REITs usually have significant depreciation expenses from their asset-intensive nature (there's no depreciation for REITs under IFRS).
- Next, we subtract interest expense, which is usually large for REITs because REITs borrow a lot to finance their operations.
- Lastly, REITs are constantly buying and selling real estate, so there's usually significant income or loss from discontinued operations (along with gains or losses on sales) reported below net income.

What is NOI and why is it an important metric?

Net operating income (NOI) equals a REIT's rental income less property-level real estate expenses such as utilities, marketing & advertising, payroll, maintenance, and property taxes. It's a measure of profit before depreciation, corporate-level SG&A, and interest expense.

$$\text{Net Operating Income (NOI)} = \text{Rental Income} - \text{Operating Expenses}$$

The NOI is an important metric because isolates property-level profitability, enabling comparisons across different properties or portfolios of companies without the noise of corporate-level expenses or leverage.

While not the same, it's the closest thing REIT financials have to gross profit. NOI is the numerator used in the calculation of a REIT or real estate property's cap rate.

Define funds from operations (FFO) and why is it important?

Funds from operations (FFO) refers to a figure commonly used by real estate investment trusts (REITs) to measure operating performance. Assuming US GAAP, the FFO formula is shown below:

$$\text{FFO} = \text{Net Income} + \text{D\&A} - \text{Gain on Sale} + \text{Non Controlling Interest} - \text{Cash to Non Controlling Interests}$$

FFO is similar to cash from operations because it excludes things like working capital changes. Another way to think of FFO is to think of it as basically levered EBITDA, with the difference between FFO and EBITDA is that EBITDA attempts to capture profitability from operations. In contrast, FFO is levered and captures the effect of non-common equity claims against the business, such as debt, preferred shares, and non-controlling interests.

For REITs valuing properties, how do regulations differ between GAAP and IFRS?

Under IFRS, REITs must value properties as “Investment Properties,” meaning no depreciation is recognized, and instead, the properties are regularly marked to market (which is recognized as a gain on the income statement). Therefore, the bridge from IFRS net income to FFO doesn't require a depreciation add-back since there was no depreciation expense on the income statement, and the revaluation gain needs to be subtracted from IFRS net income to get to FFO.

What is the difference between FFO and AFFO?

Adjusted FFO, or Cash Available for Distribution (CAD), takes FFO and subtracts capital expenditures from FFO to get closer to free cash flow (though still missing the working capital adjustments).

$$\text{Adjusted Funds from Operations (AFFO)} = \text{FFO} + \text{Non Recurring Expenses} - \text{Capex}$$

Read More → [Funds from Operations \(FFO\)](#)

Walk me through a NAV valuation.

The most common real estate valuation is a valuation driven by the net asset value (NAV) model.

NAV Valuation Process

- 1. Value NOI-Generating Assets:** This step involves identifying the NOI generating assets from the real estate portfolio and then dividing it by a cap rate assumption that represents that cap rate for the cumulative portfolio. The cap rate assumption is the most important part of the valuation. If distinct NOIs and cap rates are available by property, region, or property type, a more detailed valuation be done.
- 2. Value Other Income Streams:** If ancillary income and management fees are not included in the NOI, these should be valued separately, which is also done by taking the income from those activities and dividing it by a cap rate assumption, which may or may not be the same as the cap rate used to value the NOI-generating real estate. In addition, any JV income should be valued separately using the same approach and added to the NAV.
- 3. Reduce NAV by Required Future Expenditures:** Next, NAV must be adjusted down by the present value of future expected capital expenditures and corporate overhead required to sustain the properties that are not included in the NOI. This is also typically done by dividing the expenditures by a cap rate assumption.
- 4. Add “Non-Operating” REIT Assets:** Any assets not already included in the NAV build up like cash and construction in progress should be added to the NAV buildup. These can be valued at book value or by applying a premium to book value (in the case of construction in progress).
- 5. Subtract Debt & Other Non-Equity Claims:** Subtract debt and other debt-like claims not yet valued – often the book value is used under the thinking that book values for these liabilities don't deviate too far from market values, but if there's reason to believe deviations exist, market values should be used.
- 6. NAV-Derived Equity Value:** The end result represents the REIT's NAV-derived equity value. This can be compared against its market cap or divided by shares outstanding to compare against its share price.

Why might the NAV be preferable to a DCF for REITs?

The NAV approach is so well suited for real estate because, unlike other industries, REIT balance sheets carry real estate assets for which there are often comparable assets with observable market values. The NAV approach recalculates a REIT's balance sheet to reflect fair market values of assets using cap rate assumptions. This is simply not feasible for most industries, so a discounted cash flow analysis is often employed instead.

What is a cap rate, and why is it important?

Capitalization rates ("cap rates") are the most common real estate multiple, which compares a measure of unlevered profitability against a property's value.

$$\text{Cap Rate} = \frac{\text{Net Operating Income (NOI)}}{\text{Property Value}}$$

For example, if a \$2 million property is generating annual NOI of \$100,000, it has a cap rate of \$100,000/\$2 million = 5.0%. It is a little different from traditional valuation multiples in that the numerator and denominator are reversed so that a high cap rate implies a low valuation (and vice versa).

All else being equal, a property with a higher cap rate relative to comparable properties suggests it's undervalued or has some combination of higher risk, lower expected future returns, or growth.

What are the most common methods for valuing REITs?

REITs Valuation Methods

- **Net Asset Value ("NAV"):** Applies to US REITs, where book values are not marked up to their FMV
- **Price to Book (P/B):** Applies to IFRS REITs because the book values reflect market value due to IFRS revaluation rules
- **Relative Valuation of REITs:** Refers to comps analysis
- **Intrinsic Value of REITs:** Used less often but can supplement the more common approaches

REIT Valuation Multiples

- **"FFO" Multiple:** FFO = Equity Value/Funds from Operations
- **"AFFO" Multiple:** AFFO = Equity Value/Adjusted Funds from Operations

Intrinsic REIT Valuation Methods

- **Dividend Discount Model (DDM):** Unlike other types of businesses, the DDM is suitable for REITs because at least 90% of a REIT's profits must be distributed to shareholders. The DDM discounts all future expected dividends using the cost of equity.
- **Discounted Cash Flow Analysis (DCF):** The traditional DCF valuation can be done for REITs as with other industries, but in practice, it's used far less often than the NAV approach.

Read More → [REIT Valuation Approaches](#)

Why is the self-storage industry known to be such a resilient performer?

The self-storage industry has been one of the most reliable, non-cyclical asset classes in real-estate. During the 2008 recession, self-storage REITs were the only real estate asset class that produced positive returns, when others such as hospitality fell 50%+. The increased urbanization has led to the recent spike in demand over the past few years in the US. Millennials are opting not to purchase homes – as shown by the decrease in new home sales. While self-storage was traditionally viewed as a place to store unneeded items that consumers were reluctant to dispose of, these facilities have become extensions of homes (i.e., extra closet space/garage).

Real Estate Private Equity (REPE)

Real Estate Private Equity (REPE) is a subset of private equity that deals with real estate assets. To learn about the recruiting process, industry, salary, main players, and career path, [read this guide](#).

What are the three methods for valuing real estate assets?

1. **Cap Rates:** Property Value = Property NOI/Market Cap Rate
2. **Comparables:** Similar transactions inform price per unit or per square foot valuations (and current market cap rates)
3. **Replacement Cost Method:** An investor would never purchase a property for more than it could be built.

Compare the cap rates and risk profiles for each of the main property types.

1. **Hotels:** Hotels trade at the highest cap rates because cash flow is driven by nightly stays (extremely short-term) and more operationally intensive activities like restaurants and conferences.
2. **Retail:** The creditworthiness of retail tenants is increasingly in question due to trends in e-commerce.
3. **Office:** Office sector is closely correlated to the broader economy but has longer-term leases.
4. **Industrial:** Industrial sector benefits from e-commerce trends and longer-term leases.
5. **Multifamily:** Multifamily is thought of as the safest asset class because no matter how the economy is performing, people will need a place to live.

Walk me through a basic cash flow pro forma for a real estate asset.

1. The top line is revenue, which will be primarily rental income but might include other revenue lines and almost always deduct vacancy and leasing incentives such as rent abatements and concessions.
2. Next, you subtract all operating expenses to get to NOI.
3. After NOI, you subtract any capital expenditures and account for the purchase and sale of a property – this will get you to unlevered cash flow.
4. Finally, to get from unlevered cash flow to levered cash flow, you subtract financing costs.

Describe the four main real estate investment strategies.

1. **Core Investments:** Core is the least risky and therefore targets the lowest returns. Core investments are typically newer properties in great locations with high occupancy and very creditworthy tenants.
2. **Core-Plus Investments:** Core-plus is similar to core but slightly riskier and may feature minor leasing upside or require small amounts of capital improvements. Core-plus is what most people think of when they hear “real estate investing.”
3. **Value-Add Investments:** Value-add investments are riskier deals, and risk can come from various places – substantial lease-up, an older property needing meaningful capital improvements, a tertiary location, or poor credit tenants.
4. **Opportunistic Investments:** Opportunistic is the riskiest area and therefore targets the highest returns. Opportunistic investments include new development or re-development.

If I paid \$100 million for a building and it has 75% leverage, how much does it need to sell for to double my equity?

\$125 million. With 75% leverage, you would invest \$25 million of equity and borrow \$75 million of debt. If you doubled your equity, you would get \$50 million (\$25 million x 2) of cash flow to equity and still need to pay down \$75 million of debt. \$50 million of equity + \$75 million of debt = \$125 million sale price.

If you had two identical buildings in the same condition and right next to each other, what factors would you look at to determine which property is more valuable?

Since the physical attributes, building quality, and location are the same, I would focus on the cash flows.

- First, I would want to understand the amount of cash flow. You can determine this by looking into what average rents are in the buildings and how occupied the buildings are. Despite the same location and quality, each building's management and leasing could vary, leading to differences in rents and occupancy.
- Second, I would want to understand the riskiness of the cash flows. To assess this, I would look at the rent roll to understand tenants' creditworthiness and the term of leases.
- Lastly, the value of the properties would be looked at. The formula for value being NOI/cap rate. NOI will be informed by the cash flows amount, whereas the cap rate will be informed by the cash flows' riskiness. The property with higher cash flow and less risk will be valued higher.

Assume you purchased a property for \$1 million at a 7.5% cap rate, had 0% NOI growth throughout the holding period, and exited at the same cap rate after three years. What is your IRR?

First, we know that the NOI/Cap Rate = Value. If a property's NOI and cap rate don't change, then the value remains the same. Because there's 0% NOI growth and after three years, we are selling the property for the same 7.5% cap rate we purchased it for, we'll sell the property for \$1 million, resulting in no terminal value profit. Since there's no terminal value profit, the only profit comes from interim NOI, which is \$1 million x 7.5%, which remains constant each year. IRR is our annual return, so IRR equals our cap rate of 7.5%.

If you purchase a property for \$1 million at a 5.0% cap rate with 60% leverage and a 5.0% fixed cost of debt, what is the cash-on-cash yield?

1. First, we know that Cash-on-Cash Yield = Levered Cash Flow/Equity Invested, and Levered Cash Flow = NOI – the Cost of Debt.
2. If 60% of leverage was used, this implies \$600k of debt and \$400k of equity invested. A \$1 million purchase price at a 5.0% cap rate implies \$50k of annual NOI. \$600k of debt at a 5.0% fixed cost implies a \$30k annual cost of debt. \$50k annual NOI - \$30k annual cost of debt = \$20k of levered cash flow.
3. Lastly, \$20k levered cash flow/\$400k equity invested = 5.0% cash-on-cash yield.

Do you have any questions for me? (REPE Context)

Roles & Responsibilities Questions

- Is there a strict delineation between the acquisitions and asset management team?
- How much exposure do junior acquisition professionals get to the legal process of executing a deal?
- How much traveling is there for junior team members?

Firm Strategy Questions

- Does the firm focus on a single risk profile (core, core-plus, value-add, opportunistic) or multiple strategies?
- Does the firm only do equity investing or both debt and equity?
- Does the firm do any development or only acquisitions?

Walk me through a back-of-the-envelope multifamily real estate acquisition model.

- This type of test is usually part of its separate section of the interview. You'll have approximately ten minutes and be provided pen and paper, or with Excel, to complete a basic modeling test.
- [Read this step-by-step tutorial](#) on building a back-of-the-envelope (BoE) multifamily real estate acquisition model.

Industrials

Why have industrial manufactures shifted their focus towards the aftermarket?

Over the past decade, there has been a clear shift in industrial manufacturers becoming more service-oriented and offering more aftermarket services, an all-encompassing term that includes maintenance services, spare parts delivery, and other value-added services. The overall cyclical nature within the manufacturing industry in terms of customer demand and market maturity has led to manufacturers attempting to mitigate the risk of cyclical fluctuations by decreasing the proportion of revenue from new equipment sales.

For industrial manufacturers, achieving organic growth through new product development and sales can be a costly, capital-intensive process. Comparatively, revenue from aftermarket service is a more predictable, recurring stream of revenue. In the latest recession, those with a greater proportion of revenue coming from services fared far better compared to those that are 90%+ dependent on new manufacturing sales. While certain manufacturers have built-out their services division in-house, most industry players have chosen the path of vertical integration through increased M&A activity.

This trend has been escalated from COVID-19, as demand for new industrial equipment has declined as their end markets reduced capital spending to preserve cash (especially in the aerospace industry).

What is the trend of servitization in manufacturing?

Simply put, servitization refers to building additional revenue streams for manufacturers from providing more service offerings, rather than depending on a third party. Some of the most common services include replacing spare parts, product repairs, maintenance, overhauls, performance monitoring, and customer service.

In the past, the focus was solely on equipment manufacturing. Sales and distribution would be handed off to 3rd party equipment vendors and customers would rely on specialists for product support. Nowadays, manufacturers are selling "systems," focusing on the outcome of their products and providing the services themselves to ensure customer satisfaction. This trend is in-line with the shift towards placing greater emphasis on after-sales service operations to improve customer retention and increase margins.

Why is there such a high level of M&A activity in the industrial manufacturing space?

Historically, industrial manufacturing has been one of the most active areas for M&A because of the constant innovation in product development. This can be attributed to the highly specialized nature of the industry in which there are hundreds of niches, which all provide differentiated capabilities.

For those reasons, existing incumbents acquire companies with proven technologies and an existing customer list rather than incurring high R&D expenses to build it in-house.

Financial sponsors are active in this industry and attempt to acquire niche players to grow through add-ons and then exit to large strategics such as Honeywell, Siemens, ABB, ASSA ABLOY, and Johnson Controls.

For most of these industrial conglomerates, recent revenue growth has become almost entirely inorganic growth. Many of these financial sponsors acquire companies under the belief that there'll be buyers when they look to exit their investment.

The more specialized niches there are within an industry, the higher M&A activity will be generally.

Which valuation multiples are most often used to value companies in industrials?

The most common multiple used valuation multiple for industrials is EV/EBITDA, especially for industrial tech. EV/(EBITDA – Capex) is often used for manufacturing companies, while EV/EBITDAR can be used for the transportation and logistics segments if equipment such as the trucks are rentals.

A few more valuation multiples specific to transportation and logistics would be EV/Total Freight Tonnage, Volume, EV/Miles Driven, and EV/Average Revenue per Mile or Weight Unit.

Let's say you're looking at an industrials company. What business and industry characteristics would you first consider?

- **End Markets:** First, the customers served and their diversification will determine how cyclical the industrials company's revenue is. The relationship between the company and their customers, and how essential their products are will determine how recurring the revenue is.
- **Supply Chain:** Next, I would consider how efficient the company's processes are by looking at how many suppliers it depends on and how well integrated it appears. Other than the manufacturing process, the distribution network would need to be examined to see how much reach the company has and opportunities for expansion by selling in more channels.
- **Product Value/Upgrade Cycles:** Then, the product or service being offered should be looked at when determining how valuable the offering is and how often upgrades (or any type of recurring service such as maintenance) would be required.
- **Competitive Landscape:** Lastly, the strength of the industry's barriers to entry and how the company's products fare compared to its competitors should be examined. These two factors will directly affect the stability in the revenue generated and the pricing power of the company over its end markets.

An industrials company has shown COVID-related impaired revenue. What metric would you look at to assess the damage and predict the pace of recovery?

The metric the interviewer is looking for is the build-up of backlog (or the lack of it). Ideally, the company's backlog should have significantly increased despite the decrease in revenue. This is because many customers have delayed their projects into the future, either due to their own decision or regulations that prevent the completion of the project.

The backlog represents customer contracts that have been secured but have not yet been earned. Throughout much of Q1 and Q2 2020, the leading industrial conglomerates saw a healthy backlog and was a key talking point their management team would focus on during the earnings calls. The reason being management could point towards this pent-up backlog as evidence to back up their statement that a swift recovery would be underway going-forward. The absence of backlog or lagging growth compared to its competitors would be viewed negatively as a sign that the company couldn't secure commitments from new customers throughout the lock-down period and/or existing customers have canceled their projects by choice or gone bankrupt.

Walk me through the stages from Industry 1.0 to Industry 4.0.

- **Industry 1.0:** Referred to as the "Industrial Revolution," this was when manufacturing moved from human labor to reliance on machines. The machines were fueled by water, steam, and coal power, and the production process became easier and faster.
- **Industry 2.0:** Often called the "Technological Revolution," this is when electrical technology entered and allowed greater production volume and more sophistication in the machines used.
- **Industry 3.0:** Known as the "Digital Revolution" and began with the early days of computers, which were massive despite the minimal computing power provided. However, this period planted the seeds for the electronics and IT infrastructure for the future stage of reliance on computer technology.
- **Industry 4.0:** This is where our society is currently at and is often described as the "Automation Revolution." Industry 4.0 is most notably characterized by the new capabilities of automation in which machines can decide and govern themselves using key innovations such as IoT, cloud computing, and big data. Other examples of developments include smart factories, sensor technology, and voice/facial recognition, to name a few.

How did Industry 4.0 help bridge the gap between OEMs and their customers?

Before the technological advancements and hyper-connectivity from Industry 4.0, the original equipment manufacturer (“OEM”) would produce replacement parts in which service contractors would then perform the planned maintenance and reactive repairs on their behalf. The result is limited visibility into the existing install base and less customer connectivity/engagement.

Industrial IoT has allowed OEMs to control uptime and monitor performance throughout the asset and service life-cycle relationship. Also, guaranteed service level agreements (SLAs) are directly put into place and enable better customer relationships.

What are some concerns surrounding Industry 4.0 that exist to this day?

- **M2M Mistrust (IoT):** Despite all the progress made in all fronts of IoT, machine-to-machine communication has its limitations in terms of reliability and has not been fully developed to have the level of performance and stability envisioned by many (i.e., 100% trust).
- **Security of Data:** The vast amount of data collection and the level of access these machines are given create concerns for IT security – especially for facilities without the most updated security infrastructures.
- **Costly Mistakes:** In manufacturing, slight mishaps or minor mistakes could be very costly and be damaging to the production process. Therefore, processes are rolled out gradually at lower volumes.

What is Industry 5.0, and what will it do for manufacturing once this next phase arrives?

The term Industry 5.0 could be best described as humans working alongside robots and smart machines. Compared to Industry 4.0, it's more about enabling humans to work efficiently with robots and machines collaboratively by leveraging new advancements in existing technologies such as IoT and big data.

Industry 5.0 is necessary because consumers demand customization in their products, and a requirement of this is the creative input from a human – this has been the driving force behind this movement of human contributions returning to manufacturing. In Industry 5.0, IoT, AI, and ML will all continue to play a significant role when handling repetitive tasks. However, humans would have increased responsibility for being the decision-maker for perception-driven design and customization.

What are co-bots, and what makes them unique?

Shorthand for “collaborative robots,” co-bots are intended to operate alongside humans. The distinction is that these co-robots and humans can safely complete tasks right next to one another in proximity, rather than a robot performing tasks and the human monitoring its performance from a safe distance.

What is the difference between customer support agreements and outcome contracts?

- **Customer Support Agreements:** A customer support agreement is an arrangement signed between the customer and service provider to outline the work, the terms and conditions surrounding its completion, and the specific costs for each type of service provided.
- **Outcome Contracts:** As its name suggests, outcome-based contracts are result-oriented and focus on their equipment quality. The contract terms and the pricing are tied to the achievement of specific, measurable performance metrics and requirements that must be met (i.e., KPI-based contracts).

What is the difference between process manufacturing and discrete manufacturing?

- **Process Manufacturing:** Process manufacturing is a production method that uses a formulaic, blending approach to producing goods (i.e., follows recipes using the ingredients and raw materials) and the type of manufacturing seen in the food, beverage, chemical, and pharmaceutical industries.

- **Discrete Manufacturing:** Discrete manufacturing is characterized by the production of lower volumes from the higher level of complexity involved. The end products are composed of identical components that could be individually counted and identified numerically (e.g., nuts, bolts, screws).

Tell me about the differences between predictive, prescriptive, and condition-based maintenance.

- **Predictive Maintenance:** Utilization of vibration, temperature, or pressure-based detection software to predict and correct asset degradation using data before the event of machine failure.
- **Prescriptive Maintenance:** Applies an additional layer of ML/AI to predictive maintenance to further improve upon the prediction accuracy and to prescribe more immediate corrective actions. Rather than predicting a potential failure alone, it comes up with specialized recommendations based on the corresponding outcomes to enhance overall efficiency.
- **Condition-Based Maintenance:** The timing of maintenance services is decided by predetermined conditions, or a certain threshold level is met. While this may sound similar to predictive maintenance, it's more about setting a contingency "rule," and to an extent, the performance deterioration may have already begun (e.g., smoke signal, less stability in temperature control, slowed cool downtime).

What are some basic unit economics for a truck transportation company you would look at to forecast revenue?

To build a revenue model for a trucking transportation company, some key historical metrics necessary first required would be the average number of shipment orders (i.e., transportation hauls) in a year, the average length of each haul in miles, and the total annual shipping revenues. Then you can back into the average revenue per mile driven. Then, to further build out the model with greater specificity, one could look at the number of clients it works with consistently, the current pipeline of shipment orders, and the number of new client contracts it could win in the years ahead. Alternatively, you could forecast the revenue on a per-driver, per-truck, or per-project basis.

What are some of the cost drivers you would look at for a truck transportation company?

The trucking industry, similar to many sub-sectors in industrials, are known for their high-cost structure.

- The most important metrics to forecast would be the costs related to the truck drivers, which includes the total number of drivers employed by the company and the compensation of the drivers (usually on a per-mile basis or hourly basis, rather than a set-base salary).
- Capex needs should then be considered, and the total number of existing trucks, the average age of the fleet of trucks, the average life-cycle of each truck, and the average annual repair costs.
- Lastly, other factors such as the sales & marketing spend (S&M), insurance costs, and rent expenses should be looked at, but these costs will not be impactful compared to the previous two.

Would you consider distribution and transportation service providers to be a low-margin business?

Historically, distribution and transportation services are known for being in the bottom-tier of margins. However, software developments such as route optimization leverage geographic information systems (GIS) to ensure the most efficient service routing should improve each transport and enable the provider to increase prices for their improved capabilities.

Other developments, such as asset location tracking, enable the real-time monitoring of parts and components through bar code/QR code scanning to improve data accuracy and more information for the client. Asset location tracking is useful for the logistics industry as it ensures the accurate identification of the inventory's location to satisfy compliance requirements if relevant (e.g., pharmaceutical drugs).

Many of these developments mentioned above serve as opportunities to improve their service offerings and differentiate themselves from competitors and improve margins.

Why does the airline industry have such low margins?

Despite providing an essential transportation service, airlines have historically suffered from razor-thin margins due to their high-fixed cost structure. The cost of purchasing new fleets is very expensive, and the constant need for fleet maintenance, repairs, and geographic footprint all cut into the margins significantly. Each airline has high variable costs (e.g., certified pilots, flight crews, key personnel, airport customer service agents, security services) to worsen the margins that have already been hampered by the highly capital-intensive nature of the industry.

Airline companies are also constrained in their ability to cut costs because of how high-stakes each expenditure is. For example, the decision to not repair a damaged plane, spend less on airline security, or reduce the number of flights could have significant reputational repercussions and wipe out a large portion of current and future bookings. There are very few costs/expenses that could be considered being “unnecessary.”

Another consideration is the competitive nature of the airline industry with various rival incumbents within the market that compete on pricing, availability of flights, and geographic reach. Therefore, airline businesses must ensure that they get their pricing strategy correct to appeal to customers, which effectively reduces their pricing power and reduces their ability to improve upon their margins.

What is the difference between fully autonomous vehicles and self-driving cars?

The terms are often used interchangeably, but a fully autonomous car would be completely self-aware and capable of making its own choices with no human supervision required (Level 5). A self-driving car can drive itself, but a human passenger is required to be ready to take control of the wheel (Level 3).

Why has the rollout of Waymo and self-driving been done at such a slow pace?

Waymo has long been considered a leading player in the self-driving technology space. Despite their head start, the project has been beset by delays and revisions to commercialization predictions. Given the regulatory challenges they face, safety has become their highest priority as their autonomous vehicles are deployed. One fatal accident could cause this venture to grind to a halt and cause reputational damage from society and mistrust in the technology being used.

What does energy management software (EMS) refer to, and what role does it play in increasing the number of so-called “smart buildings”?

Energy management software (EMS) refers to applications that track and measure energy usage, identify leakages/waste, and is the core technology that enables smart buildings. For example, EMS can enable the building owner to optimize and adjust the HVAC system, lighting controls, and energy usage based on simulation modeling and demand analysis. EMS allows the building owner to conserve energy, reduce their utility bills, and comply with sustainability regulations that depend on the city/state.

What does industrial access control involve, and what value does it provide to end-users?

Industrial access control systems manage people's entry into or out of specific areas within facilities while restricting unauthorized entry for security purposes. The mounting security concerns regarding industrial applications have created the necessity for adopting access control technologies to protect their most valuable assets. The solutions provided include a range of electronic access control devices, electronic or mechanical locks, and integrated surveillance systems to oversee the passage of people and assets through buildings and facilities. These assets protected can be tangible, in addition to intangible assets held by companies (often in

the technology vertical). Thus, industrial access control involves expensive equipment and safety hazards, but it also protects internal documents, data, and other intangibles belonging to industrial companies.

What is the difference between positive displacement pumps and centrifugal pumps?

- **Positive Displacement Pumps:** Positive displacement pumps move fluids by trapping a confined volume of the liquid and forcing this trapped amount into the discharge port's suction. The most common methods seen are rotary, reciprocating, or diaphragm pumps. The fluid movement is generated by two (or three) spindles that move in opposing directions and thus displace the liquid. A key distinction is that the flow rate will remain constant, even with a change in pressure.
- **Centrifugal Pumps:** Centrifugal pumps are used to transport fluids using the rotational energy generated by a spinning motor/engine. Once the fluid nears the impeller of the pump, the movement is accelerated. As the fluid nears this critical point, the pressure and velocity at which the fluid is pushed past the impeller increases. To summarize, centrifugal pumps pass on velocity onto the liquid to result in outlet pressure.
- **Distinction:** The primary distinction is how centrifugal pumps transmit velocity to the liquid, resulting in increased pressure for the outflow. In contrast, positive displacement pumps trap the liquid and transfer it from a suction-like discharge port. For centrifugal pumps, the pressure is first created and then flow results, but with positive displacement pumps, flow is first created, and then pressure is created.

What are the two most common types of positive displacement pumps?

1. **Reciprocating Pumps:** Based on the expansion, contraction, or reciprocating of a chamber to impact a pump's pressure to draw fluids and guide them through the discharge outlet.
 - **Diaphragm Pump:** functions similar to the human diaphragm, in which, as the diaphragm expands, the fluid is drawn into the pump and then discharged when it expands.
 - **Piston Pump:** consists of a reciprocating mechanism in which the first stroke of the pump causes suction of the fluid and the chamber of the cylinder expands, and then as the pump force reverses, the outlet opens, allowing the fluid out from the high-pressure built up.
 - **Plunger Pumps:** has an appearance similar to piston pumps, but the high-pressure seal is stationary, and a smooth cylindrical plunger slides through the seal (i.e., more suited for higher pressures). In a plunger pump, the high-pressure seal is stationary, and thus the plunger slides through the seal, allowing the pump to be used at higher pressures.
2. **Rotary Pumps:** Utilize rotating chambers that trap fluid and then transport it from the intake side to the pump's discharge side.
 - **Screw Pumps:** Uses two screws that pressurize fluids and move them in a system based on rotation against the other – the screws take in the fluid then push it out from the other side from the pressure
 - **Peristaltic Pumps:** Traps fluid in a flexible hose, and then the hose is pressurized (“pinched”) between the rollers and the inside wall of the housing – the rollers will revolve around a central axis, pushing out the fluid through the tubes and out the discharge outlet (i.e., like squeezing toothpaste)
 - **Vane Pumps:** Sliding vanes press up against the wall of the cavity when the rotor spins, creating a sealed chamber in which the fluid will be flown out
 - **Progressive Cavity Pumps:** Transfer fluids through specifically shaped cavities once the rotor spins bi-directional (i.e., cork-screw motion) and is best suited when the liquid's viscosity is high
 - **Gear Pumps:** Rigid rotating gears inside a chamber push the fluid out towards as the gears spin (“revolutions”), and the trapped fluid will be carried by the rotation and tight meshing of the gears – making these pumps well suited for high viscosity fluids

- **Lobe Pumps:** Functions similar to gear pumps, but the rotors are specifically designed to come close to touching to drive each other's movement and for items to flow through without being crushed (hence, its usage for food processing)

List a few manufacturing KPIs you would look at to measure the efficiency of a factory.

- **Throughput:** The throughput KPI measures the production capabilities of a machine, line, or plant (i.e., how much can be produced over a specified period, usually an hour or day).

$$\text{Throughput} = \frac{\text{Total Number of Units Produced}}{\text{Time}}$$

- **On-Time Delivery (OTD):** The OTD measures the percentage of orders delivered on-time – often more related to logistics performance.

$$\text{On Time Delivery (OTD)} = \frac{\text{Work Orders Delivered on Original Schedule}}{\text{Original Schedule Work Orders Due}}$$

- **Capacity Utilization:** Capacity utilization measures the proportion of potential output that's being used. The target is for the machine/factory to run close to 100% capacity as possible.

$$\text{Capacity Utilization} = \frac{\text{Actual Utilization}}{\text{Total Production Capacity}}$$

- **First Pass Yield (FPY):** The FPY KPI metric calculates the percentage of products manufactured properly to specification the first time through the process (i.e., these don't require any rework or become scrap).

$$\text{First Pass Yield Rate (FPY)} = \frac{\text{Quality Units}}{\text{Total Units Produced}}$$

- **Machine Downtime Rate:** The machine downtime rate is a basic KPI that provides a high-level view of how operations function in terms of both scheduled downtime and unscheduled downtime – a target will be set by the factory based on the scheduled downtime.

$$\text{Machine Downtime Rate} = \frac{\text{Downtime Hours}}{(\text{Downtime Hours} + \text{Operational Hours})}$$

What are some ongoing trends in the plastics industry?

- **Shift towards ESG:** The foremost trend in the plastics industry has been the increased focus on Environmental, Social, and Corporate Governance (ESG). OEMs and suppliers, whether by their own choice or due to regulations, have been assuming more sustainability responsibilities to leave a lesser impact on the environment by committing to zero-waste initiatives and the use of recycled materials.
- **Biodegradable Plastics:** One trend has been the increased usage of biodegradable plastics such as Bioresins, made from plant-based materials and thus naturally more compostable.
- **Pro-degradant Concentrates:** Another development gaining traction is conventional thermoplastics becoming more biodegradable. For example, straws are known for being very durable, which is beneficial for suppliers and consumers, but not for the environment. By having more pro-degradant concentrates added to them, these thermoplastics are becoming more eco-friendly.

What is reinforced plastic, and why are they increasingly being used?

Reinforced plastics serve as an alternative to traditional metal. Historically, the automotive industry was the leading end market; however, reinforced plastics are now gaining traction with medical, defense, and construction applications due to being lightweight yet durable with design versatility.

Most notably, carbon, glass, metal, and graphite-reinforced based plastics have been embraced as substitutes.

What impact will the trend of electric vehicles (EVs) have on the polymer market?

The imminent shift towards electric vehicles, led by Tesla, is expected to impact global polymer demand positively. Automakers prefer polymers over traditional metals to develop lighter, more energy-efficient electric vehicles. These lightweight materials not only make the vehicle conserve more energy, but they enhance the design and are manufacturable (i.e., easily molded and more versatile) for engineers. Using plastics in the automotive industry has been around for decades, but with different requirements and newer developments (e.g., smaller, more compact) in plastics, the role polymers play will become more significant as the industry finds more ways to use them.

For instance, advanced polymers such as polyphthalamide (PPA) and polyamide (PA) are expected to be the materials of choice as they can better endure the higher temperatures associated with the internal systems in EVs and support their new technologies (e.g., sensors, monitors).

What is additive manufacturing, and what benefits does it provide?

Additive manufacturing (AM), also known as 3D printing, is a technological advancement in industrial production that enables the efficient creation of lighter yet durable parts and systems through layering. AM uses computer-aided-design (CAD) and 3D object scanners to create a customizable, precise product quickly without a deterioration in quality despite the reduced cycle time.

However, the term “rapid prototyping” has become closely aligned with AM, as that’s been its primary use case so far. While the turnaround times are short (~1-2 weeks), only low-volume production runs can be performed at once. Despite the promise shown, most of the work done has been for designing and prototyping relatively small plastic parts or components.

What are the differences between injection molding and urethane casting?

- **Injection Molding:** In injection molding, the molded product is created by injecting molten plastic materials into a mold and then cooling and sliding them. The cavity comes together with the core to form a void into which the melted plastic is injected to form the plastic part (i.e., to fill the “cavities”).
- **Urethane Casting:** During urethane casting, the cavity in the silicone mold forms the cavity by pouring the liquid (the “cured silicone”) around a master pattern (the design model for the end parts to be made). Injection modeling is better suited for higher temperatures and thousands of part production cycles per day (therefore used for mass production at lower costs).
- **Distinction:** A silicone mold is not durable enough to last for high volume production but can be produced more quickly than injection molding – thus, it’s more commonly used for prototype manufacturing. In terms of part quality, both methods can produce robust parts that have better durability and impact resistance than those produced by additive manufacturing. The deciding factor comes down to the volume being produced and the stage at which the product development is.

How does a maritime/shipping company generate revenue?

A maritime/shipping company either owns or operates vessels that service the global commodity (crude/refined oil, coal, iron ore) and trade markets. A shipping company earns daily rental rate voyages secured with a counterparty such as oil majors, commodity mining companies, and finished goods suppliers.

The agreements can be for extended voyages (time charters and bareboat) or short voyages (spot charters):

1. **Time Charter:** A time charter contract defines a fixed daily rate over a defined period, usually one to three years. This provides a predictable and defined revenue stream for the shipping company for a specified period. In time charters, the charterer (e.g., the counterparty like an oil major) usually pays voyage expenses (such as port and fuel costs for the vessel).

2. **Bareboat Charter:** A bareboat charter is a long-term time charter, typically lasting longer than five years. As the name implies, the charterer gets a bareboat in bareboat charters and covers virtually all expenses tied to operating the ship, including the vessel fuel, bunkers, port dues, and vessel operating expenses (e.g., day-to-day operations, maintenance, crewing, and insurance). The vessel owner receives daily charter hire payments and is responsible only for the vessel's capital costs.
3. **Short Voyages (Spot Charter):** While the time and bareboat charters provide the shipping company with a long-term fixed payment, spot charters are based on a fixed daily rate (usually based on the weight of cargo) for a specific journey from a loading port to a discharge port. Spot charters are usually much shorter than time charters, lasting 90-120 days, as the term is specifically tied to a small defined number of voyages. In contrast to long-term or bareboat charters, spot charters are a relatively more volatile source of revenue since the daily rental rate is agreed upon based on the marketplace's current dynamics and could change. In addition, unlike longer time or bareboat charter, the shipping company usually covers voyager expenses such as port and fuel costs, which drive the daily rate higher. From the charterer's perspective, spot charters are full-service short-term rentals, while a time charter is typically a one to three-year lease where the charterer has some responsibility to cover expenses. The bareboat charter is a long-term lease where the charterer is almost like an owner, covering all expenses and getting a no-frills empty boat.

How do the financials of a maritime/shipping company differ from a traditional company?

A maritime company's financials are similar to those in other industries, but there are a few differences.

- First, depreciation and amortization are typically separately identified on the income statement, whereas in other industries, these line items are usually baked within operating expenses like COGS, SG&A, or R&D expense. Here, it's relatively easy to derive a top-down EBITDA, but you would need to remove the effect of non-recurring charges, if applicable.
- Second, taxes are minimal to non-existent because companies are typically domiciled in "tax havens" where they're exempt from federal taxes. However, if there are specific jurisdictional taxes related to transportation income that the company has to pay, they'll show up on the income statement. For example, various jurisdictions (like the US) change a tonnage tax on the amount of cargo freighted within their territories.

Walk me through a shipping company's income statement.

A shipping company's income statement is typically broken down by revenue and operating and non-operating expenses. Revenues are typically generated from various activities (i.e., time charter revenue, spot charter revenue, or pooled charter revenue) and adjusted for ownership percentages.

The operating expenses of a shipping company typically consist of:

- **Voyage Expenses:** Port and fuel costs and transport taxes in certain applicable jurisdictions.
- **Vessel Operating Expenses:** Crew costs, provisions, deck and engine stores, lubricating oil, insurance, maintenance, and repairs incurred.
- **G&A:** Salaries, legal costs, and stock-based compensation.
- **D&A:** Related to newly acquired vessels and other fixed assets.
- **Other Expenses:** FX gains or losses and goodwill impairment are captured here as well.

Then, non-operating expenses will consist of net interest expense and other expenses.

Walk me through a shipping company's balance sheet.

- **Assets:** For a shipping company, the vessels and vessels under construction comprise the largest assets on the balance sheet, along with cash and cash equivalents.

- **Liabilities:** Current and long-term debt represents the largest liabilities. A key characteristic to be aware of is that shipping companies carry little to no working capital.
- **Equity:** Consists of common stock and treasury stock, preferred stock, retained earnings, and other comprehensive income.

How do you value a shipping company?

The most common valuation models employed to value a shipping company include:

- Net Asset Value (NAV)
- Enterprise Value Multiples (e.g., EV/EBITDA, EV/EBIT)
- Equity Value Multiples (e.g., Price to Cash Flow, Price/NAV)

Walk me through a NAV valuation for a shipping company.

The most common maritime valuation is the net asset value (NAV) model, which involves:

1. **Value EBITDA-Generating Assets:** This part refers to valuing the vessels. The market values of vessels are readily available in industry trade journals based on selling/purchasing activity in the vessel marketplace. To value vessels, start by computing the actual steel value based on market benchmarks, which will represent the bulk of the valuation. Then, analysts typically conduct a discounted cash flow analysis on a company's future time-charter profile by calculating the difference between the daily rate fixed on each vessel and the forward curve of expected time charter rates, as published in industry trade journals. The typical WACC used is 10%, but the PV of the embedded contract value is added to the steel value to obtain a "charter adjusted" value per vessel. This represents the "enterprise value" based solely on vessel value.
2. **Subtract Debt & Other Non-Equity Claims Not Yet Valued:** When subtracting debt and debt-like claims, the book value is often used under the thinking that book values for these liabilities don't deviate too far from the market values. But if there's reason to believe deviations exist, market values should be used.
3. **Add Non-Operating Assets:** Any non-operating assets not already included in the NAV build up like cash and construction in progress should be added to the NAV buildup. These can be valued at book value or the projected market value in the case of construction in progress.
4. **Ending Valuation:** The resulting value represents the NAV-derived equity value of the shipping company. This can be compared against the company's market cap or divided by shares outstanding to compare against the shipping company's current share price to calculate Price/NAV. A company trading at a premium to NAV is a positive for stock price performance.

Why might the NAV be preferable to a DCF for shipping companies?

The NAV approach is suited for shipping companies because, unlike other industries, shipping company balance sheets carry real assets for which there are often comparable assets with observable market values. The NAV approach is essentially recalculating a shipping company's balance sheet to reflect fair market values of assets. This is simply not a feasible option for most industries, which is why a discounted cash flow analysis is often employed for them instead.

What are some factors that may affect the valuation of a shipping company?

Valuation is influenced by Daily Revenue, Charter Rates, and Vessel Asset Values.

These are influenced by two factors:

1. The supply and demand of the underlying commodities being shipped
2. The supply and demand of actual vessels

Supply and demand dynamics of the underlying commodity and/or products being transported:

- Crude oil and refined oil product markets
- Liquefied natural gas (LNG) markets
- Dry bulk commodities such as coal, iron ore, and grain
- Global freight transport through containerships
- Supply and demand dynamics of existing vessels “on the water” and vessels currently being built (i.e., “on the global order book”)

Ships typically take 18 months to construct, and there typically is a frenzy of oversupply in the marketplace when underlying commodity demand is strong, which produces a supply lag, and makes timing (i.e., when the vessel owner bought, constructed, and finally deployed the vessel on voyages) a crucial element in this space.

In the years leading up to the financial crisis, there was a major uptick in ship orders that increased the order backlog, and once those ships were ready to be deployed, the financial crisis hit, and ships were left idling “on-the-water” and were also for lack of a better term, financially underwater.

Oil & Gas (O&G)

What are the three major subsectors that make up O&G? Tell me a bit about each.

1. **Upstream:** Upstream consists of Exploration and Production (E&P) companies and Oil-Field Service (OFS) companies. OFS companies run drilling rigs and prepare seismic and geological/geophysical data to support E&Ps, which actually end up producing the oil and gas for sale.
2. **Midstream:** Midstream companies are primarily involved in the transportation (pipelines, rail, etc.) and storage of hydrocarbons (crude and refined product terminals, gas, and NGL storage facilities). Processing gas into natural gas liquids is also considered midstream, unlike the refining of crude.
3. **Downstream:** Downstream refers to refining crude into motor gasoline ("mogas"), distillates, and others.

How do you value each of the subsectors of O&G?

- OFS and refining are typically valued similarly to the standard company: EV/EBITDA, P/E.
- Midstream might be valued on an EV/EBITDA basis, but it's complicated because many midstream companies are organized as Master Limited Partnership (MLPs) and pay Incentive Distribution Rights (IDRs) to the General Partner. The "GP burden" should be included as part of Net Debt in the numerator of EV/EBITDA. Otherwise, midstream companies are valued using a dividend discount model (or distribution discount model for MLPs).
- While E&Ps can be looked at on an EV/EBITDA or EV/EBITDAX basis, the predominant way of valuing E&Ps is via a Net Asset Value Model. Essentially, it's a long-term DCF (30+ years into the future) where the E&P's reserves are depleted to zero, and cash flows are calculated over that long-term time frame. You would then include any balance sheet adjustments (net debt, non-operating assets, asset retirement obligations, hedge assets/liabilities, and G&A costs).

Name a few O&G industry specific multiples beyond the typical multiples.

- **EV/EBITDAX:** EBITDAX consists of an adjustment to EBITDA by adding back exploration expenses for successful efforts companies.
- **EV/DACF:** DACF is an abbreviation for debt-adjusted cash flow, calculated as cash from operations plus after-tax interest expense (could be before or after working capital changes).
- **EV/Daily Production:** This calculation results in a dollar figure and the denominator is based on either BOE or CFE production (aka EV/flowing barrel).
- **EV/Reserves:** In the US, this is usually calculated off of Proved Reserves, so it would technically be EV/1P. Canadian IFRS allows proved + probable reserves in filings, so the calculation would be EV/2P.
- **Acreage Value:** On many transactions, the most valuable asset may be undeveloped acreage. Say a Permian Basin operator acquires an operator nearby: it'll be worth some amount for its current production (flowing production), but a lot of the purchase price is the undeveloped acreage (for the Permian, you may see prices for \$30,000/acre up to over \$60,000/acre, but note that each region has different metrics).

Describe the difference between full cost accounting and successful efforts accounting for E&Ps, and then the impact on the income statements and balance sheets of identical companies.

There are a few major differences between the two:

- Successful efforts accounting expenses geological/geophysical (G&G) costs as they're incurred and expensing exploratory dry holes immediately.
- Development dry holes are still capitalized under successful efforts.
- Full cost companies capitalize G&G costs, as well as capitalize exploratory dry holes.

Assuming the companies are identical, successful efforts will show lower initial profits since they're expensing both G&G and exploratory dry holes immediately. However, in later years, full cost companies should show lower profits since both exploratory dry holes and G&G costs are capitalized (making DD&A higher, as well as a greater likelihood of an impairment/write-down).

Describe the characteristics that make O&G different from more traditional industries, such as consumer staples.

- **Commodity Pricing:** O&G companies are "price-takers" and have little control in pricing (indirect control based on hedging policy and where to sell production).
- **Depleting Assets:** Oil/Gas fields suffer from a natural decline element, so companies must continuously drill to offset this decline.
- **Asset Intensive:** In an M&A environment, there should be little to no Goodwill recognized as Upstream/Midstream/Downstream assets are all easily identifiable.
- **Cyclical:** O&G goes through severe boom and bust moments, so it's difficult to predict.
- **SFAS 69:** An unique aspect for E&Ps is that they're required to value their E&P operations per SFAS 69.

For oil & gas companies, what is the SMOG disclosure requirement?

SFAS 69 requires the public disclosure of a standardized measure of discounted future cash flows from proved oil and gas reserves quantities – this is often referred to as the standardized measure of oil and gas, or SMOG.

The SMOG has several limitations because it uses historical pricing and historical costs (which cannot be escalated), excludes hedging and G&A, and companies are required to use a 10% discount rate. Companies can only value Proved Reserves and don't include any non-E&P assets like midstream or refining assets.

Describe the different categories of reserves.

- **Proved Reserves (1P):** 90% chance that the indicated reserve is there or larger
- **Probable Reserves (1P+Probable Reserves=2P):** 50% chance the probable reserves are there or larger
- **Possible Reserves (2P+Possible Reserves=3P):** 10% chance the possible reserves are there or larger

For reserves, how do 10-Ks under GAAP differ from IFRS filings?

Official GAAP filings (10-Ks) only allow Proved reserves to be shown (non-official reports like Investor Relations may show 3P reserves); IFRS allows companies to show 2P reserves.

What are some differences between the financial statements for an E&P company vs. a regular company?

- Realized and/or unrealized gains on hedges
- Instead of D&A, E&P companies show DD&A (depletion, depreciation & amortization)
- Asset retirement obligations are common (essentially environment remediation when it comes time to plug and abandon a well or wells)
- Exploration expense: shown for successful efforts since these companies expense exploratory dry holes and G&G costs

What are some metrics that would be looked at in an O&G M&A model?

- Aside from accretion/dilution to EPS (for the most part, only applicable to large companies), the focus would be accretion or dilution to NAV.
- Debt-adjusted per share growth refers to production growth that excludes capital structure (i.e., neutral to how the company is financed). Typically, debt would be converted to shares assuming a quarter-end share price for this calculation. This metric allows comparison between companies that use debt vs. equity to finance operations.

- Reserves to Production ratio (R/P) is calculated as the reserves at the end of the year divided by the production during that year. Quick metric to determine how many years of proved reserves are left.
- Production replacement ratios: Ideally, a company can replace its production. Defined as new reserves divided by production. New reserves may or may not include acquisitions of reserves. It's better for a company to find its own reserves vs. acquiring reserves.
- F&D stands for finding and development costs, and its calculation is similar to Production Replacement ratios (except it's in dollars rather than percentages). The numerator is based on a company's total costs (whether capitalized or expensed) divided by new reserves. It can also be calculated by excluding acquisitions in the numerator and excluding reserves from purchases in the numerator.

Would an E&P company be a good candidate for an LBO? Why or why not?

No, due to the lack of control over prices and unpredictable cash flows, an E&P company would not be an ideal candidate for an LBO.

What are IDCs, and what type of companies can use them?

IDCs stand for intangible drilling costs, which can be immediately expensed for tax purposes for non-integrated companies (lack of refining, for simplicity).

What is the energy equivalency between a barrel of oil and natural gas in cubic feet?

One barrel of oil has the energy equivalence of approximately 6,000 cubic feet of gas (6 mcf). Using that relationship, you could convert back and forth between a barrel of oil equivalent (BOE) and an mcf (thousand cubic feet equivalent).